SNF 79

ANALYSIS OF MUSSEL BEDS IN THE LITTLE MISSOURI AND SALINE RIVERS, BLUE MOUNTAIN, OZARK AND DARDANELLE LAKES, ARKANSAS

Christopher Lee Davidson

156 Pages August 1997

The objectives of this study were to map the areal extent and determine selected community parameters for the mussel beds of the Little Missouri and Saline rivers, Blue Mountain, Ozark and Dardanelle lakes in Arkansas.

APPROVED:

7/10/97	George L Harp
Date	Dr. George L. Harp, Adviser
7/2/97	John L. Harris
Date	Dr. John L. Harris, Co-adviser
7/7/97	Jerry L. Farris
Date	Dr. Jerry L. Farris
10 Jul 97	V. Rick McDaniel
Date	Dr. V. Rick McDaniel, Department Chair
Date	Dean of Graduate School

ANALYSIS OF MUSSEL BEDS IN THE LITTLE MISSOURI AND SALINE RIVERS, BLUE MOUNTAIN, OZARK AND DARDANELLE LAKES, ARKANSAS

A Thesis

Presented to

the Faculty of the Graduate School

Arkansas State University

In Partial Fulfillment

of the Requirements for the Degree of

MASTER OF SCIENCE

Department of Biological Sciences

by

CHRISTOPHER L. DAVIDSON

August 1997

An Abstract

of

ANALYSIS OF MUSSEL BEDS IN THE LITTLE MISSOURI AND SALINE RIVERS, BLUE MOUNTAIN, OZARK AND DARDANELLE LAKES, ARKANSAS

August 1997

The primary purpose of this research project was to define and map the location and areal extent of mussel beds in the Little Missouri and Saline rivers, Blue Mountain, Ozark and Dardanelle lakes. Secondary objectives included determining community structure, population estimates, size frequency distributions, percent legal harvestable mussels of commercially valuable species, survivorship of native mussels exposed to fouling by the introduced zebra mussel in Ozark and Dardanelle lakes and age-growth relationships for two commercially valuable species. This study provides baseline data, which completes a statewide investigation by Rust (1993), Harris et al. (1993), Christian (1995), Posey (1997) and this report, as no investigation existed of these parameters on these three reservoirs and two rivers.

River and reservoir substrates were explored with the aid of an oil-less surface air compressor with a breathing line attached to a regulator. Three levels of sampling intensities were implemented in the rivers: qualitative notes were taken for areas with mussel densities < 10 mussels/m^2 with no limitation to area; minor beds were categorized by areas with mussel densities ≥ 10 mussels/m^2 with an overall area <500 m^2; major beds were

categorized by areas with ≥ 10 mussels/m^2 with an overall area > 500 m^2. Minor bed samples were determined by five non-random samples from the areas with the greatest density. Major beds consisted of 10-25 samples, which were determined by computer generated random numbers. Sampling size consisted of a 1 m^2 quadrat. The same general levels of sampling intensity existed for the reservoirs with one exception: Both minor and major beds were determined to be areas with mussel densities averaging ≥ 1 mussel/m^2.

A total of 548 individuals was collected from four minor beds in the Little Missouri River, and 34 species were identified. *Pleurobema pyramidatum* dominated with 42% of the total. Mean density was 27.4 mussels/m^2. A total of 3,278 individuals was collected from five major and five minor beds in the Saline River. *Amblema plicata* dominated with 35% of the total. Mean density for major and minor beds was 28.8 and 27.2 mussels/m^2, respectively.

No beds were defined in Blue Mountain Lake. One live specimen of *Anodonta suborbiculata* was collected during this study. A total of 44 individuals was collected from three major beds in Ozark Lake. Eleven species were identified. *Quadrula quadrula* dominated with 64% of the total. Mean density was 0.68 mussel/m^2. A total of 323 individuals was collected from five major and one minor bed in Lake Dardanelle. Eighteen species were identified. *Plectomerus dombeyanus* dominated with 47% of the total. Mean density for major and minor beds was 3.8 and 1.0 mussels/m^2, respectively.

Age at minimum legal harvest size (69.85 mm) for Saline River *Amblema plicata* was calculated to be 19.1 years. Age at minimum legal harvest size (69.85 mm) for the Ozark Lake *Quadrula quadrula* was calculated to be 14.7 years. The same species from Lake Dardanelle reached minimum legal harvest size at 15.4 years.

No legally harvestable mussels were collected from the Little Missouri River. Percent legal harvestable mussels from the Saline River was 0.15%. Of the mussels collected from Ozark and Dardanelle lakes, 16.0 and 14.5% were of minimum legal harvest size, respectively.

ACKNOWLEDGEMENTS

I express my greatest gratitude to Dr. George L. Harp and Dr. John L. Harris for their invaluable advice over the last two years. Dr. Harp has been a great mentor both in and out of the classroom. Dr. Harris also deserves a special thanks for his technical advice in the field and for his patience and forgiveness on that miserable afternoon when the lower unit of the boat motor mysteriously disappeared to the black depths of the Arkansas River (Luckily, Santa Claus brought him another lower unit later that summer). However, a few prayers were said and the lost lower unit was laid to rest, with great despair.

I would also like to thank Dr. Jerry Farris for his unyielding support and time in teaching me the basics of thin-section age-growth techniques for mussels. I hope Dr. Farris appreciated my lesson on diving the White River as much as I enjoyed his lessons (It was difficult to convince him that there really was not a White River Monster on the bottom of the river with him). I also appreciate all the help J. D. Wilhide contributed to the thin-sectioning of the mussels. Also, Alan Christian was a great help in determining the von Bertalanffy growth models.

I would like to thank Bill Posey for teaching me everything I know about blackwater diving. I would also like to thank Bill for being patient on those days that I moaned about there really not being an Arkansas-Louisiana state line on the Ouachita River (We had to complete the Ouachita River before I could start my research). I would also like to thank Brady Baker and Patrick Daniel for assisting in the field.

I am especially thankful to my wife for her support through all of graduate school. I am also thankful to the funding agencies who made this project possible, Arkansas Game and Fish Commission, U. S. Fish and Wildlife Service and U. S. Army Corps of Engineers, and anybody else that I may have forgotten that helped with this research project.

CLD

TABLE OF CONTENTS

ANNOTATION.....i

ABSTRACT.....iv

ACKNOWLEDGEMENTS.....vii

TABLE OF CONTENTS.....ix

LIST OF TABLES.....xiii

LIST OF FIGURES.....xvi

CHAPTER

I. INTRODUCTION.....1

Rivers Studied.....2

Little Missouri River.....2

Saline River.....5

Lakes Studied.....6

Blue Mountain Lake.....6

Ozark and Dardanelle Lakes.....7

Unionacea.....8

Natural History.....8

Environmental Stresses.....10

Habitat Manipulation.....12

Exploitation.....13

Exotic Species.....16

Statement of the Problem.....17

Purpose of Study.....19

II. METHODS AND MATERIALS.....20

Field Methods.....20

Statistical Treatment.....23

Curation.....24

Age Determinations.....25

Growth Model.....28

Study Area.....28

Little Missouri River.....28

Saline River.....29
Blue Mountain Lake.....29
Ozark Lake.....29
Lake Dardanelle.....29

III. COMMUNITY ESTIMATES.....43
Little Missouri River.....43
Results.....43
Minor Beds.....43
Endangered Species.....48
Discussion.....48
Saline River.....49
Results.....49
Major Beds.....49
Minor Beds.....50
Endangered Species.....50
Discussion.....50
Blue Mountain Lake.....58
Results.....58
Ozark Lake.....58
Results.....58
Major Beds.....59
Endangered Species.....63
Discussion.....63
Lake Dardanelle.....64
Results.....64
Major Beds.....64
Minor Beds.....67
Follow-up.....67
Endangered Species.....71
Discussion.....71

IV. SIZE ANALYSIS AND AGE DETERMINATIONS.....73
SIZE ANALYSIS
Little Missouri River.....73
Results.....73
Discussion.....76
Saline River.....76
Results.....76
Discussion.....78
Ozark Lake.....81
Results.....81
Discussion.....81
Lake Dardanelle.....81
Results.....81

Follow-up....86
Discussion....86
Recruitment....92
AGE DETERMINATIONS....93
Saline River....93
Results....93
Discussion....93
Ozark Lake....102
Results....102
Discussion....111
Lake Dardanelle....111
Results....111
Discussion....120

V. MANAGEMENT RECOMMENDATIONS....121

LITERATURE CITED....123

Appendix 1.1 Physical parameters, species composition, percent legal harvestable mussels and population and community estimates for the minor beds of the Little Missouri River, Arkansas, 1996....128

Appendix 1.2 Physical parameters, species composition, percent legal harvestable mussels and population and community estimates for the major beds of the Saline River, Arkansas, 1996....132

Appendix 1.3 Physical parameters, species composition, percent legal harvestable mussels and population and community estimates for the minor beds of the Saline River, Arkansas, 1996....138

Appendix 1.4 Physical parameters, species composition, percent legal harvestable mussels and population and community estimates for the major beds of Ozark Lake, Arkansas, 1996....143

Appendix 1.5 Physical parameters, species composition, percent legal harvestable mussels and population and community estimates for the major beds of Lake Dardanelle, Arkansas, 1996....146

Appendix 1.6 Physical parameters, species composition, percent legal harvestable mussels and population and community estimates for the minor beds of Lake Dardanelle, Arkansas, 1996.....151

Appendix 1.7 Physical parameters, species composition, percent legal harvestable mussels and population and community estimates for a follow-up on selected major beds of Lake Dardanelle, Arkansas, 1996.....152

Appendix 2.0 Scientific nomen and common names of Unionacea known to occur in Arkansas, 1997.....155

LIST OF TABLES

Table Page

1. Commercial shell taking licenses, mussel shell harvest size & value of harvest from 1981-1995 (Burnley, 1996).....14

2. Arkansas Game and Fish Commission's minimum legal dimensions in mm for harvest of commercial mussel species in Arkansas, 1996, and parameters measured for all species. Equivalent inch measurements are shown parenthetically.....15

3. Unionacea collected from study areas of the Little Missouri and Saline rivers and Blue Mountain, Dardanelle and Ozark lakes, Arkansas, 1995-1997.....44

4. Relative abundance, in percent of total, of species from Little Missouri River minor beds.....46

5. Diversity and other indices for the minor and major of the Little Missouri and Saline rivers.....48

6. Relative abundance, in percent of total, of species from Saline River major beds.....50

7. Relative abundance, in percent of total, of species from Saline River minor beds.....51

8. A comparison of Arkansas mussel beds in the Little Missouri, Saline, Ouachita, St. Francis, White, Cache, Spring and Black rivers, Arkansas.....57

9. Relative abundance, in percent of total, of species from Ozark Lake major beds.....60

10. Diversity and other indices for the major beds of Dardanelle and Ozark lakes.....61

11. Number of adult *Dreissena polymorpha* attached to live Unionacea collected from study sites on Ozark Lake, Arkansas, 1996.....62

12. Relative abundance, in percent of total, of species from Lake Dardanelle major beds.....67

13. Number of adult *Dreissena polymorpha* attached to live Unionacea collected from study sites on Lake Dardanelle, Arkansas, 1996.....69

14. Relative abundance, in percent of total, of species from Lake Dardanelle major Beds 1-3M, 1996-1997.....70

15. Size frequency distribution of selected mussel species, Little Missouri River, Arkansas, 1996.....74

16. For selected mussel species from the Little Missouri River, Arkansas, 1996, mean values ($\bar{x}$) and standard deviations (S) for the measurements used by the Arkansas Game and Fish Commission to determine legal size.....75

17. Percent legal harvestable mussels, commercial species only, in the Little Missouri and Saline Rivers, Arkansas, 1996.....77

18. Depth frequency distribution of selected mussel species, Saline River, Arkansas, 1996.....79

19. For selected mussel species from the Saline River, Arkansas, 1996, mean values ($\bar{x}$) and standard deviations (S) for the measurements used by the Arkansas Game and Fish Commission to determine legal size.....80

20. Depth frequency distribution of selected mussel species, Ozark Lake, Arkansas, 1996.....82

21. For selected mussel species from the Ozark Lake, Arkansas, 1996, mean values ($\bar{x}$) and standard deviations (S) for the measurements used by the Arkansas Game and Fish Commission to determine legal size.....83

22. Percent legal harvestable mussels, commercial species only, in Ozark Lake and Lake Dardanelle, Arkansas, 1996.....84

23. Size frequency distribution of selected mussel species, Lake Dardanelle, Arkansas, 1996.....85

24. For selected mussel species from the Lake Dardanelle, Arkansas, 1996, mean values ($\bar{x}$) and standard deviations (S) for the measurements used by the Arkansas Game and Fish Commission to determine legal size.....87

25. Depth frequency distribution of selected mussel species in major Beds 1-3M, Lake Dardanelle, Arkansas, 1996-1997.....88

26. Percent legal harvestable mussels, commercial species only, in major Beds 1-3M, Lake Dardanelle, Arkansas, 1996-1997.....91

27. Length, depth, width, dry weight and age of selected *Amblema plicata* used for thin-sectioned age determinations from the Saline River, Arkansas.....94

28. Length, depth, width, dry weight and age of selected *Quadrula quadrula* used for thin-sectioned age determinations from the Ozark Lake, Arkansas.....103

29. Length, depth, width, dry weight and age of selected *Amblema plicata* used for thin-sectioned age determinations from the Lake Dardanelle, Arkansas.....112

LIST OF FIGURES

Figure Page

1. Selected drainages of Arkansas sampled during the present study.....3

2. Minor (m) bed locations from the Little Missouri River, Arkansas.....31

3. Major (M) and minor (m) bed locations from the Saline River, Arkansas.....33

4. Commercial beds (Δ) surveyed from Blue Mountain Lake, Arkansas.....35

5. Major (M) bed locations and commercial beds (Δ) surveyed from Ozark Lake, Arkansas.....37

6. Major (M) bed locations and commercial beds (Δ) surveyed from Ozark Lake, Arkansas.....39

7. Major (M) and minor (m) bed locations and commercial beds (Δ) surveyed from Lake Dardanelle, Arkansas 41

8. Description of the typical location of mussel beds in a river system (as seen in Posey, 1997).....52

9. Description of a typical mussel bed located in a bay of Lake Dardanelle.....65

10. Scatterplot of depth and age correlations for *Amblema plicata*, Saline River, Arkansas.....96

11. Scatterplot of length and age correlations for *Amblema plicata*, Saline River, Arkansas.....98

12. Scatterplot of width and age correlations for *Amblema plicata*, Saline River, Arkansas.....100

13. Scatterplot and growth curve, predicted limits of depth and age correlations for *Quadrula quadrula*, Ozark Lake, Arkansas.....105

14. Scatterplot and growth curve, predicted limits of length and age correlations for *Quadrula quadrula*, Ozark Lake, Arkansas.....107

15. Scatterplot and growth curve, predicted limits of width and age correlations for *Quadrula quadrula*, Ozark Lake, Arkansas 109

16. Scatterplot and growth curve, predicted limits of depth and age correlations for *Quadrula quadrula*, Lake Dardanelle, Arkansas.....114

17. Scatterplot and growth curve, predicted limits of length and age correlations for *Quadrula quadrula*, Lake Dardanelle, Arkansas.....116

18. Scatterplot and growth curve, predicted limits of width and age correlations for *Quadrula quadrula*, Lake Dardanelle, Arkansas.....118

Chapter I

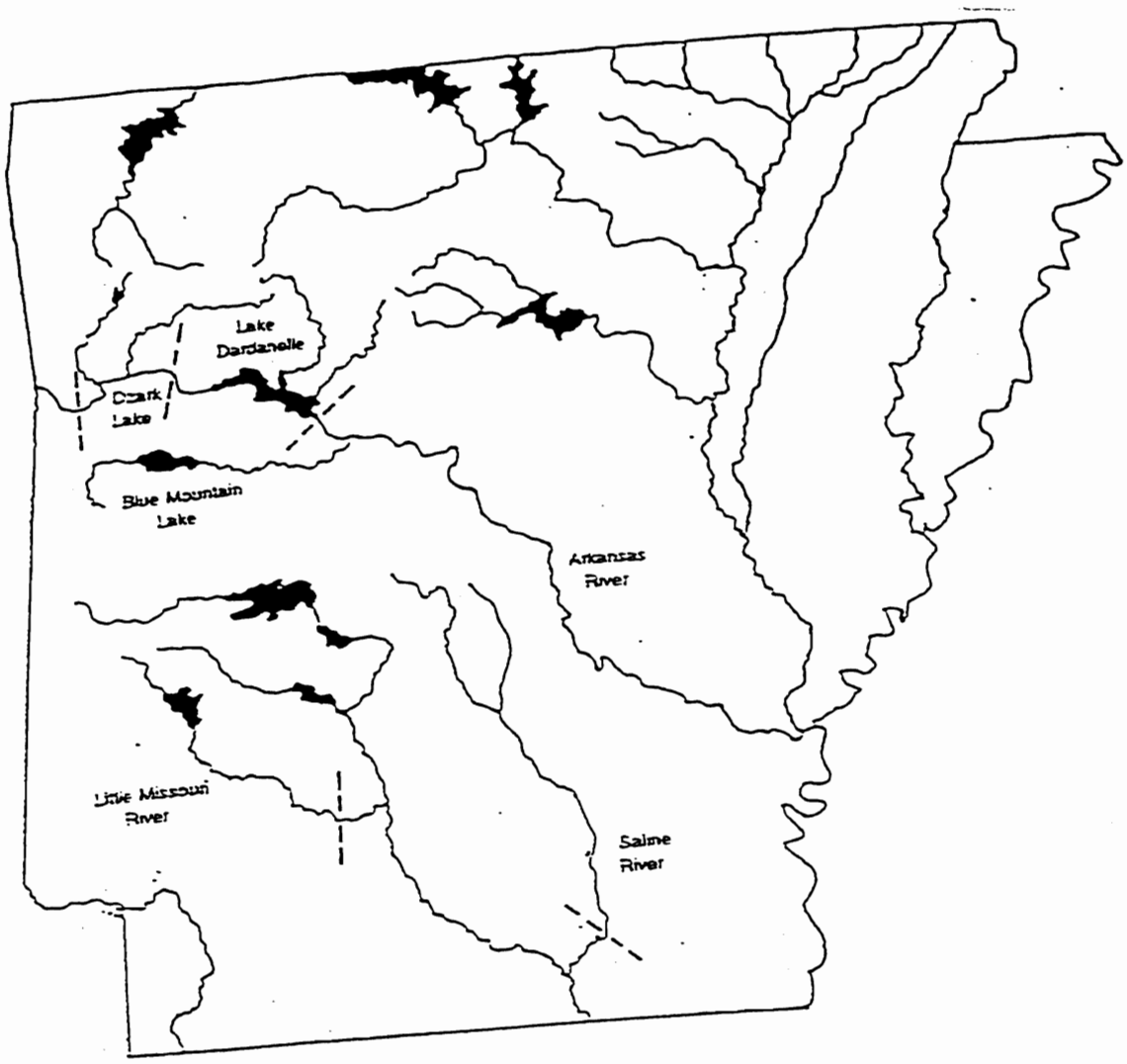
INTRODUCTION

Arkansas covers approximately 13,765,182 hectares (ha), of which 45.9% is devoted to agriculture. The state has six major river basins (Red, Ouachita, Arkansas, White, St. Francis and Mississippi) with a total of 140,188 river and stream kilometers (km), and there are 208,196 ha of lakes, reservoirs and ponds (ADPCE, 1994). The varying geography and topography which provide numerous habitats that contrast between ecoregions have contributed to the diversity of the aquatic fauna in Arkansas (Gordon, 1980; Robison and Buchanan, 1988).

Factors determining the distribution of aquatic fauna may be grouped into three general categories: historical, physical and biological. Historical factors include geological and climatic changes, while physical factors are exemplified by composition of dissolved and suspended organic and inorganic substances, substrate type and gaseous content. Biological factors include food and reproduction as well as the biological interrelationships among different species (Gilbert, 1980). In Arkansas all three have interacted to produce a diverse fish (Robison and Buchanan, 1988) and molluscan (Gordon, 1980; Gordon et al., 1994) fauna.

Freshwater mussels (Suborder Unionacea) reach their greatest diversity in North America with about 297 recognized taxa. Currently, 213 taxa (71.7%) are considered endangered, threatened or of special concern. Mussels often serve as indicator organisms in freshwater ecosystems (Williams et al., 1993). Due to the imperilment of Unionacea, efforts are being made

Figure 1. Selected drainages of Arkansas sampled during the present study.


Lake
Dardanelle
Ozark
Lake
Blue Mountain
Lake
Arkansas
River
Little Missouri
River
Saline
River

remainder. The primary land use in this drainage basin is forestry. Robison and Buchanan (1988) reported 80 species of fishes from the Little Missouri River. Wheeler (1918) documented the Mollusca of Clark County, Arkansas; however, no references were made specifically to the Little Missouri River. Gordon et al. (1979) reported 11 species of Unionacea from the Little Missouri River.

Saline River

The Saline River flows southward before reaching its confluence with the Ouachita River near the Arkansas-Louisiana state line (Figure 1). The total drainage basin is 8,417.5 km^2 (USGS, 1979). The Saline River begins on the east end of the Ouachita Mountains as four distinctive branches (South, North, Middle and Alum forks). These drainages converge north of Benton, Arkansas, just before the Saline River enters the Gulf Coastal Plain. The river basin is gently rolling, and some lands adjacent to larger streams are swampy. As the river enters the Gulf Coastal Plain it flows over less resistant rocks of the Cretaceous and Tertiary Ages. Surface rocks consist of sandstone, shale and chert (Croneis, 1930).

The U.S. Army Corps of Engineers (USACE) operates a series of navigational structures on the lower Ouachita River. The most downstream structure occurring in Arkansas, Felsenthal Lock and Dam, is located approximately 20.8 river km downstream of the confluence of the Saline River with the Ouachita River. This structure provides a 6.0 m lift for 104 river km upstream on the Ouachita River and a 1.6 m fish and wildlife pool used to manage the 162.5 km^2 Felsenthal National Wildlife Refuge. The study area for the Saline River lies within the boundaries of Felsenthal NWR. Physical characteristics of the river are affected by the lift provided by Felsenthal Lock and Dam. Robison and Buchanan (1988) reported 104 species of fishes from the Saline River. Gordon et al. (1979) reported 35 species of Unionacea from

the Saline River. Posey et al. (1996) reported 36 species of Unionacea from the lower Ouachita River in Arkansas.

Lakes Studied

Blue Mountain Lake

Blue Mountain Dam is located on the Petit Jean River at mile 74.4, 1.2 km southwest of Waveland, Arkansas (Figure 1). There are about 1,264 km^2 in the drainage basin above the dam (USGS, 1970). The predominant geologic formation within the Petit Jean River drainage is the upper Atokan. Surface rocks in the Arkansas River Valley are primarily sandstone and shale with alluvial deposits along the larger waterways. The terrain is level to mountainous with erosional and folded hills, cuestas, hogbacks and buttes. The higher elevations of the headwater area are dominated by the Euders-Mountainburg Association, a well-drained, gently sloping to steep, deep and shallow, loamy and gravelly or stony soil on hills and mountains. The valleys in the upper drainage are dominated by the Leadvale Association, a moderately well-drained, nearly level to gently sloping, deep, loamy soil, on old stream terraces in broad valleys. The narrow floodplain of the main stream channel is dominated by the Barling-Spadra Association, a moderately well-drained to well-drained, level to nearly level, deep, loamy soil on floodplains and terraces. Dominant vegetation includes shortleaf-pine and upland hardwoods. Land use is evenly distributed between forestlands (52%) and agricultural lands (45%) (ADPCE, 1987).

Blue Mountain Lake was one of several USACE projects constructed in the Arkansas River Basin for the primary purpose of flood control. Construction of the outlet works was initiated in 1940, and completion of the overall project was accomplished in June 1947 at a cost of $4.7 million. Surface areas of the lake at top of the flood-control pool and the conservation pool are 4,453 and 1,178 ha, respectively. Flood-control and conservation pool storage capacities are

233,000 and 25,000 acre feet. Shoreline lengths at top of flood-control pool and conservation pool are 80 and 142.4 km, respectively (USACE, 1974). No records exist for Unionacea in Blue Mountain Lake.

Ozark and Dardanelle Lakes

Locks and dams are major units in the McClellan-Kerr Arkansas River Navigation System which includes the Arkansas River and its tributaries in Arkansas and Oklahoma. This system was authorized by the River and Harbor Act of 1946 and modified by subsequent acts of Congress. The navigation system includes several multiple-purpose lakes located in western Arkansas and eastern Oklahoma, a channelized navigation route from Catoosa, Oklahoma (18.8 km east of Tulsa) to the Mississippi River, and bank stabilization and channel rectification works to control meanderings of the river. The canalized navigation route includes 13 low-head navigation locks and dams and four high-head locks and dams for navigation and power (USACE, 1987). Within Arkansas, the Arkansas River Valley is a gently undulating plain that varies in width from 40-56 km. Surface rocks consist of sandstone and shale of Carboniferous Age, and nearly all of them belong to the Pennsylvanian series (Croneis, 1930).

Ozark Lake is formed on the Arkansas River by Ozark Lock and Dam at Ozark, Arkansas, at Arkansas River mile (ARM) 256.8 and extends upstream to J.W. Trimble Lock and Dam at ARM 292.8 (Figure 1). Ozark Lock and Dam construction was initiated in June, 1965, and completed for navigation in November, 1969. J.W. Trimble Lock and Dam construction was initiated in April, 1966, and completed for navigation in August, 1969. Channel length between the two locks and dams is 57.6 river km, minimum channel depth is 2.75 m. At normal navigation pool, shoreline length is 276.8 km with a surface area of 4,291.5 ha. The total drainage basin above Ozark Lock and Dam is 389,917 km^2 (USACE, 1987). Stoeckel et al. (1996) reported 11 species

of Unionacea from the Mulberry River, a tributary that enters the Arkansas River at Ozark Lake. Gordon (1985) reported 17 species of Unionacea from Frog Bayou, also a tributary which enters the Arkansas River at Ozark Lake. No published records exist for Unionacea in Ozark Lake.

Lake Dardanelle is a 13,800 ha lake located on the Arkansas River and is formed by Dardanelle Lock and Dam at ARM 207.1 (Figure 1). The mean depth of the lake is 4.3 m. The total drainage basin above Dardanelle Lock and Dam is 398,005 km^2. Lake Dardanelle is separated from Ozark Lake by Ozark Lock and Dam at ARM 256.8. Channel length between the two locks and dams is 79.5 km. Lake Dardanelle has a storage capacity of 420,900 acre feet at top of the navigation pool and a surface area of 12,591 ha. The geology and topography of the Lake Dardanelle area is similar to that of Ozark Lake. Dardanelle Lock and Dam was completed in 1964 and is a major unit in the McClellan-Kerr Arkansas River Navigation System (Harris, 1992). Harris (1992) reported 14 species of Unionacea from Lake Dardanelle at ARM 218.6.

Unionacea

Natural History

Unionacea feed primarily on diatoms, other algae, protozoans, plant and animal debris, ova and spermatozoa of other animals and mold and mold spores both in lentic and lotic environments. Mussels are usually embedded in the substrate. Mobility is via a muscular organ called the foot, and degree of mobility is species-dependent (Oesch, 1984). Mussels may occur as solitary organisms or in concentrations called beds in a variety of physical and chemical environments (Oesch, 1984; Rust, 1993; Williams et al., 1993; Christian, 1995).

Mussel anatomy consists of two general components: the outer protective shell and the internal visceral mass. The outer shell contains two symmetrical valves composed completely of non-living material. Mussel shells consist of four layers: 1) a thin proteinaceous outer layer

called the periostracum; 2) the prismatic layer, a thin layer composed of vertical prisms of calcium carbonate in the mineral form of calcite; 3) the nacre, the thickest layer, composed of calcite and aragonite in thin plates which lie parallel to the shell surface; 4) The innermost layer, called the hypostracum, is a secretion of the major muscles at their points of attachment. The three outer layers are secreted by the mantle. The two valves are held together by a stout proteinaceous ligament, which constantly exerts pressure on the valves. Adductor and abductor muscles are used in opening and closing the valves. The inside of the valves contains two pairs of teeth near the ligament, the pseudocardinals on the anterior end and the lateral teeth. These teeth vary in development and are absent in some species, providing important taxonomic characteristics (Oesch, 1984).

The internal visceral mass contains the digestive, nervous, excretory, endocrine, respiratory, circulatory and reproductive systems. The mantle surrounds the visceral mass and is attached to the shell by the dorsal and pallial muscles (Oesch, 1984).

Unionacean species are usually dioecious; however, there are reports of several species being monoecious (Harris and Gordon, undated). Eggs are produced in ovaries and released into the suprabranchial chamber from which they move into the cloaca. Here their direction is reversed and they pass forward into the gill plates (lamellae) which act as brood pouches. This process continues for approximately 10 days, and up to 0.5 million eggs may be produced. Spermatozoa are released by males into the water column, enter the female via the incurrent siphon, traverse the mantle chamber and pass through ostia into the water tubes of the gill plates where fertilization occurs. Each fertilized egg develops into an intermediate larval stage called a glochidium. Glochidia are stored in the female's gills where an oxygen supply is provided. Glochidial expulsion occurs through the exhalent aperture. Then they are dispersed into

the water column (Oesch, 1984; Cummings and Mayer, 1992).

Glochidia are of two types, hooked and hookless. The hooked type is an external parasite on the fins of fishes, whereas the hookless type is an internal parasite which attaches to fish gills. At the time of attachment, both types of glochidia possess a mouth, intestine, adductor muscle, rudiments of a foot, heart and other organs. Glochidia may range in size from 0.005 mm x 0.052 mm in *Cumberlandia monodonta* to 0.290 mm x 0.355 mm in *Cyclonaias tuberculata*. While encysted on the host, the glochidia undergo a metamorphosis (duration for metamorphosis is species-specific), drop to the substrate and, if the substrate is suitable, begin an independent life. The period of attachment varies from 1-25 weeks and is dependent on water temperature, species of glochidia and the host fish (Oesch, 1984; Meglitsch and Schram, 1991; Cummings and Mayer, 1992). The glochidia of two species, *Strophitus undulatus* and *Utterbackii imbecillis*, are not parasitic.

Unionacea in Arkansas were first reported by Call (1895). The known diversity is 74 species (Posey et al., 1996). Eight species are on the Federal endangered species list; *Arkansia wheeleri, Epioblasma florentina curtisi, Epioblasma turgidula, Lampsilis abrupta, Lampsilis streckeri, Lampsilis powelli, Potamilus capax* and *Quadrula fragosa*. Harris and Gordon (1987) also discussed rare and endangered species. Distributional data have been provided by Call (1895), Wheeler (1918), Gordon (1980), Gordon et al. (1979, 1993/1994), Harris and Gordon (1987), Rust (1993), Christian (1995) and Posey (1997). Prior to research by Rust (1993), Christian (1995) and Posey (1997) there were no baseline data available concerning mussel densities of major commercial beds in Arkansas.

Environmental Stresses

Environmental stresses may be categorized as either natural or imposed. All aquatic environments cope with natural environmental stresses. For example, freshwater fish must

maintain their body fluid constancy by osmo-regulation activity of the kidneys which excrete excess water but at the price of energy expenditure. Low dissolved oxygen concentrations may mean that an organism has to indulge in greater respiratory activity. Other natural stresses may include extremes of temperature, discharge with related changes in depth or velocity, turbidity and unstable substrate (Hellawell, 1986).

In addition to natural stresses, aquatic environments encounter anthropogenically imposed stresses. Within the Arkansas River and Petit Jean River basins, agriculture occupies a large percentage of the land use (ADPCE, 1987). In the Little Missouri and Saline river basins, forestry occupies a large percentage of the land use. Many times these practices result in increased erosion which increases turbidity. The subsequent reduction in photosynthesis results in lower O_2 levels. It may also interfere with collecting mechanisms of filter feeders. Also, erosion results in introduction of materials into the water column, which may change benthic communities by loss of interstitial species, reduction of diversity, increased number of a few adventitious species and result in unstable substrate. The application of pesticides to fields and clear-cut areas may add these toxins to nearby lakes or streams via over-spray or indirectly by run-off. The result may be changes in water quality which may cause acute or chronic toxicity to organisms (Hellawell, 1986).

Organic enrichment has been observed in Ozark, Dardanelle and Blue Mountain lakes where waterfowl add large quantities of feces to the water. This often leads to closure of swimming areas in the summer due to high bacterial counts (ADPCE, 1994). Additionally, treated municipal and industrial effluents are released into all the rivers and lakes studied. In May, 1996, a tornado destroyed the Fort Smith, Arkansas, sewage treatment plant, leading to 3,785,000 L of raw sewage per day entering the Arkansas River approximately five km upstream of J.W.

Trimble Lock and Dam (Arkansas Dept. of Health, pers. comm.). This enrichment may lead to reduced O_2 levels, increased turbidity and the release of methane and hydrogen as the sulphide matter decomposes (Hellawell, 1986). The local fauna may be affected to varying degrees depending on the nature of the effluent and environmental conditions (Justus and Harp, 1992).

Habitat Manipulation

The construction of reservoirs has many potential ecological consequences. Dams may interfere with the downstream drift of many aquatic macroinvertebrates. Many physicochemical and habitat modifications occur due to changes from lotic to lentic conditions (Hellawell, 1986). It has been documented that dams affect mussels by inundating riverine habitat, eliminating host fish populations, changing water quality and altering daily and seasonal hydrographs (Layzer et al., 1993; Williams et al., 1993; Layzer and Madison, 1995).

In addition to the series of 17 locks and dams constructed on the Arkansas River, one dam each has been constructed on the Little Missouri, Saline and Petit Jean rivers. Lake Greeson Dam on the Little Missouri River was a hypolimnetic release until 1995 when it was converted to an epilimnetic release. It is still too early to determine what effects this conversion will have on the Little Missouri River. It is likely that hypolimnetic releases completely alter the aquatic fauna for several km downstream (Hynes, 1979; Petts, 1984; Hellawell, 1986). Petts (1984) stated that epilimnetic releases may improve the volume and reliability of the planktonic food supply, thus increasing biomass of filter-feeding organisms below the dam.

Channelization has been documented to: 1) increase soil erosion due to loss of bank vegetation and elimination of marshes, swamps and floodplain lakes; 2) increase siltation; 3) alter flow regimes; 4) increase temperature maxima; 5) remove food sources; and 6) lower the water table (less recharge). Loss of microhabitats may result in reduced diversity or the benthic fauna

may be removed (Hynes, 1979; Hellawell, 1986). In regard to fish, siltation negatively affects the survival rate of eggs, area of spawning and nesting ground, number and kinds of food organisms, visibility of sight feeders, habitat diversity and substrate stability (Ritchie, 1972). Channelization has had a profound influence on mussel populations in Ozark and Dardanelle lakes. The USACE maintains a navigation channel through these lakes, and shell takers have reported mussel beds being covered by dredge material.

Exploitation

As evidenced from Native American middens, mussels were used for food, tools and ornamental objects. Later, the manufacture of buttons by early settlers represented an important commercial fishery in which shells were used. With the advent of plastics after World War II, it became more cost efficient to use these materials, and mussels lost their value. A summary of the history of commercial shelling in the United States and Arkansas is provided by Rust (1993).

Thick shells characterize some North American Unionacea species which led to the revitalization of the commercial shell industry. Today, the commercial value of mussels revolves around their use in the cultured pearl industry. The resurgence has resulted in prices as high as $6.00 per pound in some locales in the United States. Japan is currently the foremost buyer of these shells. Japan bought 9000 short tons of shells in 1991, but purchases have averaged 4500 short tons in recent years (Williams et al., 1993). The number of licenses sold and harvest in Arkansas have generally followed the same trends (Table 1).

A description of methods used to collect mussels is provided by Rust (1993), Christian (1995) and Posey (1997). The size of harvestable shells in Arkansas is regulated by the Arkansas Game and Fish Commision (AG&FC) (Table 2).

Table 1. Commercial Shell taking licenses, mussel shell harvest size & value of harvest from 1981-1995 (Burnley, 1996).

Year	Resident Licenses	Non-Resident Licenses	Tons Harvested	$Value
1981	144	5	105.5	45,365
1982	268	11	10.0	4,300
1983	143	0	10.0	4,300
1984	109	9	22.5	9,900
1985	70	7	22.5	10,800
1986	60	6	141.5	99,926
1987	136	15	103.0	68,566
1988	151	75	223.0	195,150
1989	263	64*	298.5	330,200
1990	687	13**	1117.0	1,854,220
1991	842	2**	982.0	1,473,063
1992	766	0	226.0	256,759
1993	289	0	191.0	278,901
1994	291	0	225.0	369,000
1995	412	0	1095.5	2,585,380

* Non-resident license fees increased from $50 to $100.

** Non-resident harvest prohibited; non-resident buyers only.

Table 2. Arkansas Game and Fish Commission's minimum legal dimensions in mm for harvest of commercial mussel species in Arkansas, 1996, and parameters measured for all species. Equivalent inch measurements are shown parenthetically.

Species	Parameter	Minimum legal size
Actinonaias ligamentina	length	101.6 (4.00)
Elliptio dilatata	length	101.6 (4.00)
Lampsilis satura	length	NC*
Lampsilis teres	length	101.6 (4.00)
Leptodea fragilis	length	NC
Ligumia recta	length	101.6 (4.00)
Plectomerus dombeyanus	length	101.6 (4.00)
Potamilus purpuratus	length	101.6 (4.00)
Quadrula cylindrica	length	101.6 (4.00)
Tritogonia verrucosa	length	101.6 (4.00)
Amblema plicata	diameter	69.85 (2.75)
Arcidens confragosus	diameter	NC
Cyclonaias tuberculata	diameter	63.5 (2.50)
Cyprogenia aberti	diameter	63.5 (2.50)
Ellipsaria lineolata	diameter	63.5 (2.50)
Fusconaia ebena	diameter	63.5 (2.50)
Fusconaia flava	diameter	63.5 (2.50)
Lampsilis abrupta	diameter	ES**/NC
	length	
	width	
Megalonaias nervosa	diameter	92.25 (3.75)
Obliquaria reflexa	diameter	57.15 (2.25)
Obovaria jacksoniana	diameter	NC
Pleurobema pyramidatum	diameter	63.5 (2.50)
Quadrula metanevra	diameter	57.15 (2.25)
Quadrula nodulata	diameter	63.5 (2.50)
Quadrula pustulosa	diameter	63.5 (2.50)
Quadrula quadrula	diameter	69.85 (2.75)
Truncilla truncata	diameter	NC

* NC = No commercial value

** Endangered species

Exotic Species

Two exotic zebra mussel species, *Dreissena polymorpha* and *Dreissena bugensis*, are presumed to have entered the U.S. at Lake St. Clair in 1985 or 1986 in the ballast water of foreign ships. By 1995, *Dreissena polymorpha* had invaded seven major river systems (the St. Lawerence, Hudson, Mississippi, Ohio, Illinois, Susquehana and Arkansas rivers), and had been found on trailered boats as far west as California (Ohio Sea Grant, 1996).

Zebra mussels can have devastating effects on ecosystems. For example, zebra mussels are capable of reaching concentrations greater than 40,000 individuals/m^2, and each adult can filter phytoplankton and zooplankton from one liter of water/day. Material not eaten is ejected in a mucous-covered ball called pseudofeces. Due to their filtering abilities, they may reduce turbidity, thereby causing a resurgence of aquatic macrophytes. There is also speculation of polychlorinated biphenyls (PCB's) and polycyclic aromatic hydrocarbons (PAH's) being bioaccumulated in zebra mussels due to their high lipid concentration. Consequently, they may concentrate contaminants to 100x greater than fish (Ohio Sea Grant, 1996).

Zebra mussels can encrust native mussels resulting in stressed individuals due to malnutrition (Ohio Sea Grant, 1996). Ricciardi et al. (1995) found that Unionacea mortality increased at *Dreissena* densities above 1,000/m^2, and Unionacea mortality was greater than 90% when *Dreissena* densities exceeded 6,000/m^2. Thus, there appears to be a critical threshold above which Unionacea mortality increases significantly until the community becomes virtually extirpated at higher densities and/or mean fouling intensities of 100 *Dreissena*/mussel. Unionacea morphologies and behaviors (e.g. burrowing activity) that are intrinsic to gender or species may potentially enhance or reduce fouling (Haag et al., 1993; Ricciardi et al., 1995).

In addition to ecosystem effects, the zebra mussel is causing a large socioeconomic impact

on the United States. Electric generation power plants, municipal water systems and industrial users with water intakes in the Great Lakes Basin alone spent over $120 million controlling zebra mussels in 1994. Other industries experiencing declines in their industry include commercial fisheries, recreation and tourism and shipping and navigation industries. Commercial fisheries have declined in some areas where zebra mussels have invaded due to shifts in the food chain and mortality in freshwater mussels. Recreation and tourism have been affected by declines in fisheries and fouling of popular recreation areas. Shipping and navigation industries have experienced increased costs in maintenance, which have negatively affected some businesses (Ohio Sea Grant, 1996).

In Arkansas, the zebra mussel was first discovered in September 1992 in Lake Dardanelle. Since that time, it has been found up the Arkansas River into Oklahoma. Zebra mussel populations have been increasing dramatically. Between 1993-1995, zebra mussel veligers increased several fold per year in Lake Dardanelle. In 1996, the mean adult settling density was over 150,000/m^2 (Stoeckel, pers. comm.). Effects of zebra mussels on water quality and ecology of Lake Dardanelle are difficult to predict because it is a different system from those previously studied (Gagen and Stoeckel, 1994/1995; Stoeckel et al., 1995).

Another exotic species, *Corbicula fluminea* (Asian clam), was introduced into the United States in the late 1930's. It has spread rapidly over the past few decades and is now found throughout the United States. It inhabits nearly all aquatic habitats, and it may compete with native mussels for habitat and food. Since *C. fluminea* does not produce a parasitic larval stage, it is suspected that man has aided in the rapid distribution of this essentially sessile organism (Oesch, 1984).

Statement of the Problem

Because of their life history characteristics, freshwater mussels are extremely susceptible to

changes within an aquatic system (Williams et al., 1993). Major changes within the study area include dams, dredging, channelization, siltation, contaminants, changes in fish community structure, exotic species and harvesting pressure. Further, in 1995, 1095.6 tons of shells were harvested and sold to Arkansas shell buyers, an increase of 388% from the 1994 harvest and 344% higher than the mean harvest of 318.2 tons from 1981-1995 (Table 1) (Burnley, 1996).

To date, baseline population estimates on Unionacea in Arkansas are available for the Black, Current, Spring, Strawberry, White, Cache, St. Francis and Ouachita rivers and Lake Chicot (Rust, 1993; Harris et al., 1993; Christian, 1995; Posey, 1997). Information is needed for remaining drainages to ensure comprehensive management of this resource. Locations of commercial quality mussel beds are needed specifically for the Little Missouri and Saline rivers, Blue Mountain Lake, Ozark Lake and Lake Dardanelle along with defining, mapping areal extent and recording physical habitat variables so that community estimates and dynamics can be determined. While the Ouachita River supports a large and diverse freshwater mussel community, most of the specimens of commercial species are not legal size (Posey et al., 1996; Posey, 1997). It is still important to establish baseline population estimates for the Little Missouri and Saline rivers in the Ouachita River drainage to gain insight into the pressures placed on mussel populations in other drainages brought about by commercial shelling. The data gathered from the Little Missouri and Saline rivers, along with those of Posey (1997) will provide a more comprehensive database for the Ouachita River drainage.

The effects of removing legal size mussels from the reproductive pool of a population are not well known. Some shell takers believe that by removing adults, space is provided for juveniles to establish themselves. However, the removal of too many reproductive adults could decrease the reproductive success. Chronological age of commercial sized shells is not well known.

Therefore, a need exists for age-growth data to allow for a more comprehensive management of this resource.

No standardized method of sampling has been developed for scientific research. For this reason, scientific research methods for large rivers and reservoirs are varied (Isom and Gooch, 1983). Line transect sampling was used by Isom and Gooch (1983) for a large river and Cvancara and Freeman (1978) for a reservoir. Rust (1993), Christian (1995) and Posey (1997) conducted their studies exclusively by diving, using an oil-less air compressor (hookah). Other methods may include brailing, snorkling, raking or wading. The method used is dependent upon the results desired. However, some sort of random sampling should be used to determine community density and dynamics

Purpose of Study

The purpose of this thesis research is to locate and define the dimensions and community structure of commercial quality mussel beds found from the mouth upriver approximately 16 km on both the Little Missouri and Saline rivers; and in Blue Mountain Lake, Ozark Lake from Lock and Dam #13 (J.W. Trimble Lock and Dam) to Lock and Dam #12 (Ozark Lock and Dam) and Lake Dardanelle from Lock and Dam #12 (Ozark Lock and Dam) to Lock and Dam #11 (Dardanelle Lock and Dam). Sampling techniques have been developed for large rivers and reservoirs by Rust (1993), Christian (1995) and Posey (1997).

Thin-section age-growth determinations will estimate the minimum harvestable age of two commercially valuable species, *Amblema plicata* (Saline River) and *Quadrula quadrula* (Ozark Lake and Dardanelle Lake). Summary statistics and community and population estimates will be performed on the collected data. Endangered, threatened and newly reported species information will be provided for potential future management of these organisms.

Chapter II

METHODS AND MATERIALS

Field Methods

A questionnaire was sent to each state licensed mussel taker in Arkansas during late 1990 to gain preliminary information on commercial mussel bed locations. Since the Little Missouri and lower Saline rivers are not commercially shelled, very little information existed about locations of commercial beds on these rivers. Two commercial shellers and one commercial shell buyer were contacted about the known location of commercial beds on Blue Mountain, Ozark and Dardanelle lakes. An oil-less air compressor (hookah) with a breathing line attached to a regulator was used to locate and define beds. Personnel and equipment were transported in a 4.9 m (16 ft.) flat-bottom boat powered by a 25 hp Yamaha outboard motor.

Searching methods differed slightly between riverine and reservoir study sites. In the rivers, initial searches for probable Unionacea habitats were conducted in an upstream to downstream fashion. Substrates were explored by touch while moving across the study sites. While searching, if the density of mussels was determined to be $>10/m^2$, a curvature bobber was sent to the surface at that locale. The head, margins and end of the bed were determined by searching and releasing additional bobbers. The end of a bed was determined when Unionacea densities diminished to an average of $< 10/m^2$. Additional

searches were conducted downstream until Unionacea densities diminished to zero. In this area, the diver approximated the width of the bed instead of floating a curvature bobber. Actual dimensions of the bed were determined with a standardized range finder.

While in the field, a record of each site surveyed was compiled on a reproduced 7.5 minute quadrat map or a USACE navigation map. Additional information such as substrate type, density of Unionacea, dimensions of the bed, species composition, water depth (determined by a Hummingbird™ Depth Finder) and river structure (e.g., meander, straightaway) was compiled on field data sheets following each dive.

Study sites were categorized and sampled by one of three methods. For areas with densities $<10/m^2$, qualitative sample notes were taken on species composition, relative abundance and general habitat.

The second level of sampling intensity was that for minor beds (mbeds). These involved semi-quantitative samples in areas with densities of $10/m^2$ or greater and occupying an area between 100-500 m^2. For each sample, all mussels were gathered from inside 1 m^2 delineated by a 2.5 cm PVC pipe quadrat weighted with sand or gravel. Five samples were collected non-randomly from areas with the highest density.

The third level of sampling intensity was for major beds (Mbeds), those consisting of densities of $10/m^2$ or greater and occupying an area greater than 500 m^2. Samples were quantitative and number taken was based on total bed area. Ten samples were taken from beds with areas of 500-999 m^2. Beds with areas between 1000-2500 m^2 was sampled by removing 1% of the area (e.g., 10-25 samples), while 25 samples were taken from beds greater than 2500 m^2. If appropriate, beds were stratified by depth, substrate or river structure. Random numbers were computer generated and utilized to select sample sites,

and the number of samples collected from each stratum was based on the proportion of the total area. A minimum of two samples was taken from each stratum.

In the reservoirs, sampling methods were somewhat modified of necessity. The first modification was with respect to qualitative information obtained. In areas which were sampled qualitatively, all Unionacea encountered were collected, brought to the surface, measured, weighed (minus any zebra mussels which may have been attached), an approximate number of attached zebra mussels recorded and the specimens returned to the substrate. These areas were known commercial beds, which had sporadic and low densities (e.g., areas which had a large area between collected specimens or no consistency in location of shells). A second modification was used with respect to quantitative sampling. Major and mbeds were designated when shells could be located in a consistent pattern, regardless of density. Information was also noted at all qualitative and quantitative sites on the degree of infestation by zebra mussels.

Shells were brought to the surface in a nylon mesh dive bag. At the surface Unionacea were identified, measured to 0.1 mm by a vernier caliper, weighed to the nearest gram with an Ohaus Model CT-6000-S analytical scale, and then returned to the substrate. Measurements were collected in accordance with legal dimensions set by the AGFC (Table 3). Length, width and depth measurements were recorded for 10% of all collected specimens of *Amblema plicata* from the Little Missouri and Saline rivers and *Quadrula quadrula* from Ozark and Dardanelle lakes. All previously described measurements, plus wet weight, were taken on specimens of endangered species. These specimens were then returned, in the live position to the substrate from which they came.

Statistical Treatment

Summary statistics calculated include total individuals, minimum and maximum densities, mean density, sample variance and standard deviation for each species and each quadrat sampled within a bed. A program developed in Lotus 1-2-3™ was used to perform these summary statistics.

Species diversities were calculated at base 2 logarithm using the AQUATIC ECOLOGY-PC program of Oak Leaf Systems, Decorah, IA. Simpson's Index of Diversity calculates the probability of randomly selecting pairs of individuals that must be drawn from a community in order to have an even chance of obtaining a pair with both individuals of the same species. The index increases from a value of 1.0 for a community containing only one species to an infinite value for a community in which every individual belongs to a different species. Simpson Dominance illustrates its relationship as one minus Simpson's Index of Diversity. A Shannon-Weiner Diversity Index (often referred to as H') describes the average degree of uncertainty of predicting the species of a given individual picked at random from a community. This uncertainty increases both as the number of species increases and as the individuals are distributed more and more equitably among the species already present. The maximum possible diversity if all species occur in equal numbers is expressed by H'max. Evenness is a function of the absolute diversity (H') divided by the H'max (Cox, 1980).

Population and community estimates were calculated by the method of Sampford (1962) which is summarized below. The total number of mussels is:

$$[1] \quad X = \sum_{o}^{i} y_i \times g_i$$

where X is the total number of mussels in a bed, i is the number of strata, y_i is the sample total (total number of organisms of encountered in the n_i sampling units) and g_i is the raising factor ($g_i = 1/f_i$, where f_i is the fraction sampled and is defined by n_i/N_i, with n_i being the number of sample units counted in the ith stratum, and N_i the total potential number of sampling units in the ith stratum) (Huebner et al., 1990).

The 95% confidence interval around the total number of Unionacea in a bed was calculated using:

$$[2]\quad x \pm \left(t \times \sqrt{\sum_{0}^{i} N_i^2 \times S^2 y_i \times \frac{1 - f_i}{n_i}} \right)$$

where S^2y_i is the sample variance computed from raw counts in the n_i sampling units in the ith stratum and t is the Student's t for effective degrees of freedom (Huebner et al., 1990).

Curation

Voucher specimens were collected for each species encountered within each river or lake, except endangered species, and curated and placed in the Unionacea Collection of the Arkansas State University Museum of Zoology. Live mussels were fixed in 10% formalin for a maximum of 48 hours, rinsed in water for eight hours twice and preserved in a final solution of 40% isopropyl alcohol.

In the laboratory, a catalog number was assigned to each specimen. Information pertaining to each specimen was entered in the Unionacea Catalog Book and a Lotus 1-2-3™ file. Soft parts were removed intact and a tag was affixed to the foot. Tags included

catalog number, date and location of collection. Once tagged, soft parts were placed in a punctured bag and stored in a labeled plastic bucket with 40% isopropyl alcohol.

The valves were assigned the same catalog number as their corresponding soft parts. The valves were washed in soapy water using a wire brush and then rinsed. Next, using a rapidograph pen and permanent ink, each valve was labeled with catalog number, site location and river or lake. Valves were then dipped in a xylene/paraffin solution, air dried and stored numerically in museum cabinets.

Age Determinations

Fifty *Amblema plicata*, 50 *Quadrula quadrula* and 48 *Q. quadrula* were collected for age determinations from the Saline River, Ozark Lake and Lake Dardanelle, respectively. Various size classes were collected to assure that as many age classes as possible were represented. A thin-sectioning technique used by Neves and Moyer (1988) was used in this study.

An initial cut was made on one valve from the umbo to the shell margin along the vector of greatest length using a Felker geological saw. A second parallel cut was made to get a 2.0 cm wide section. The cut side which was to be mounted on the slide was first polished using a 400 grit aluminum oxide to provide a smooth mounting surface. A perpendicular cut was made on sections which were too large to fit on the 27 x 46 mm petrographic slide. These sections were then mounted on separate slides. Loc-tite Ultraviolet Curing Adhesive was used to bond wedges to the petrographic slide, and a diamond tip pen was used to label slides. Newly bonded slides were placed in a UV box for 20 minutes at 350-380 nm to hard cure the Loc-tite. Next, the cured slide was attached to a vacuum sealed cutting arm of an

Ingram thin-section saw and ground to a thickness between 200-380 microns. This thin-section cut allowed for a workable size to be used in the Ingram grinder.

Once the desired thickness had been reached with the grinder, each slide was hand polished with 400 then 600 aluminum oxide grit. A Minimet® 1000 Grinder/Polisher with a special thin-section attachment was then used for the final polishes. Each slide was polished with Metadi II® 6 micron and 4 micron diamond compound grit, and 0.3 micron alpha alumina Micropolish®, in that order.

A compound microscope with a video monitor hook-up was used to view the slides. Annuli were counted as they exited from the interior layers to the exterior periostracum. Each annulus is considered to be one year of growth.

Shell depth and age determinations were paired, and correlation coefficients (r) were calculated for *A. plicata* (Saline River) and *Q. quadrula* (Ozark Lake and Lake Dardanelle) specimens. The correlation coefficient indicates the type of linear relationship (positive or negative) and the strength of the relationship. The correlation coefficient used is defined as:

$$[3] \quad r = \frac{n\sum xy - \left(\sum x\right)\left(\sum y\right)}{\left[\sqrt{n\sum x^2 - \left(\sum x\right)^2}\right]\left[\sqrt{n\sum y^2 - \left(\sum y\right)^2}\right]}$$

where n was the number of individuals thin-sectioned, x was the depth in mm and y was the age in years. This correlation coefficient (r) was then squared to determine the amount of explained variability (Witte, 1993).

The least squares prediction equation was used to predict the age at minimum legal harvest size for the species selected. It was defined by the following equation:

$$[4] \quad Y' = bX + a$$

where Y' represented the predicted value; X represented the minimum legal harvestable size. In solving for b, the following equation was used:

$$[5] \quad b = \frac{S_y}{S_x}(r)$$

where S_y represented the standard deviation for all observations along Y, and S_x represented the standard deviation for all observations along X. The equation used for a was:

$$[6] \quad a = \overline{Y} - b\overline{X}$$

Where $\overline{Y}$ and $\overline{X}$ referred to the means for all observations along Y and X. The standard error of prediction ($S_{x/y}$), which is a rough measure of the average amount of predictive error, was determined using the following equation:

$$[7] \quad S_{x|y} = S_y\sqrt{1-(r)^2}$$

and the predictive error value was expressed in terms of $\pm$ to Y' ($Y' \pm S_{x/y}$) (Witte, 1993).

Growth Model

To describe growth patterns, total length, depth and width at age data were fitted to the von Bertalanffy asymptotic growth model:

$$[8] \quad L_t = L\infty\left(1 - e^{-K(t-t_0)}\right)$$

where L_t is the size (i.e. length, depth or width) of the individual at time t, $L\infty$ is the theoretical maximum size of the individual, K is the growth constant which is used as a location parameter determining where the predicted growth curve begins on the X-axis and t_0 is the theoretical time when the individual is at zero length (Kohler and Hubert, 1993).

In the von Bertalanffy equation, the highest rate of growth is predicted to be at the very beginning of life. A plot of this relationship shows a progressive increase in body size that aymptotically approaches $L\infty$. This is based on the theory that the rate of change in size per unit of time will decline and approach zero as a mussel or fish nears its maximum possible size.

Study Area

Little Missouri River

The Little Missouri River was surveyed from the mouth of the river upstream approximately 16 km (Figures 1-2). Field work was conducted in June 1996, requiring 10 man days to complete. No commercial shelling takes place on the Little Missouri River due to "stunted" individuals which do not meet size restrictions set forth by the AG&FC (Table 2).

Saline River

Saline River research was conducted from 0.4 km downstream of Godfrey, Arkansas, to approximately 1.6 km upstream from its confluence with the Ouachita River (Figures 1 and 3). Field work was conducted during July 1996, requiring 34 man days to complete. The lower 1.6 km was not surveyed due to inundation caused by Felsenthal Lock and Dam. This survey area is part of Felsenthal National Wildlife Refuge. Permission was granted by the Felsenthal NWR headquarters to research areas within the refuge normally closed to commercial shell takers.

Blue Mountain Lake

Four known commercial mussel beds were surveyed in Blue Mountain Lake (Figures 1 and 4). Field work was conducted in June 1996, requiring four man days to complete. Prior to the survey, shell takers told us there were not many shells left in Blue Mountain Lake due to regular lake drawdowns, which leave shells exposed to dry conditions and easy harvesting by commercial shellers.

Ozark Lake

Seven known commercial mussel beds were surveyed on Ozark Lake (Figures 1, 5-6). Field work was conducted during four man days in September 1995 and 10 man days in May 1996.

Lake Dardanelle

Seventeen known commercial mussel beds were surveyed on Lake Dardanelle (Figures 1 and 7). Field work was conducted during eight man days in September 1995, 26 man days in May-June 1996 and six man days in May 1997. Although commercial mussel beds are

known to exist in certain tributaries of the Arkansas River on Lake Dardanelle, these areas were not considered in this study since they are not considered part of Lake Dardanelle.

Figure 2. Minor (m) bed locations from the Little Missouri River, Arkansas.

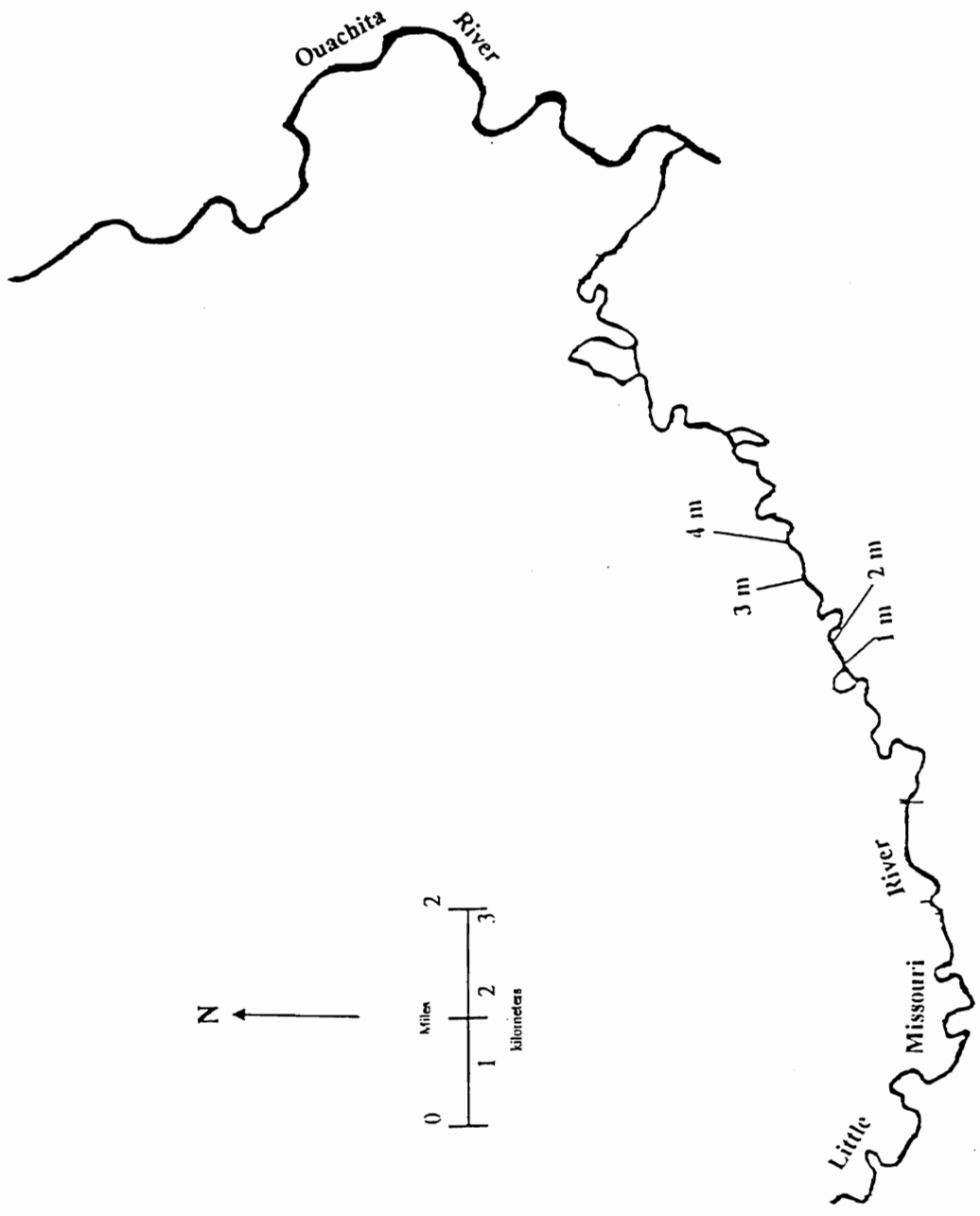

Ouachita River
4 m
3 m
2 m
1 m
Missouri River
Little
N
0
1
2
Miles
2
3
Kilometers

Figure 3. Major (M) and minor (m) bed locations from the Saline River, Arkansas.

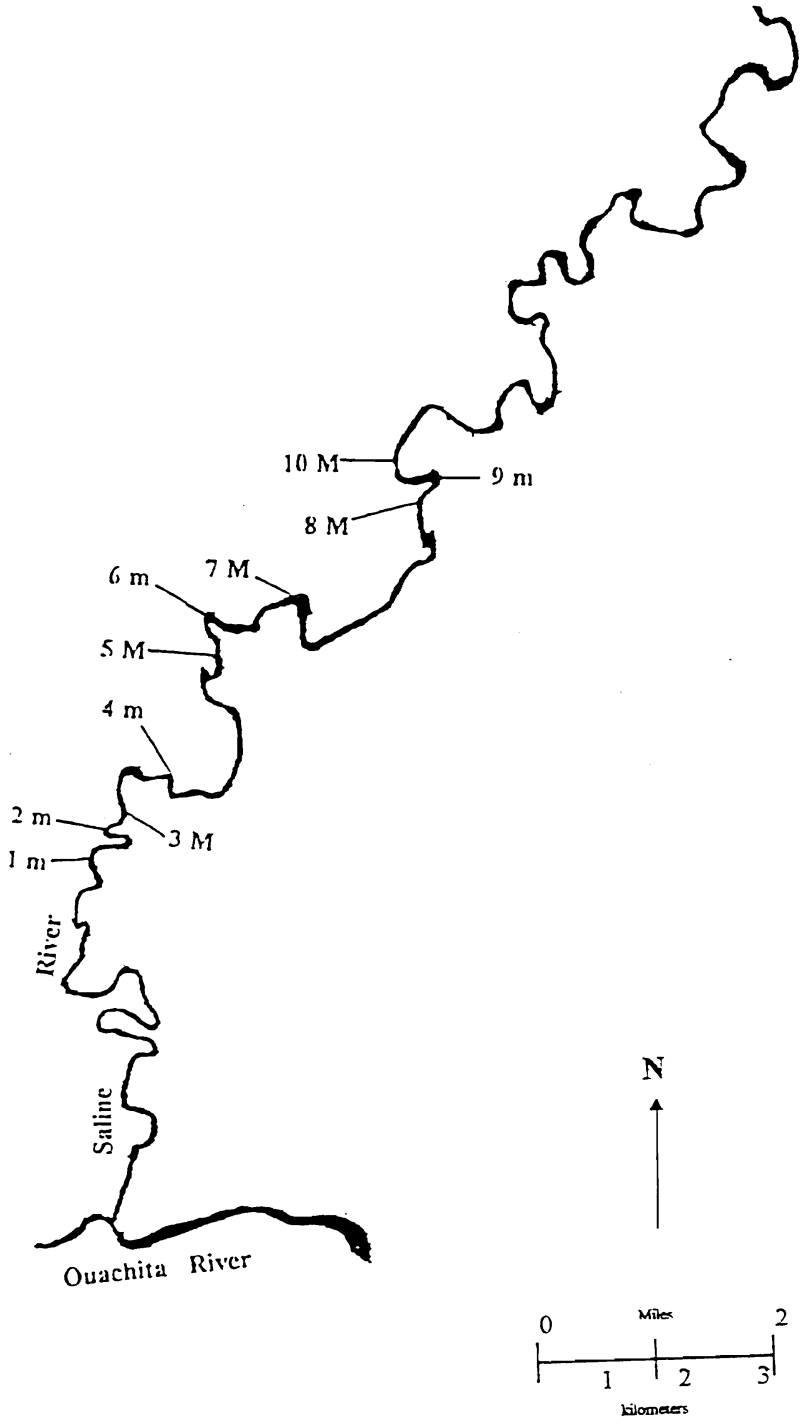

10 M
9 m
8 M
6 m
7 M
5 M
4 m
2 m
3 M
1 m
River
Saline
Ouachita River
N
0
Miles
2
1
2
3
kilometers

Figure 4. Commercial beds (Δ) surveyed from Blue Mountain Lake, Arkansas.

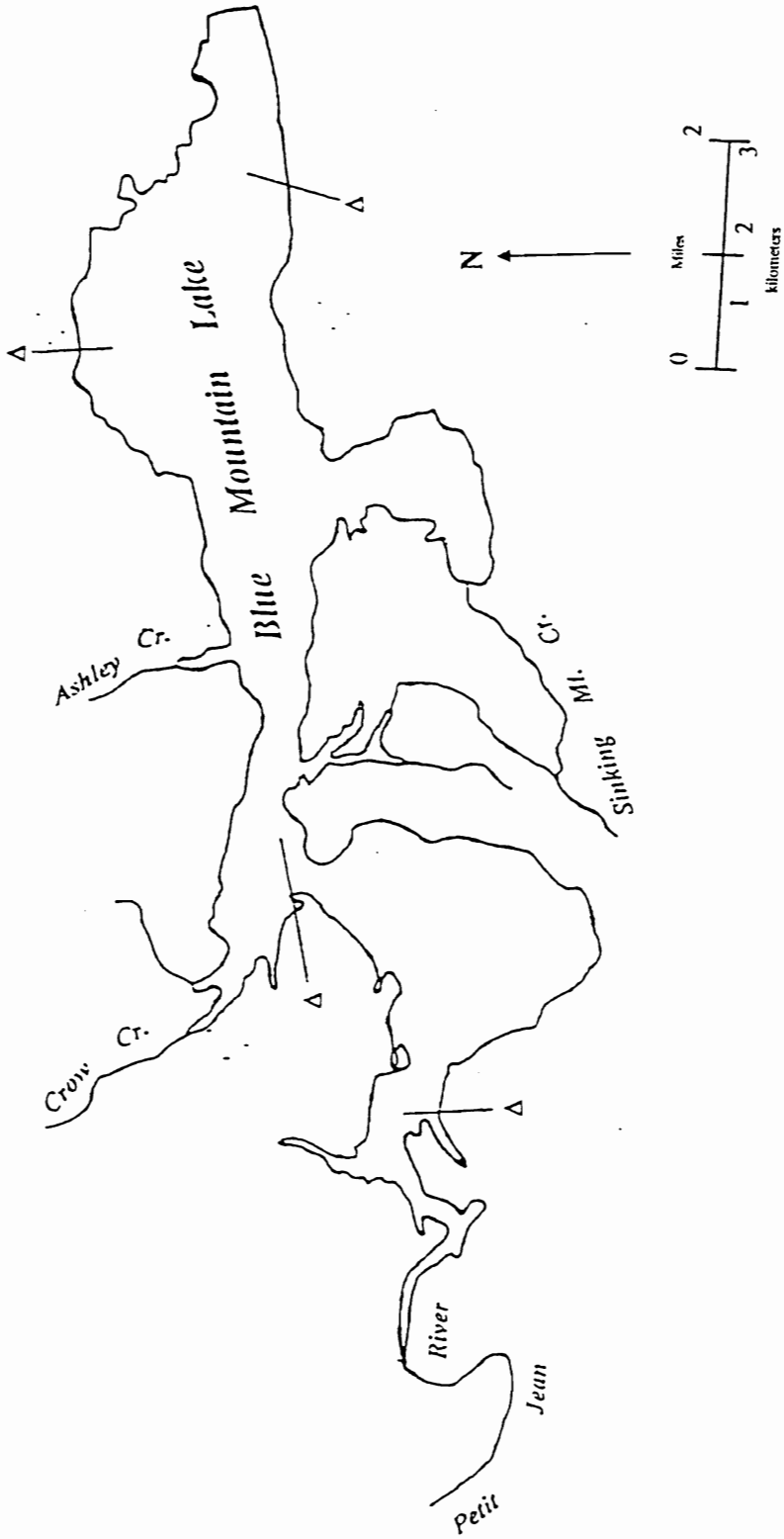

Blue Mountain Lake
Ashley Cr.
Crow Cr.
Sinking Mt. Cr.
Petit Jean River
N
0
1
2
Miles
2
3
kilometers

Figure 5. Major (M) bed locations and commercial beds (Δ) surveyed from Ozark Lake, Arkansas.

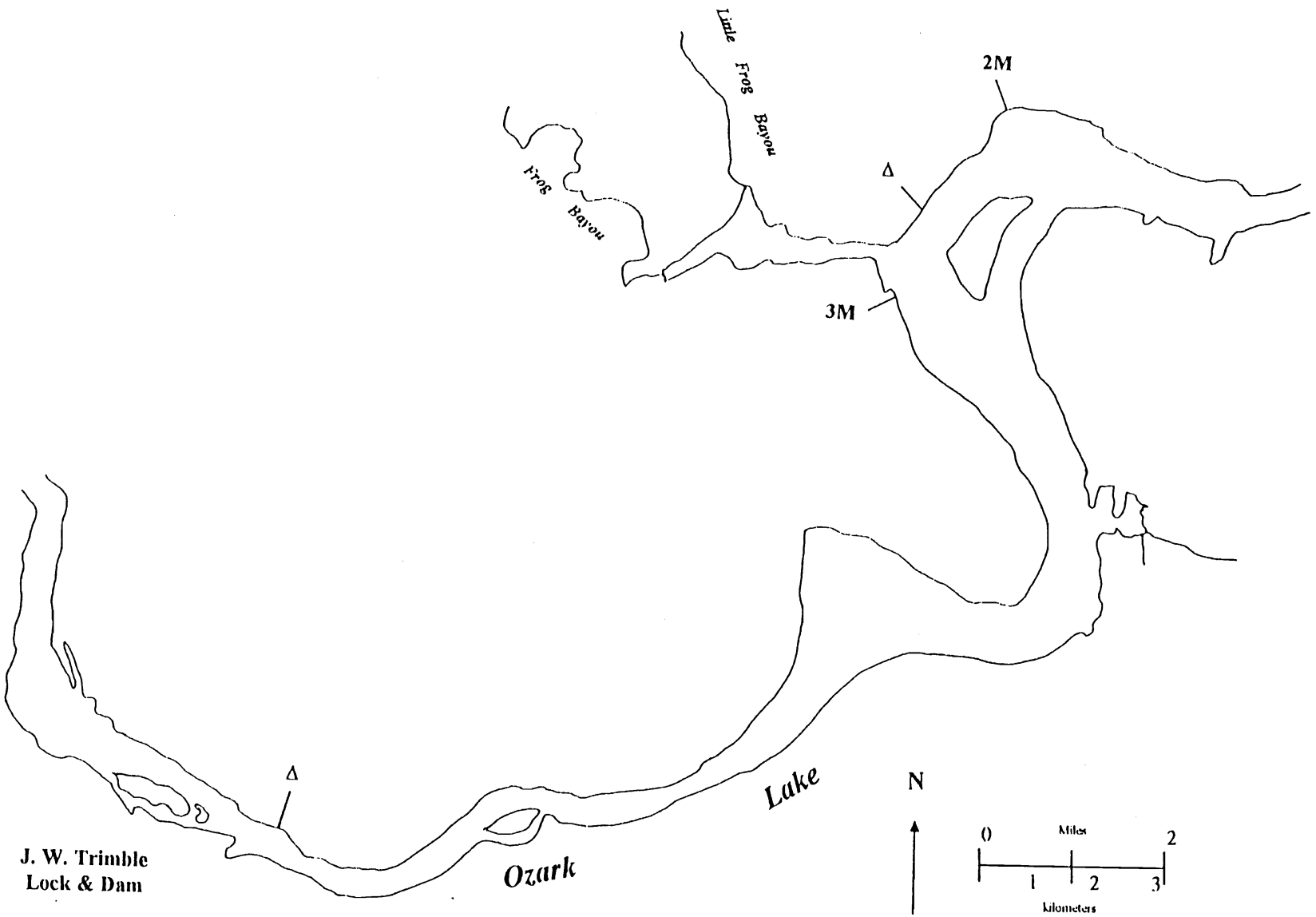

Little Frog Bayou
Frog Bayou
2M
3M
Δ
Δ
J. W. Trimble
Lock & Dam
Ozark
Lake
N
0
Miles
2
1
2
3
kilometers

Figure 6. Major (M) bed locations and commercial beds (Δ) surveyed from Ozark Lake, Arkansas.

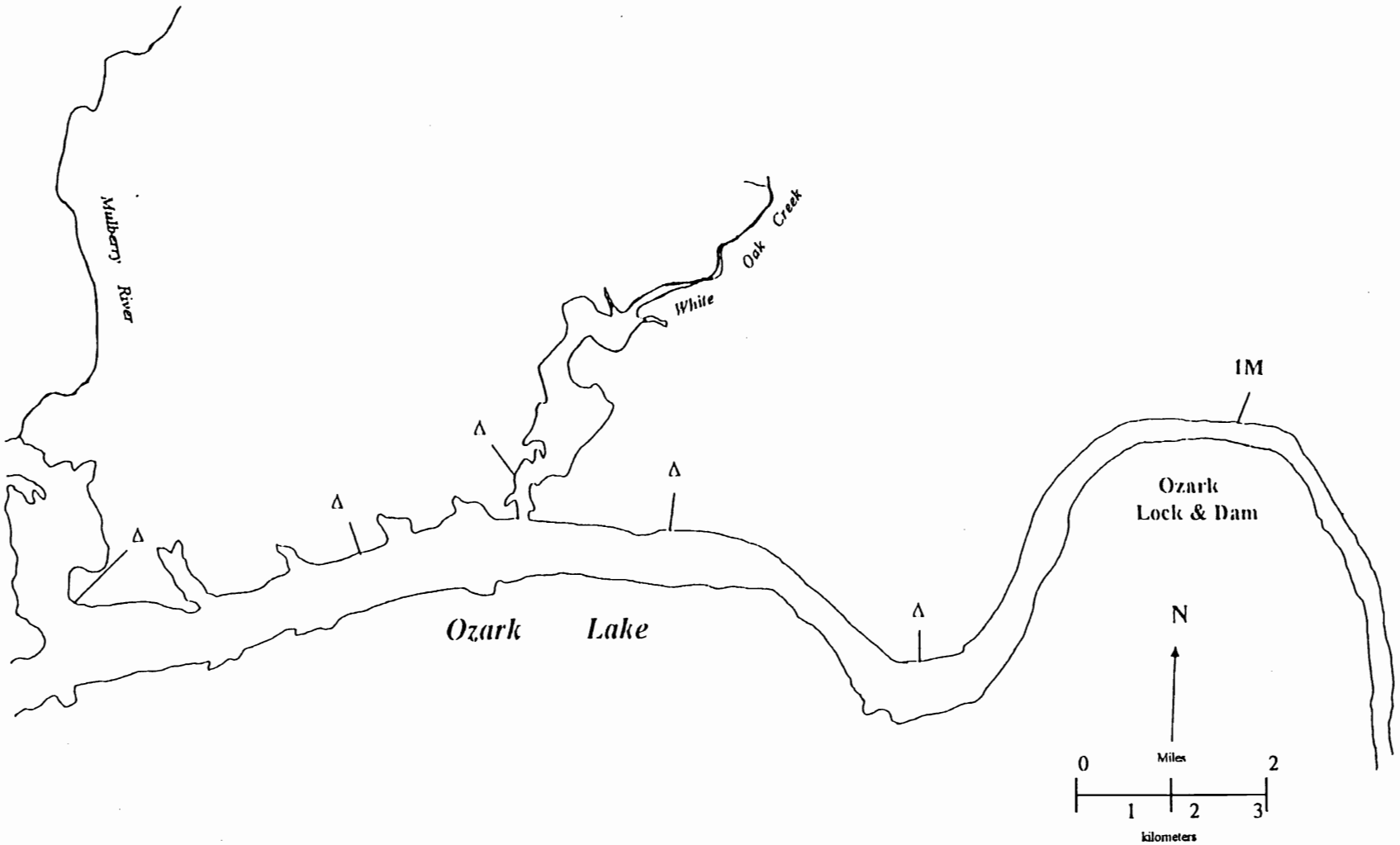
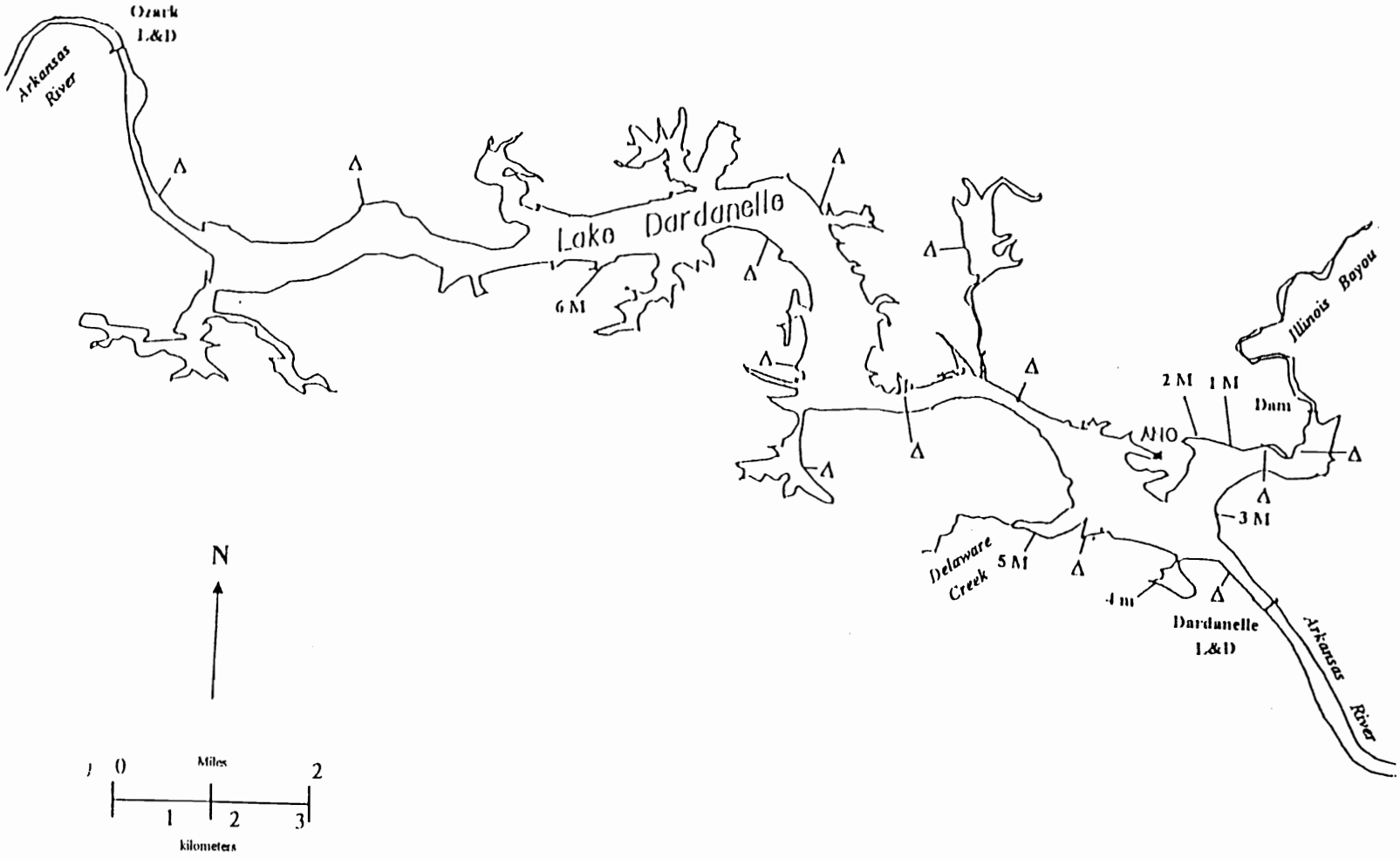

Mulberry River
White
Oak Creek
1M
Ozark
Lock & Dam
Ozark Lake
N
0
Miles
2
1
2
3
kilometers

Figure 7. Major (M) and minor (m) bed locations and commercial beds (Δ) surveyed from Lake Dardanelle, Arkansas.


Ozark
L&D
Arkansas
River
Lake Dardanelle
6 M
Illinois Bayou
2 M
1 M
Dam
M10
3 M
Delaware
Creek
5 M
4 m
Dardanelle
L&D
Arkansas River
N
0
Miles
2
1
2
3
kilometers

Chapter III

COMMUNITY ESTIMATES

Little Missouri River

Results

Four mbeds were defined in the 16 km surveyed, and thirty-four species were identified (Table 3). Of these, 23 species were found in the mbeds (Table 4), and the remaining individuals were found in various concentrations throughout the area.

Bed 4m was located in a backwater area at the mouth of an abandoned channel along the left-hand descending bank. Beds 3m and 2m were located in eddies above or below shoal areas. Bed 1m was located below the mouth of a creek. Substrates in all beds consisted of gravel and sand, or silt over gravel and sand. Numerous concentrations of Unionacea were found in the study area. Most concentrations were either too small to quantify or had densities of < 10 Unionacea/m^2. These were located above or below shoal areas or on clay banks. Water depths for the study area varied from 1.0-5.0 m.

Minor Beds

Twenty 1-m^2 samples contained 548 individuals. Mean mBed densities ranged from 20.0-38.8 with an overall mean density of 27.4 Unionacea/m^2. Bed areas varied from 150-400 m^2 (Appendix 1.1).

Pleurobema pyramidatum was the most abundant species (Table 4) and was numerically dominant in all mbeds. Diversity values were somewhat higher than those for the Saline

Table 3. Unionacea collected from study areas of the Little Missouri and Saline rivers and Blue Mountain, Dardanelle and Ozark lakes, Arkansas, 1995-1997.

Species	Little Missouri River	Saline River	Blue Mountain Lake	Lake Dardanelle	Ozark Lake
Actinonaias ligamentina	X*	X	-	-	-
Amblema plicata	X	X	-	X	X
Anodonta suborbiculata	-	X	X	X	X
Arcidens confragosus	-	-	-	X	X
Cyprogenia aberti	X	-	-	-	-
Ellipsaria lineolata	X	X	-	-	-
Elliptio dilatata	X	X	-	-	-
Fusconaia ebena	X	X	-	-	-
Fusconaia flava	X	X	-	X	-
Lampsilis abrupta	X	R**	-	-	-
Lampsilis hydiana	X	X	-	-	-
Lampsilis satura	X	X	-	-	-
Lampsilis siliquoidea	X	-	-	-	-
Lampsilis teres	X	X	-	X	R
Lasmigona complanata	-	-	-	X	-
Lasmigona costata	X	-	-	-	-
Leptodea fragilis	X	X	-	X	X
Ligumia recta	X	R	-	-	-
Megalonaias nervosa	X	X	-	X	-
Obliquaria reflexa	X	X	-	X	X
Obovaria jacksoniana	X	X	-	-	-
Plectomerus dombeyanus	X	X	-	X	-
Pleurobema pyramidatum	X	X	-	-	-
Potamilus ohiensis	-	-	-	X	X

Table 3. cont'd

Species	Little Missouri River	Saline River	Blue Mountain Lake	Lake Dardanelle	Ozark Lake
Potamilus purpuratus	X	X	R	X	R
Ptychobranchus occidentalis	X	X	-	-	-
Pyganodon grandis	X	X	-	X	X
Quadrula cylindrica	X	-	-	-	-
Quadrula fragosa	X	-	-	-	-
Quadrula metanevra	X	X	-	X	-
Quadrula nodulata	-	X	-	-	-
Quadrula quadrula	X	X	-	X	X
Quadrula pustulosa	X	X	-	X	-
Strophitus undulatus	X	X	-	-	-
Toxolasma glans	X	-	-	-	-
Toxolasma lividus	X	X	-	X	X
Tritogonia verrucosa	X	X	-	-	-
Truncilla donaciformis	-	X	-	-	-
Truncilla truncata	X	X	-	-	-
Utterbackii imbecillis	-	X	-	X	-
Villosa lienosa	X	-	-	-	-
Number of species	34	31	2	18	11

*X = Live specimen

**R = Relict; Included in species totals.

Table 4. Relative abundance, in percent of total, of species from Little Missouri River minor beds.

Species	Number of Individuals	% of Total
Actinonaias ligamentina	5	0.91
Amblema plicata	54	9.85
Cyprogenia aberti	2	0.36
Ellipsaria lineolata	8	1.46
Elliptio dilatata	53	9.67
Fusconaia ebena	1	0.18
Fusconaia flava	53	9.67
Lampsilis satura	3	0.55
Lampsilis teres	3	0.55
Ligumia recta	1	0.18
Megalonaias nervosa	6	1.09
Obovaria jacksoniana	2	0.36
Plectomerus dombeyanus	9	1.64
Pleurobema pyramidatum	230	41.97
Potamilus purpuratus	9	1.64
Ptychobranchus occidentalis	1	0.18
Quadrula cylindrica	1	0.18
Quadrula metanevra	23	4.20
Quadrula pustulosa	77	14.05
Strophitus undulatus	2	0.36
Toxolasma lividus	2	0.36
Tritogonia verrucosa	1	0.18
Truncilla truncata	2	0.36
Total	548	100.00

River (Table 5).

Endangered Species

Two endangered species were collected, three individuals of *Lampsilis abrupta* (LMR 7.3, 7.8 and 9.5) and one individual of *Quadrula fragosa* (0.4 km upstream of the river mouth). *Quadrula fragosa* was recently reported by Posey et al. (1996) as a new state record, extending its range substantially southward. This is only the second known site from which this species has been discovered in Arkansas and serves to further document this species' range.

Discussion

Pleurobema pyramidatum was numerically dominant in both the Little Missouri River and Segment 4 of the Ouachita River. However, *Pleurobema pyramidatum* composed 8.0% more of the total in the Little Missouri River. Several factors could be responsible for the distribution and abundance of *Pleurobema pyramidatum.* It does seem that this species prefers higher current velocities. *Quadrula pustulosa* was more abundant than *Fusconaia flava* in the Little Missouri River, which was the reverse for species composition reported by Posey et al. (1996) for Segment 4 of the Ouachita River. In the Little Missouri River, *Fusconaia ebena* was only represented by one individual, while Posey et al. (1996) reported *Fusconaia ebena* to comprise 7.3% of the total in mbeds of Segment 4.

The mean density for mbeds in the Little Missouri River was similar to that found in the Saline River. This was also true for mbed mean density reported by Posey et al. (1996) for Segment 4 (includes confluence of the Little Missouri River) of the Ouachita River. These three rivers share one characteristic; none are commercially harvested for shells. In other ways, they differ substantially. For example, the lower Saline River does not consist of alternating riffle-pool biocies, and the watershed basin is larger for both the Ouachita and Saline rivers.

Table 5. Diversity and other indicies for the minor and major beds of the Little Missouri and Saline rivers.

Little Misssouri River

Bed	Simpson Diversity	Simpson Dominance	Shannon Diversity	Hmax'	Evenness
1*m*	0.644	0.356	2.209	3.697	0.597
2*m*	0.816	0.184	2.754	3.697	0.745
3*m*	0.778	0.222	2.814	3.907	0.720
4*m*	0.809	0.191	2.787	3.807	0.732

Saline River

Bed	Simpson Diversity	Simpson Dominance	Shannon Diversity	Hmax'	Evenness
Minor beds					
1*m*	0.594	0.406	1.591	3.000	0.530
2*m*	0.677	0.323	2.106	3.584	0.588
4*m*	0.673	0.327	2.143	3.807	0.563
6*m*	0.743	0.257	2.508	3.697	0.678
9*m*	0.691	0.309	2.252	3.697	0.609
Major beds					
3*M*	0.702	0.298	2.405	4.086	0.589
5*M*	0.751	0.249	2.628	4.245	0.619
7*M*	0.742	0.258	2.555	3.907	0.654
8*M*	0.753	0.247	2.598	4.000	0.650
10*M*	0.850	0.150	3.339	4.697	0.711

Saline River

Results

A total of five Mbeds and five mbeds was defined in the 16 km surveyed. Thirty-one species were identified from the study area. Of these, 27 species were found in either Mbeds (27 species), mbeds (19 species) (Tables 3, 6-7) or in areas outside of the beds (four species).

Unionacea were found scattered throughout most of the study area. Beds were found immediately below or within bendways (Figure 8). Beds usually began at the bottom of either the inside or outside descending banks and in a few beds extended from bank to bank. Several species of Unionacea could be found on the banks, but densities were low and sporadic. The substrate consisted of gravel or cobble with sand in all beds.

Approximately 6.4 km from the confluence, the river began spreading laterally and becoming more swamp-like. In this region, the substrate changed to primarily silt over sand, occasionally with some pea gravel. Unionacea abundance was much lower from this point to the mouth. A few concentrations were located in this region, but many of the mussels were dead. Siltation was the only discernable cause of mortality.

Major Beds

Ninety 1-m^2 samples yielded 2,597 individuals. Mean Mbed densities ranged from 15.3-39.5 with an overall mean density of 28.8 mussels/m^2. Total bed areas ranged from 900-6,000 m^2 with a mean of 2,500 m^2 (Appendix 1.2).

Amblema plicata was the dominant species for all Mbeds and dominated in three Mbeds. The second-most dominant species was *Quadrula pustulosa,* which was dominant in two Mbeds (Table 6). Estimated community numerical standing crop (CNSC) ranged from

Table 6. Relative abundance, in percent of total, of species from Saline River major beds.

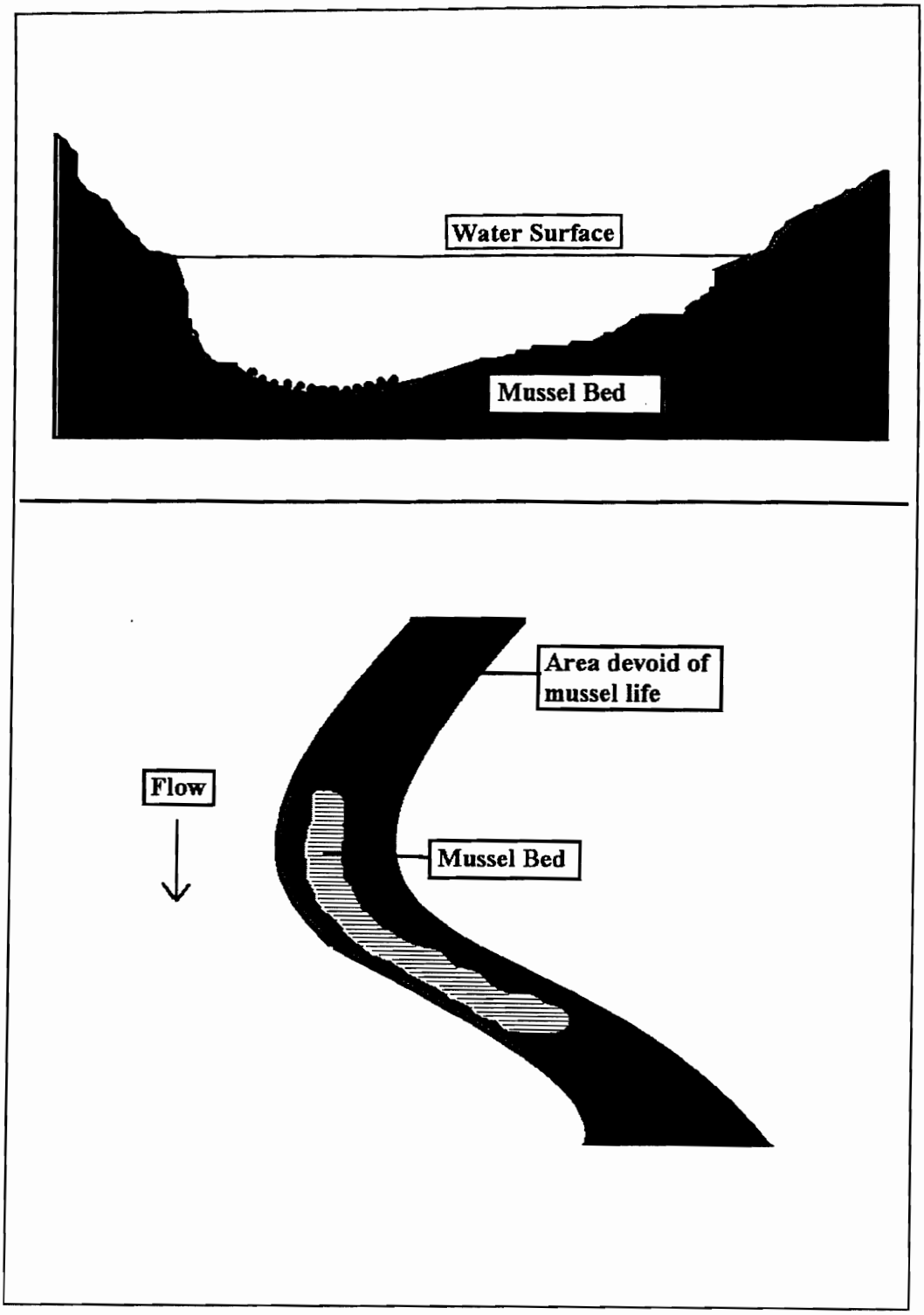
Species	Number of Individuals	% of Total
Actinonaias ligamentina	8	0.31
Amblema plicata	843	32.46
Ellipsaria lineolata	37	1.42
Elliptio dilatata	3	0.12
Fusconaia ebena	158	6.08
Fusconaia flava	291	11.21
Lampsilis hydiana	11	0.42
Lampsilis satura	28	1.08
Lampsilis teres	12	0.46
Leptodea fragilis	11	0.42
Megalonaias nervosa	67	2.58
Obliquaria reflexa	27	1.04
Obovaria jacksoniana	43	1.66
Plectomerus dombeyanus	68	2.62
Pyaganodon grandis	4	0.15
Pleurobema pyramidatum	72	2.77
Potamilus purpuratus	14	0.54
Ptychobranchus occidentalis	10	0.39
Quadrula metanevra	12	0.46
Quadrula nodulata	2	0.08
Quadrula pustulosa	718	27.65
Quadrula quadrula	42	1.62
Strophitus undulatus	3	0.12
Tritogonia verrucosa	62	2.39
Truncilla donaciformis	4	0.15
Truncilla truncata	46	1.77
Utterbackii imbecillis	1	0.04
Total	2597	100.00

Table 7. Relative abundance, in percent of total, of species from Saline River minor beds.

Species	Number of Individuals	% of Total
Actinonaias ligamentina	1	0.15
Amblema plicata	300	44.05
Ellipsaria lineolata	2	0.29
Fusconaia ebena	9	1.32
Fusconaia flava	35	5.14
Lampsilis hydiana	2	0.29
Lampsilis satura	7	1.03
Lampsilis teres	2	0.29
Leptodea fragilis	3	0.44
Megalonaias nervosa	15	2.20
Obliquaria reflexa	8	1.17
Obovaria jacksoniana	19	2.79
Plectomerus dombeyanus	16	2.35
Potamilus purpuratus	3	0.44
Quadrula nodulata	1	0.15
Quadrula pustulosa	237	34.80
Quadrula quadrula	9	1.32
Tritogonia verrucosa	7	1.03
Truncilla truncata	5	0.73
Total	681	100.00

*( ) = Number of individuals

Figure 8. Description of the typical location of mussel beds in a river system (from Posey, 1997).


Water Surface
Mussel Bed
Area devoid of mussel life
Flow
Mussel Bed

24,500±13,600 (Bed 3M) to 237,486±42,385 (Bed 10M) Unionacea per bed. Diversity values were moderate (Table 5).

Minor Beds

The mbeds ranged from 300-455 m^2 in area. Twenty-five samples yielded 681 individuals. Mean mbed densities ranged from 21.6-39.2 mussels/m^2 with an overall mean density of 27.24 mussels/m^2 (Appendix 1.3).

Amblema plicata was the most abundant species and was dominant in four mbeds. *Quadrula pustulosa* was the second-most abundant species and was dominant in 6m (31.5% of total). Diversity indices were moderate (Table 5).

Endangered Species

No live specimens of any endangered species were collected during this study. One relict of *Lampsilis abrupta* was collected approximately 7.2 km upstream of the confluence.

Discussion

Mean density for Mbeds, Mbed area and diversity values all were similar to those reported by Posey (1997) for the Felsenthal Pool of the Ouachita River. I suggest that these similarities result from similar habitat and environmental conditions (e.g. geology, floodplain and water quality) created by Felsenthal Lock and Dam.

Mean density for mbeds was similar to that of the Mbeds in the Saline River. Posey (1997) reported a greater mean density in the mbeds of the Felsenthal Pool of the Ouachita River. However, only two mbeds were sampled and the mean density of one of those was 75.6 Unionacea/m^2. A low sample number can affect the mean density value.

Amblema plicata was the dominant species in this study. Posey (1997) found *Amblema plicata* to be the dominant species in the Felsenthal Pool of the Ouachita River. Differences

occur between the two rivers when the second/third-most dominant species are compared. In the Saline River, *Quadrula pustulosa* was the second-most dominant species in both Mbeds and mbeds, followed by *Fusconaia flava*. Posey (1997) reported *Fusconaia ebena* to be the second-most abundant species in both Mbeds and mbeds, followed by *Quadrula pustulosa*. No reason is apparent. Oesch (1984) stated that *Q. pustulosa* and *F. ebena* both use *Pomoxis annularis* (white crappie) as a host fish. Several species of catfish are also reported to serve as host for *Q. pustulosa*. The numerous backwater areas of the lower Saline River may provide catfish with a more widespread suitable environment in which they can thrive, thereby increasing the number of likely hosts for *Quadrula pustulosa*. Competitve exclusion, habitat discontinuities and factors in the physical or biological environment that vary in parallel with the gradient may also explain the dissimilarity in species composition (Oesch, 1984).

The Saline River survey area was too small to detect any longitudinal zonation in species composition. Rust (1993), Christian (1995) and Posey et al. (1996) reported a shift in species as one moves from the upper reaches towards the mouth of a river. Sixty-nine individuals of *Pleurobema pyramidatum* were found in bed 10M. Only eight individuals of *Pleurobema pyramidatum* were found in the remaining nine beds. This suggests that a shift might be occurring in species composition, and bed 10M is near the end of this zonation. If a shift in species composition is occurring, I suggest that factors associated with a change in gradient, change in flow regime or a loss of host fish could be responsible for the decline of this species from the remaining study area. Posey et al. (1996) suggested that the occurrence and abundance of *Pleurobema pyramidatum* may be influenced by current velocity. This may be true, but Bed 10M is characterized by sluggish water, and this environment continues for many kilometers upstream. Therefore, it seems likely that factors other than

current velocity or the combined influence of several factors (e.g. host fish, substrate size or interspecific competition) also affect the distribution of this species.

Table 8 shows comparisons between five commercially harvested rivers (St. Francis, White, Cache, Spring and Black rivers) and three rivers which are not commercially harvested (Little Missouri, Saline and Ouachita rivers) in Arkansas. The Ouachita, White, Cache and Black rivers were similar in sample size, yet the Ouachita River had a higher mean density/m^2, higher species richness and Shannon-Weiner diversity. Perhaps beds which are not harvested for mussels typically have a greater number/m^2, higher species richness and greater diversity. However, this may also be natural or differences between the river systems. This hypothesis may be substantiated by comparing the Little Missouri and Saline rivers to the White and Black rivers. The Little Missouri and Saline rivers, which have smaller watersheds, have fewer species than the White and Black rivers, yet the Shannon-Weiner diversity values are comparable among the four rivers. This suggests that there is more stability in the Little Missouri and Saline rivers because the individuals are more evenly distributed among species. Unionacea in the Little Missouri, Saline and Ouachita rivers are less likely to be extirpated as a result of some catastrophic impact or disturbance.

When the Little Missouri and Saline rivers are compared to St. Francis and Spring rivers, which have comparable sample sizes, the differences are not quite as clear. Species richness and diversity values are moderate for the Little Missouri, Saline and St. Francis rivers, yet high for the Spring River. This suggests that variables other than commercial shelling may be acting on these mussel beds (e.g. agricultural non-point runoff).

Based on this information, I suggest that rivers which are not commercially harvested may have the potential to support a more stable mussel community. Although, there are too many unknown variables acting on these beds to conclude that commercial shelling is

Table 8. A comparison of Arkansas mussel beds in the Little Missouri, Saline, Ouachita, St. Francis, White, Cache, Spring and Black rivers, Arkansas (Rust, 1993; Christian, 1995; Posey, 1997).

	Ouachita Drainage			St. Francis	White River Drainage			
	LMR*	SR	OR	SFR	WR	CR	SPR	BR
Mean No./m^2	27.40	28.50	27.70	18.12	15.81	19.62	6.63	20.92
Species. Richness	23	27	35	22	34	26	25	31
Shannon-Weiner Diversity	2.641	2.413	3.151	2.122	2.644	2.488	3.546	2.487
No. of Beds	4	10	61	10	51	38	6	48
Drainage Area (km^2)	5,447	8,418	17,411	13,466	75,520	5,227	6,356	22,165
Survey Area (river km)	16	16	248	210	433	68	18	265

*LMR = Little Missouri River
SR = Saline River
OR = Ouachita River
SFR = St. Francis River
WR = White River
CR = Cache River
SPR = Spring River
BR = Black River

reducing mussel diversity and species richness in Arkansas rivers, it does appear that commercial shelling leads to a lower mean density in Arkansas rivers.

Blue Mountain Lake

Results

Four commercial mussel beds were surveyed in this study area. Substrates included silty clay, sand, deep silt and organic matter, in that order. One live specimen, *Anodonta suborbiculata*, was found on the northeastern end of the lake. Also, one relict, *Potamilus purpuratus*, was found near the channel approximately 0.4 km west of the dam.

Historically shellers have reported harvesting good numbers of *Potamilus purpuratus*. Regular drawdowns of the lake to the river channel have left mussels exposed to dry conditions and easy harvest by shellers, and few mussels remain.

Ozark Lake

Results

Interviews with commercial shellers identified seven mussel beds in Ozark Lake. Of these, five were qualitatively sampled and two were quantitatively sampled. One quantitative bed (Arbuckle Island area, ARM 277.0) was separated into two smaller beds due to its large area. Of the eleven species identified, five were from the Mbeds (Table 3 and 9).

Bed 3M was located at the mouth of a backwater bay near the southwestern end of Arbuckle Island (ARM 278.9). A narrow inlet supplied water to the bay. This inlet consisted of a gravel/sand substrate. This substrate continued into the bay approximately 50 m before changing to silt over clay. The inlet provided moderate current which would alternate directions of flow every 30 minutes with a slack water period separating the

Table 9. Relative abundance, in percent of total, of species from Ozark Lake major beds.

Species	Number of Individuals	% of Total
Arcidens confragosus	3	6.67
Obliquaria reflexa	8	17.78
Pyganodon grandis	4	8.89
Potamilus ohiensis	1	2.22
Quadrula quadrula	29	64.44
Total	45	100.00

alternating currents. The change in substrate marked the boundary of the bed. Water depths ranged from 0.5 to 1.0 m.

Bed 2M was located on the northern end of Arbuckle Island (ARM 277.0). The total area of this bed was difficult to determine due to low densities and a wide distribution in the area. Unionaceans were located along a descending bank on the northern bank of the bay. Water depth varied from 1.0 to 5.5 m. Substrates in this area consisted of slate/gravel near the shoreline to silty clay on the descending bank. The mussels were located in the area where the silty clay began to near the bottom of the bank.

Bed 1M was located on a shallow flat which bordered the main channel. Depths varied from 1.0 to 3.0 m on the flat to 17 m in the channel. The substrate consisted of rip-rap on the bank to silty clay on the flat. The greatest abundance occurred near the rip rap, which was also the deepest section of the flat.

Qualitative sites had a variety of substrates which included silty clay or sand. Abundance in these areas was extremely low with sporadic concentrations. Unionacea were usually located on shallow flats or banks which were free of heavy siltation.

Major Beds

Sixty-five 1-m^2 samples contained 44 individuals. Mean bed area and density were 4,767 and 0.68 mussel/m^2, respectively (Appendix 1.4). *Quadrula quadrula* was dominant in all Mbeds and diversity indices were low (Table 9 and 10). The CNSC values were low. Significant concentrations of zebra mussels were present at Beds 1M and 2M and qualitative sites near the main channel. None were found in Bed 3M, White Oak Bay and the bays near the mouth of the Mulberry River (Table 11).

Table 10. Diversity and other indicies for the major beds of Dardenelle and Ozark lakes.

Lake Dardanelle

Bed	Simpson Diversity	Simpson Dominance	Shannon Diversity	Hmax'	Evenness
1*M*	0.620	0.380	1.651	2.807	0.588
2*M*	0.648	0.352	1.810	2.807	0.645
3*M*	0.383	0.617	1.086	2.319	0.468
4*M*	0.083	0.917	0.249	1.000	0.249
5*M*	0.666	0.334	1.000	1.000	1.000

Ozark Lake

Bed	Simpson Diversity	Simpson Dominance	Shannon Diversity	Hmax'	Evenness
1*M*	0.627	0.373	1.492	2.000	0.746
2*M*	0.533	0.467	1.375	2.000	0.688
3*M*	NC*	NC	NC	NC	NC

*NC = Not calculated

Table 11. Number of adult *Dreissena polymorpha* attached to live Unionacea collected from study sites on Ozark Lake, Arkansas, 1996.

Frequency of *Dreissena polymorpha*	Site (Mbed) 278.9(3)	277.2	277.0(2)	MR*	WOB	267.2	258.2(1)
0	14**	61	14	10	51	15	10
1-10		2	10			12	8
11-20							7
21-30							1
31-40							2
41-50							
> 50							6

*MR = Bays near mouth of Mulberry River
WOB = White Oak Bay
** Number of live Unionacea collected.

Endangered Species

No endangered species were collected during this survey.

Discussion

Low diversity found in this study area is not surprising. Lotic environments provide mussels with a greater number of microhabitats and a greater diversity of fishes as potential glochidial hosts. Lakes often are subject to fluctuating water quality and serve as a sink for pollutants. Specialists which have not evolved in these conditions often become stressed and die. The species found in this study are considered generalists and are able to adapt to the lentic conditions.

Harvesting reduces numerical abundance. Typically, habitats with a few generalists have high relative abundance as a result of reduced competition. During this study, it was noted that when shells were removed from the substrate a small pocket would remain in the substrate. At all the sites sampled, numerous similar pockets were discovered. Shallow waters have allowed commercial shellers easy access to shells. A second cause for low abundance may be a result of continual dredging by the USACE. Shellers have reported large beds being covered by several meters of dredge material. This destroys entire mussel communities and may have an adverse effect on reproduction of satellite populations.

Ricciardi et al. (1995) reported that the extirpation of Unionacea communities occurs within 4-5 years after the establishment of dreissenid densities of 6000/m^2. If *Dreissena polymorpha* populations reach this density, the author believes that a reduction in already low numbers of Unionacea will occur as a result of fouling. Tucker and Atwood (1995) reported that Swan Lake, a backwater of the Illinois River, supported a diverse Unionacea fauna but did not support large numbers of *D. polymorpha*. While these habitats may not be free of *D.*

polymorpha, the bay areas of Ozark Lake may serve as natural refugia.

Lake Dardanelle

Results

Seventeen commercial mussel beds were identified by commercial shell takers. Eleven qualitative,. five Mbeds and one mbed were defined. Eighteen species were identified, ten of these from Mbeds (Table 3 and 12).

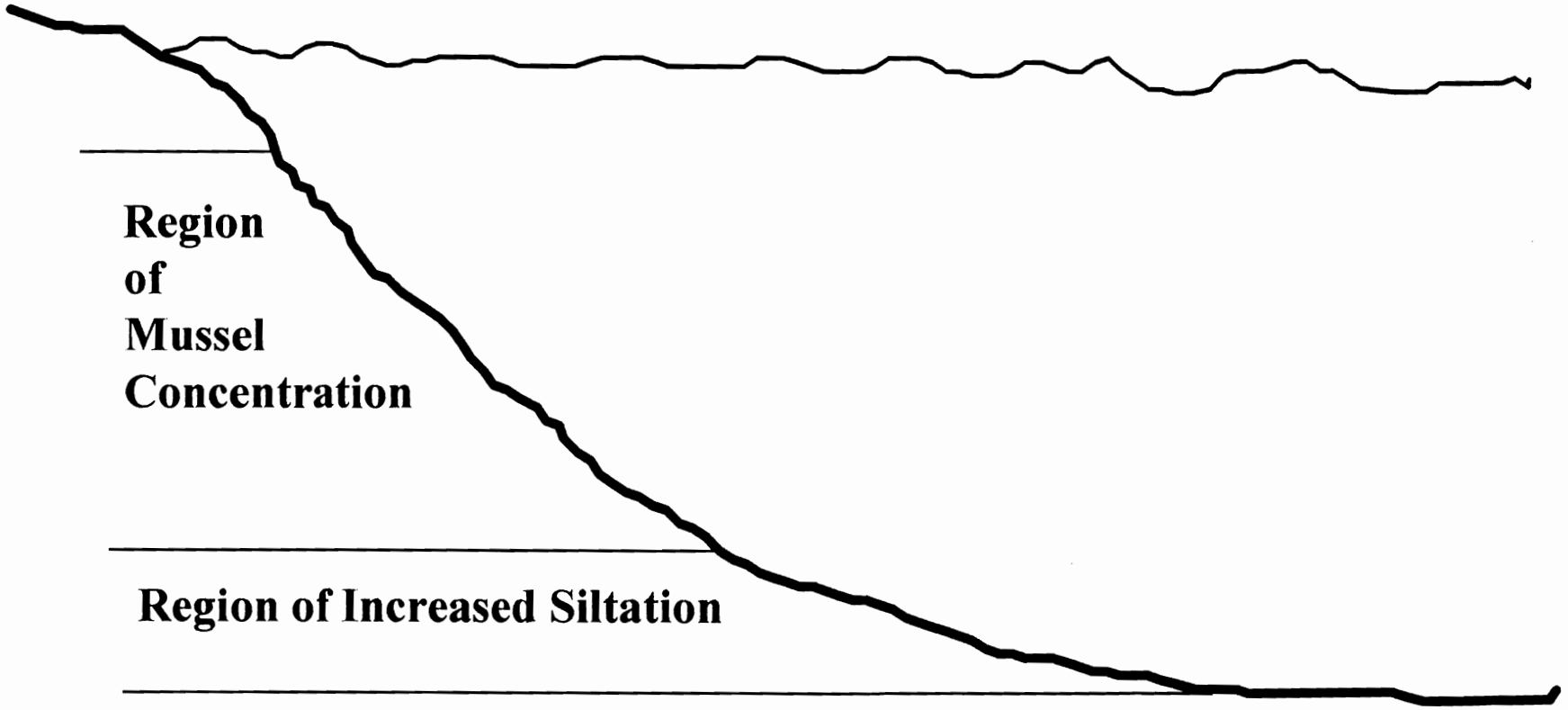
All survey beds were located on descending banks or at the bottom of the bank in a transition zone (Figure 9). Water depths ranged from 1.0 to 7.0 m. Beds 1M, 2M and 3M were located on banks next to the channel of the Illinois Bayou, and 6M was located next to a creek channel. Beds 4m and 5M were located along dikes on Arkansas State Highway 22. These two beds were at the bottom of the bank in a transition zone from riprap to silt.

Location for qualitative sites ranged from shallow flats to descending banks along the shoreline of bays. The shallow flats had a silty clay substrate, and banks had a variety of substrates which included silty clay, sand, gravel and riprap, in that order. While several of these sites had moderate populations of Unionacea, mussel occurrence was sporadic, making quantification extremely difficult.

Major Beds

Eighty-four 1 m^2 samples yielded 318 individuals. The number of species per bed varied from 2-7 with a mean of 4.6. Mean bed densities ranged from 0.27-9.0 with an overall mean of 3.8 mussels/m^2. Total area per bed ranged from 940-2,125 m^2 (Appendix 1.5). *Plectomerus dombeyanus* was the most abundant species and was dominant in two Mbeds. *Quadrula quadrula* was the second-most abundant species and was dominant in three Mbeds (Table 12). It was the only species present in all beds.

Figure 9. Description of a typical mussel bed located in a bay of Lake Dardanelle.


Region
of
Mussel
Concentration
Region of Increased Siltation

CNSC ranged from 400±490 (Bed 6M) to 12,074±3,679 (Bed 2M) Unionacea per bed. Shannon-Weiner diversity indices were very low (Table 10).

Dreissena polymorpha appeared to be seriously impacting Beds 1M, 2M and 3M, while Beds 5M and 6M appeared to be only moderately impacted (Table 13). Shoal Bay, Piney Bay and Delaware Bay north of Arkansas State Highway 22 were the only qualitative sites that appeared to be free of dreissenid fouling.

Minor Beds

Five samples yielded five individuals from Bed 4m, with a mean density of 1.0 mussel/m^2. Total bed area was 100 m^2 (Appendix 1.6). *Dreissena polymorpha* fouling appeared to be seriously impacting mussels in Bed 4m (Table 13).

Follow-up

In 1997, Beds 1-3M were resampled to determine survivorship of mussels in response to fouling caused by zebra mussels. Fifty-four 1 m^2 samples yielded 242 individuals. Species richness was similar for Bed 1M and 2M both years, while drastically reduced for Bed 3M (Appendix 1.5 and 1.7). Species composition varied drastically for Bed 1M, while mean density remained similar (Table 14). Species composition was similar for Beds 2-3M, with only slight variations in Bed 2M. Mean density was reduced for Beds 2-3M by 25% and 38%, respectively.

Samples from Beds 1-3M yielded 9 (8%), 12 (11%) and 21 (32%) dead mussels, respectively. *Quadrula quadrula* comprised the majority (56% and 38%) of dead mussels in Beds 1M and 3M, respectively. *Obliquaria reflexa* comprised the majority (42%) of dead shells in Bed 2M. Dead mussels were not collected in 1996 samples.

Table 12. Relative abundance, in percent of total, of species from Lake Dardanelle major beds.

Species	Number of Individuals	% of Total
Arcidens confragosus	1	0.31
Fusconaia flava	5	1.55
Leptodea fragilis	2	0.62
Obliquaria reflexa	21	6.52
Plectomerus dombeyanus	155	48.14
Pyganodon grandis	6	1.86
Quadrula pustulosa	8	2.48
Quadrula quadrula	124	38.51
Total	322	100.00

Table 13. Number of adult *Dreissena polymorpha* attached to live Unionacea collected from study sites on Lake Dardanelle, Arkansas,1996.

Frequency of *Dreissena polymorpha*	Site (Mbed)									
	233.0(6)	227.2	SB*	223.3	221.0	DB(5)	DSPD(4)	DSPR(3)	IB3.6(1)	IB4.3(2)
0	10**		28		3	37	7		1	
1-10	7		1	3	5	48	23	2	2	4
11-20	2	4		17	5	14	28		2	5
21-30	1	1		6	16	20	10	9	26	29
31-40					4	8	11	3	39	17
41-50				2	1	2	9	12	23	40
>50	4					9	12	53	31	53

*SB = Shoal Bay
DB = Delaware Bay
DSPD = Dardanelle State Park-Dardanelle
DSPR = Dardanelle State Park-Russellville
IB# = Illinois Bayou plus river mile number
** = Number of live Unionacea collected

Table 14. Relative abundance, in percent of total, of species from Lake Dardanelle major Beds 1-3M, 1996-1997.

Species	Bed 1M		Bed 2M		Bed 3M	
	1996	1997	1996	1997	1996	1997
Amblema plicata	NC*	0.9	NC	NC	NC	NC
Anodonta suborbiculata	NC	NC	NC	1.1	NC	NC
Arcidens confragosus	NC	NC	0.8	NC	NC	NC
Fusconaia flava	2.0	NC	2.4	2.2	NC	NC
Leptodea fragilis	NC	NC	NC	1.1	1.4	NC
Lampsilis teres	NC	0.9	NC	NC	NC	NC
Lasmigona complanata	NC	0.9	NC	NC	NC	NC
Obliquaria reflexa	8.2	0.9	8.7	3.2	2.9	NC
Plectomerus dombeyanus	39.8	66.7	48.4	54.8	77.1	84.1
Pyganodon grandis	1.0	NC	0.8	2.2	2.9	NC
Quadrula pustulosa	1.0	1.9	5.6	10.7	NC	NC
Quadrula quadrula	46.9	27.6	33.3	24.7	15.7	15.9
Totals	98.9	99.8	100.0	100.0	100.0	100.0

* NC = Not collected

Endangered Species

No endangered species were encountered during this survey.

Discussion

Since Lake Dardanelle, like Ozark Lake, is part of the McClellan-Kerr Navigation System, it has suffered similar anthropogenic perturbation. The greater mussel diversity in this lake may result from Lake Dardanelle having a greater lake area, which provides a greater number of microhabitats. It has a greater number of bays, which increases the shoreline area. Exploitation appears to be a probable cause of low abundance at most sites. Shallow flats are accessible and have been exploited the most, while deep banks along creek channels are less accessible and appear to be the least impacted.

The presence of three species, *Quadrula pustulosa, Quadrula metanevra* and *Fusconaia flava*, along the old Illinois Bayou river channel is somewhat surprising. Oesch (1984) stated *Q. metanevra* prefers clear water, swift current and a gravel bottom in streams, while *F. flava* prefers moderate current with a gravel or sand substrate. *Q. pustulosa* is typically found in lotic environments and is capable of adapting to most any substrate except loose, shifting sand. These three species are probably remnants of old populations in the Illinois Bayou. Host fishes for all these species occur in Lake Dardanelle, so it appears that these species have been able to adapt marginally to the lentic environment in Lake Dardanelle.

Several variables (e.g. commercial shelling and fouling by zebra mussels) may be acting on Beds 2M and 3M, thereby leading to the reduction of species richness and mean density. If this decline continues at the same rate, these beds will be extirpated within two-three years. Bed 1M has similar characteristics to those of Beds 2-3M, and its resilience is unexplained.

Dreissena polymorpha veliger densities have drastically increased over the past few years in Lake Dardanelle (Gagen and Stoeckel, 1994/1995). High densities of adult populations have appeared over the last couple of years, and the maximum impact is yet to be seen.

Chapter IV

SIZE ANALYSIS AND AGE DETERMINATIONS

SIZE ANALYSIS

Determining the size and age structure of mussel populations is an integral part of providing a successful plan for their management. Size analysis provides information on harvest pressure, recruitment and growth. Age determination enables resource managers to analyze age structure of the community and thereby establish minimum legal harvest size (MLHS) restrictions, which will allow for protection of reproductive populations. Currently, age-growth data are inadequate for regulatory agencies to knowledgeably set harvest limits. The limits are of necessity rather arbitrary.

Size structure will be discussed for the Little Missouri and Saline rivers and Dardanelle and Ozark lakes. The two specimens collected from Blue Mountain Lake do not allow analysis of size and age structures for this lake. Determined age was correlated to size for two species, *Quadrula quadrula* (Dardanelle and Ozark lakes) and *Amblema plicata* (Saline River).

Little Missouri River

Results

Shell depth interval was calculated for six commercial species from the Little Missouri River (Table 15). *A. plicata* was represented by the greatest number of depth intervals, while *Quadrula metanevra* was represented by the fewest. Both mean shell depth and length measurements were below the MLHS set by the AGFC for all species (Table 16). No legally

Table 15. Size frequency distribution of selected mussel species, Little Missouri River, Arkansas, 1996.

Depth or Length (mm) increment	A. p.*	E. d.**	F. f.	P. p.	Q. m.	Q. p.
0-9.9						
10-19.9	3		1	2		1
20-29.9	7		11	4		3
30-39.9	5		30	93	13	46
40-49.9	27	4	11	131	10	28
50-59.9	13	29		2		
60-69.9	1	20				
70-79.9						
80-89.9						
Total	56	53	53	232	23	78

* A. p. = *Amblema plicata*
E. d. = *Elliptio dilatata*
F. f. = *Fusconaia flava*
P. p. = *Pleurobema pyramidatum*
Q. p. = *Quadrula metanevra*
Q.q. = *Quadrula pustulosa*
** = Length frequency

Table 16. Mean ($\bar{x}$) and standard deviations (S) of measurement used to determine legal size for six species from the Little Missouri River, 1996.

Species	(N)	$\bar{x}$ length (mm)	S	$\bar{x}$ depth (mm)	S	$\bar{x}$ weight (g)	S
Amblema plicata	54			42.6	11.7	56.0	35.4
Elliptio dilatata	53	58.2	5.1			37.7	9.6
Fusconaia flava	53			34.8	5.9	31.7	13.9
Pleurobema pyramidatum	230			40.2	4.1	46.6	13.1
Quadrula metanevra	23			41.1	1.8	56.7	10.0
Quadrula pustulosa	77			38.4	4.1	35.6	12.0

(N) = Number of Individuals

harvestable shells were collected during this study (Table 17). Mean shell depths were similar for all species, but mean shell weights were not.

Discussion

Since the Little Missouri River is not commercially harvested, the size class distribution should not be skewed. Population size structures indicate that the mussel community in this river is skewed toward larger individuals. Nevertheless, on average, individuals are smaller than those in the other rivers (Rust, 1993; Christian, 1995; Posey, 1997).

This observation is similar to that reported by Posey (1997) for the Ouachita River. This indicates that the growth maxima for mussels in this system are smaller than those for mussels in other drainages. Posey (1997) proposed that other factors may be affecting growth of Unionacea in the Ouachita River (e.g. dredging and industrial discharges). The Little Missouri River has not been dredged, so these factors can be ruled out for this river. An alternate hypothesis was proposed by Christian (1995) that deals with decreased size as densities increase and resources decrease, which results in greater resource competition. I do not believe that this is the case with the Little Missouri River. This river did not support large beds nor were the densities extremely high within these beds. I propose that the "stunting" of the mussel community in the Little Missouri River is a result of natural influences characteristic of this drainage basin (i.e. physicochemical parameters). I will develop this idea further in the following Saline River discussion and explain why I do not believe that industrial discharges are a significant factor.

Saline River

Results

Shell depth interval was calculated for five commercial species (Table 18). *Megalonaias nervosa*

Table 17. Percent legal harvestable mussels, commercial species only, in the Little Missouri and Saline rivers, Arkansas, 1996.

	Little Missouri River	Saline River
Major Beds	NA	0.19(5)*
Minor Beds	0.00(0)	0.00(0)

*( ) = Number of individuals

and *Amblema plicata* were widely distributed with respect to shell depth frequencies. Mean shell depth and weight were not similar for commercial species (Table 19). They were greater for all species than those reported for the Little Missouri River. Mbeds produced the only legally harvestable mussels (Table 17).

Discussion

The greater range in mussel size for specimens from the Saline River compared with those from the Little Missouri River may be the result of a larger sample size, which reflects larger beds and greater abundance of mussels in the Saline River. Nevertheless, the mussel community still exhibited the same stunted size that characterized mussels in the Little Missouri and Ouachita rivers. Similarly, like the Little Missouri and Ouachita rivers, the Saline River is not commercially harvested for mussels. Since high numerical standing crops occurred in the Saline River, the hypothesis proposed by Christian (1995) relating to resource limitation can not be overlooked as a possible factor leading to the "stunting". However, the uniformity of a "stunted" mussel community within the lower Ouachita River drainage basin (Little Missouri, Saline and Ouachita rivers) leads me to question the hypothesis of an anthropogenic influence as the factor leading to this phenomenon, as proposed by Posey (1997).

While industrial discharges may have profound effects on downstream aquatic biota, the degree of the impact is determined by effluent type. It still seems unlikely that one or even several industrial discharges could have such a profound effect on these three rivers.

This would mean that a similar discharge or ones having similar impacts would have to occur in the Little Missouri, Saline and Ouachita rivers, which are over 160 river km apart. In addition, the dilution factor from the water over such an enormous range would reduce the impact the farther the effluent is transported downstream. Therefore, I believe that natural influences

Table 18. Depth frequency distribution of selected mussel species, Saline River, Arkansas, 1996.

Depth (mm) increment	A. p.*	F. e.	F. f.	M. n.	Q. p.
0-09.9		1			1
10-19.9	32	4	8	5	22
20-29.9	43	4	13	9	58
30-39.9	42	3	131	3	191
40-49.9	278	65	167	1	570
50-59.9	635	84	9	17	43
60-69.9	113	6		15	
70-79.9	3			15	
80-89.9				3	
90-99.9					
Total	1146	167	328	68	885

* A. p. = *Amblema plicata*
F. f. = *Fusconaia ebena*
F. f. = *Fusconaia flava*
M. n. = *Megalonaias nervosa*
Q. p. = *Quadrula pustulosa*

Table 19. Mean ($\bar{x}$) and standard deviation (S) of measurements used to determine legal size for selected species from the Saline River, 1996.

Species	(N)	$\bar{x}$ length (mm)	S	$\bar{x}$ depth (mm)	S	$\bar{x}$ width (mm)	S	$\bar{x}$ weight (g)	S
Amblema plicata	1143	62.5	16.8	50.6	10.0	33.0	8.2	88.5	39.9
Fusconaia ebena	167			49.6	8.9			91.9	33.6
Fusconaia flava	326			39.3	6.7			46.1	18.9
Megalonaias nervosa	82			61.7	23.0			190.4	135.4
Quadrula pustulosa	955			40.7	7.9			48.2	21.9

(N) = Number of Individuals

related to this drainage basin (e.g. physicochemical parameters and/or fluvial processes) are the primary causes for the "stunting" of these communities.

Ozark Lake

Results

Shell depth interval was calculated for three commercial species (Table 20). *Quadrula quadrula* was represented by the greatest number of depth intervals. *Amblema plicata* was only represented by one individual, which was of MLHS. The mean shell depth values were below MLHS (Table 21). The percent legally harvestable mussels (PLHM) in Mbeds was moderate (Table 22). This percentage was considerably higher (30.6%) when both qualitative and quantitative sites are included. This difference was mainly due to 70.5% of the commercial species in White Oak Bay (qualitative site) being of MLHS.

Discussion

The heavy harvest of *Q. quadrula* over the past decade has likely skewed the distribution toward smaller mean sizes. Widespread and severe anthropogenic activities (e.g. dredging, pollution and navigational projects) undoubtably have likely further altered the mussel community structure (e.g. decreased growth in response to resource limitation, stress related to increased sedimentation). The invasion and continued development of the zebra mussel population will almost certainly further negatively impact the native mussel community.

Lake Dardanelle

Results

Shell depth interval was calculated for six commercial species from Lake Dardanelle (Table 23). *Plectomerus dombeyanus* occurred in the greatest number of size intervals. The size intervals represented by *Obliquaria reflexa* and *Quadrula quadrula* were the same as those in Ozark Lake,

Table 20. Depth frequency distribution of selected mussel species, Ozark Lake, Arkansas, 1996.

Depth increment (mm)	A. p.*	O. r.	Q. q.
0-9.9			
10-19.9		4	4
20-29.9		4	4
30-39.9		5	11
40-49.9		5	9
50-59.9		3	28
60-69.9			50
70-79.9			41
80-89.9			14
90-99.9			
100-109.9	1		
Total	1	21	161

* A. p. = *Amblema plicata*
O. r. = *Obliquaria reflexa*
Q.q. = *Quadrula quadrula*

Table 21. Mean ($\bar{x}$) and standard deviation (S) of measurements used to determine legal size for selected species from the Ozark Lake, 1996.

Species	(N)	$\bar{x}$ length (mm)	S	$\bar{x}$ depth (mm)	S	$\bar{x}$ width (mm)	S	$\bar{x}$ weight (g)	S
Obliquaria reflexa	8			33.6	12.9			52.2	79.8
Quadrula quadrula	29	70.4	18.6	61.9	15.7	37.4	9.0	133.9	85.3

Table 22. Percent legal harvestable mussels, commercial species only, in Ozark Lake and Lake Dardanelle, Arkansas, 1996.

	Ozark Lake	Lake Dardanelle
Major Beds	16.0(6)*	14.5(46)
Minor Beds	NA	0.0(0)

*( ) = Number of individuals

Table 23. Size frequency distribution of selected mussel species, Lake Dardenelle, Arkansas, 1996.

Depth or Length (mm) increment	F. f.*	M. n.	O. r.	P. d.**	Q. p.	Q. q.
0-9.9						
10-19.9			5			1
20-29.9	1		15	1		12
30-39.9	4		32	10		27
40-49.9	1		21	26		61
50-59.9			3	50	8	135
60-69.9	1			37		165
70-79.9				41	1	66
80-89.9				35		5
90-99.9				38		
100-109.9				26		
110-119.9		1		27		
120-129.9				13		
130-139.9				2		
Total	7	1	76	306	9	472

* F. f. = *Fusconaia flava*
M. n. = *Megalonaias nervosa*
O. r. = *Obliquaria reflexa*
P. d. = *Plectomerus dombeyanus*
Q. p. = *Quadrula pustulosa*
Q.q. = *Quadrula quadrula*
** = Length frequency

however, the mean shell depth was greater for Lake Dardanelle *Obliquaria reflexa*, while the value was less for *Quadrula quadrula* (Table 24). The percent legally harvestable mussels (PLHM) was similar to that for Ozark Lake.

In both Dardanelle and Ozark lakes, recruitment appeared low based on the results in Tables 20 and 22. Heavy harvest on both of these lakes has reduced densities in most areas to < 1 Unionacea/m^2. Pockets in the substrate remain after mussels are removed, and these are the only evidence that these lakes once supported greater mussel densities.

Follow-up

Size distributions in Beds 1-3M were similar for both 1996 and 1997 (Table 25). PLHM was reduced by 15% and 51% for Beds 1M and 3M, respectively (Table 26). The number of harvestable mussels collected from Bed 2M was similar, but a lower density in 1997 resulted in 63% increase in the PLHM.

Discussion

The narrow size distribution of *F. flava* and *Q. pustulosa* resulted from their scarcity in lentic environments. *Megalonaias nervosa* is known to occur at high standing crops in other lentic environments, so its scarcity in Lake Dardanelle can not be explained. *Obliquaria reflexa* and *Q. quadrula* were both skewed toward smaller size classes, which may be reflective of high harvest pressure or reduced growth rates. As with Ozark Lake, Lake Dardanelle has been heavily and harvested over the past decade. Lake Dardanelle is also anthropogenically influenced (e.g. dredging, navigational projects and pollution) and has been colonized by zebra mussels. Unlike Ozark Lake, Lake Dardanelle consists of shallow flats, which have provided shell takers with easy "pickings". However, there was no significant difference in mean sizes of *Obliquaria reflexa* and

Table 24. Mean ($\bar{x}$) and standard deviation (S) of measurements used to determine legal size for selected species from the Ozark Lake, 1996.

Species	(N)	$\bar{x}$ length (mm)	S	$\bar{x}$ depth (mm)	S	$\bar{x}$ width (mm)	S	$\bar{x}$ weight (g)	S
Fusconaia flava	5			40.5	11.3			56.14	59.21
Obliquaria reflexa	21			35.9	8.5			40.57	27.59
Plectomerus dombeyanus	155	78.1	23.9					94.69	80.50
Quadrula pustulosa	8			45.4	5.6			59.08	21.99
Quadrula quadrula	124	73.4	NA	58.3	11.7	42.7	NA	136.29	67.68

(N) = Number of Individuals

Table 25. Depth frequency distribution of selected mussel species in major Beds 1-3M, Lake Dardenelle, Arkansas, 1996-1997.

Depth or Length increment (mm)	Bed 1M									
	F. f.*		O. r.		P. d.**		Q. p.		Q. q.	
	1996	1997	1996	1997	1996	1997	1996	1997	1996	1997
0-9.9										
10-19.9										
20-29.9			2	1	1				1	
30-39.9	1		5		1			2	2	
40-49.9	1		1			1	1		12	5
50-59.9					1	4			14	15
60-69.9					2	4			14	6
70-79.9					6	15			3	2
80-89.9					9	27				1
90-99.9					8	11				
100-109.9					6	6				
110-119.9					4	3				
120-129.9						1				
130-139.9					1					
Total	2		8	1	38	72	1	2	46	29

Table 25. Cont'd

Depth or Length increment (mm)	Bed 3M					
	O.r.		P. d.		Q. q.	
	1996	1997	1996	1997	1996	1997
0-9.9				1		
10-19.9	1					
20-29.9						
30-39.9	1		2	1	1	
40-49.9			4	1		
50-59.9			13	4	4	5
60-69.9			3	10	5	2
70-79.9			7	5	1	
80-89.9			6	5		
90-99.9			4	5		
100-109.9			4	2		
110-119.9			10	2		
120-129.9			1	1		
130-139.9						
Total	2		54	37	11	7

* F. f. = *Fusconaia flava* Q. p. = *Quadrula pustulosa* ** = Length frequency
O. r. = *Obliquaria reflexa* Q. q. = *Quadrula quadrula*
P. d. = *Plectomerus dombeyanus*

Table 26. Percent legal harvestable mussels, commercial species only, in major Beds 1-3M, Lake Dardanelle, Arkansas, 1996-1997.

	Bed 1M	Bed 2M	Bed 3M
1996	14.6(14)*	11.2(15)	22.4(15)
1997	12.4(12)	18.3(17)	11.4(5)

*( ) = Number of individuals

Quadrula quadrula for the two lakes. It still remains to be seen if Lake Dardanelle's native mussel fauna can withstand the incredible zebra mussel densities (mean adult settling density of 153,000/m^2 for 1996) now present in the lake (Gagen and Stoeckel, pers. comm.).

The low PLHM's (Table 22) reflect high harvest pressure and easy shelling in this reservoir. PLHM data reported by Christian (1995) for the Cache and White rivers were 34.9% and 19.3%, respectively. Predictably, there appears to be a direct correlation between catch per unit effort and the degree of difficulty in harvest. The reduction in PLHM in Bed 3M would seem to suggest that commercial shelling may be responsible for reductions in the mean density.

Recruitment

Recruitment in the Little Missouri River is of concern. Few juveniles were found in the Little Missouri River, which would suggest low recruitment for this river. Christian (1995) reported that we tend to bias our sampling effort in the adult habitat, while neglecting to sample juvenile habitat which is still unknown. The small size and low water levels of this river allowed me to thoroughly search all habitats in the study area, and juveniles were not located outside of adult habitats. Also, the "stunting" of individuals possibly suggests that the few small individuals collected might not have been juveniles.

As previously mentioned, it appears that the Saline River is exposed to the same factors which has led to the "stunting" in the Little Missouri River. Several mussels that appeared to be juveniles were found in the Saline River, which leads me to believe that there is recruitment in this river. However, we must ask the same question about the "stunting" and whether these individuals are truly juveniles, or if they are "stunted" adults. A detailed investigation on the age-growth and age at reproductive maturity of these individuals should help answer this question.

Heavy harvest pressures and negligible recruitment may have reduced population numbers below the minimum requirement for a viable community. Zebra mussel fouling is also negatively impacting the

native mussels, and the combined effects may effectively extirpate mussels from these lakes in the near future.

AGE DETERMINATIONS

Age and growth data are relatively sparse for freshwater mussels. Rust (1993), Christian (1995) and Posey (1997) have provided baseline age-growth data for commercial mussel species in Arkansas with respect to their study rivers. In this study, age-growth determinations were made for *Amblema plicata* in the Saline River and *Quadrula quadrula* in Ozark and Dardanelle lakes. All the major drainage basins which are commercially shelled in Arkansas, except the Red River Drainage, now have baseline age-growth data for the most commercially valuable species in their respective drainage basins. These data should clarify whether MLHS restrictions are providing adequate protection for these mussel populations in these drainages.

Saline River

Results

Amblema plicata ranged in depth from 20.8-59.1 mm. Age determinations ranged from 6 to 16 years (Table 27, Figures 10-12). Von Bertalanffy growth parameters were used to show predicted growth patterns for *A. plicata* inhabiting the lower Saline River. However, some unexplainable error prevented these data from being statistically valid, and therefore they are not shown for this species at this time. The correlation coefficient (r) was calculated to be 0.95, and r^2 was calculated to be 0.91. Age at 69.85 mm (2.75 in.), MLHS, was predicted to be 19.13 years with a standard error of prediction of 0.81 year.

Discussion

The high r^2 value indicates a strong correlation between depth and age. The unexplained 9% of the variability could result from the specimens coming from different locales within the study area. I hypothesize that the depth at which the organism is located in a lake may lead to different growth rates.

Table 27. Length, depth, width, dry weight and age of selected *Amblema plicata* used for thin-sectioned age determinations from the Saline River, Arkansas.

Number	Length	Depth	Width	Shell Weight	Age
1	52.2	39.8	27.5	26	12
2	53.6	42.5	39.5	32	12
3	51.8	39.9	39.4	30	12
4	47.3	38.8	28.6	23	11
5	52.1	43.0	28.4	28	13
6	54.0	43.4	29.7	29	12
7	52.1	39.8	26.7	25	11
8	35.0	28.9	20.4	11	8
9	55.2	43.4	28.8	31	12
10	38.8	32.4	23.6	15	10
11	74.9	59.1	37.7	77	15
12	46.4	38.2	25.4	22	11
13	53.3	44.2	29.3	33	12
14	47.6	36.8	22.5	18	11
15	46.3	37.5	25.9	22	11
16	52.6	40.4	27.3	27	12
17	74.9	56.8	35.6	65	15
18	78.7	57.1	35.9	75	16
19	55.3	41.2	27.7	31	15
20	51.5	39.2	26.8	26	12
21	46.8	38.0	26.9	25	13
22	66.5	49.2	35.5	65	15
23	67.4	50.0	32.4	48	14
24	63.2	50.5	32.8	40	14
25	52.2	40.1	25.6	22	12
26	68.5	51.4	29.7	49	14
27	50.7	41.1	27.9	27	13
28	66.7	49.7	36.5	64	14
29	52.0	41.7	27.5	28	11
30	47.2	39.0	27.2	26	11
31	68.8	50.3	33.5	49	14
32	63.4	48.8	29.9	42	13
33	66.3	47.5	32.6	45	13
34	62.9	50.2	35.5	55	14
35	30.3	25.6	20.0	8	7
36	38.4	30.1	22.6	15	9
37	63.3	48.2	29.7	46	14

Table 27. Cont'd

38	68.2	52.6	33.2	60	14
39	64.4	51.8	35.5	56	15
40	49.0	35.6	27.4	22	10
41	63.9	46.6	33.1	52	13
42	56.0	41.0	29.4	34	12
43	37.9	31.3	21.6	13	8
44	45.6	37.2	25.5	21	10
45	30.7	23.8	18.1	7	7
46	26.6	22.7	16.6	6	6
47	26.6	21.6	17.1	6	7
48	25.1	20.8	15.0	5	6
49	29.0	24.0	17.5	7	8
50	27.2	22.5	15.8	6	7

Figure 10. Scatterplot of depth and age correlations for *Amblema plicata*, Saline River, Arkansas.

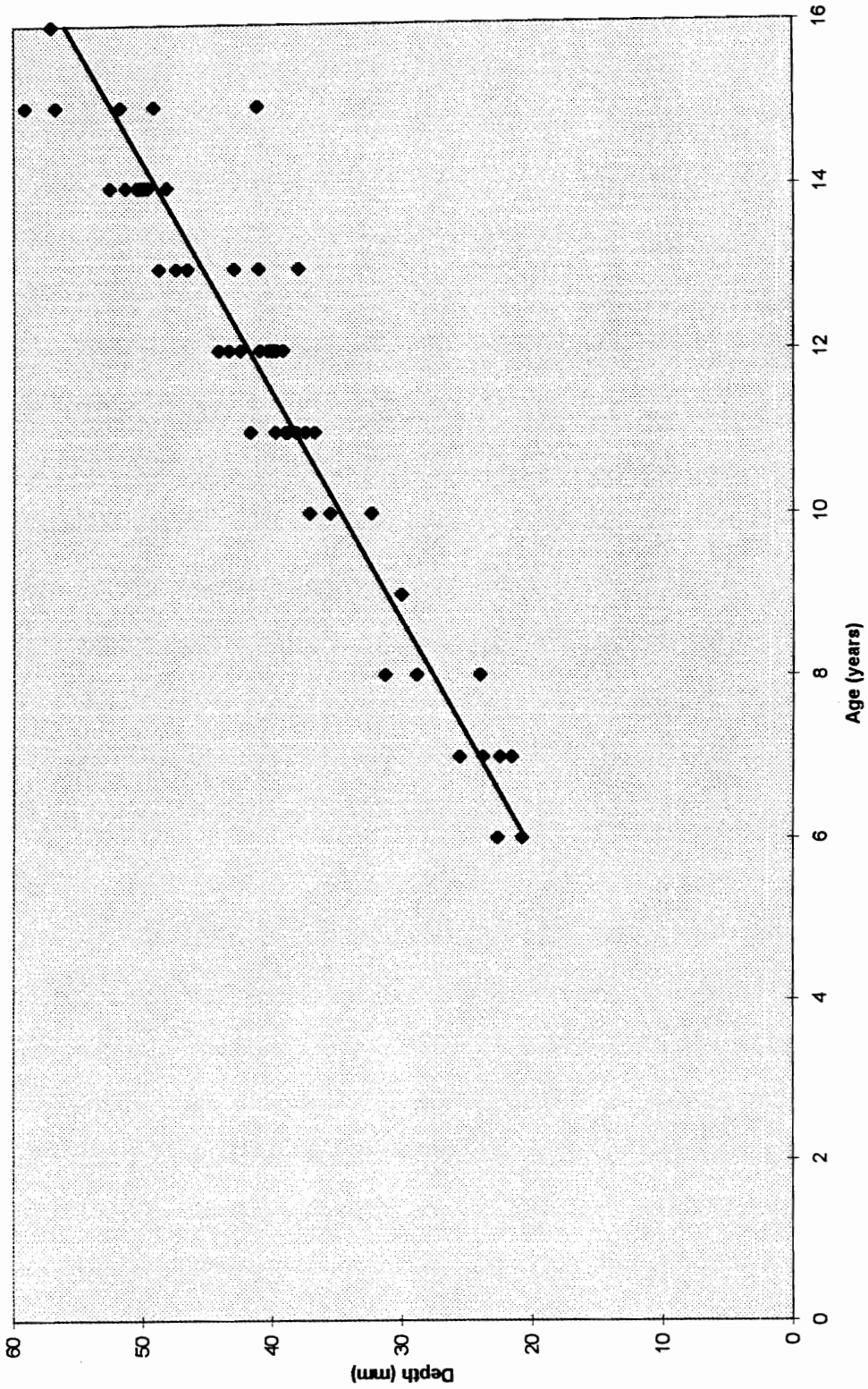

Amblema plicata
Depth (mm)
0
10
20
30
40
50
60
Age (years)
0
2
4
6
8
10
12
14
16

Figure 11. Scatterplot of length and age correlations for *Amblema plicata*, Saline River, Arkansas.

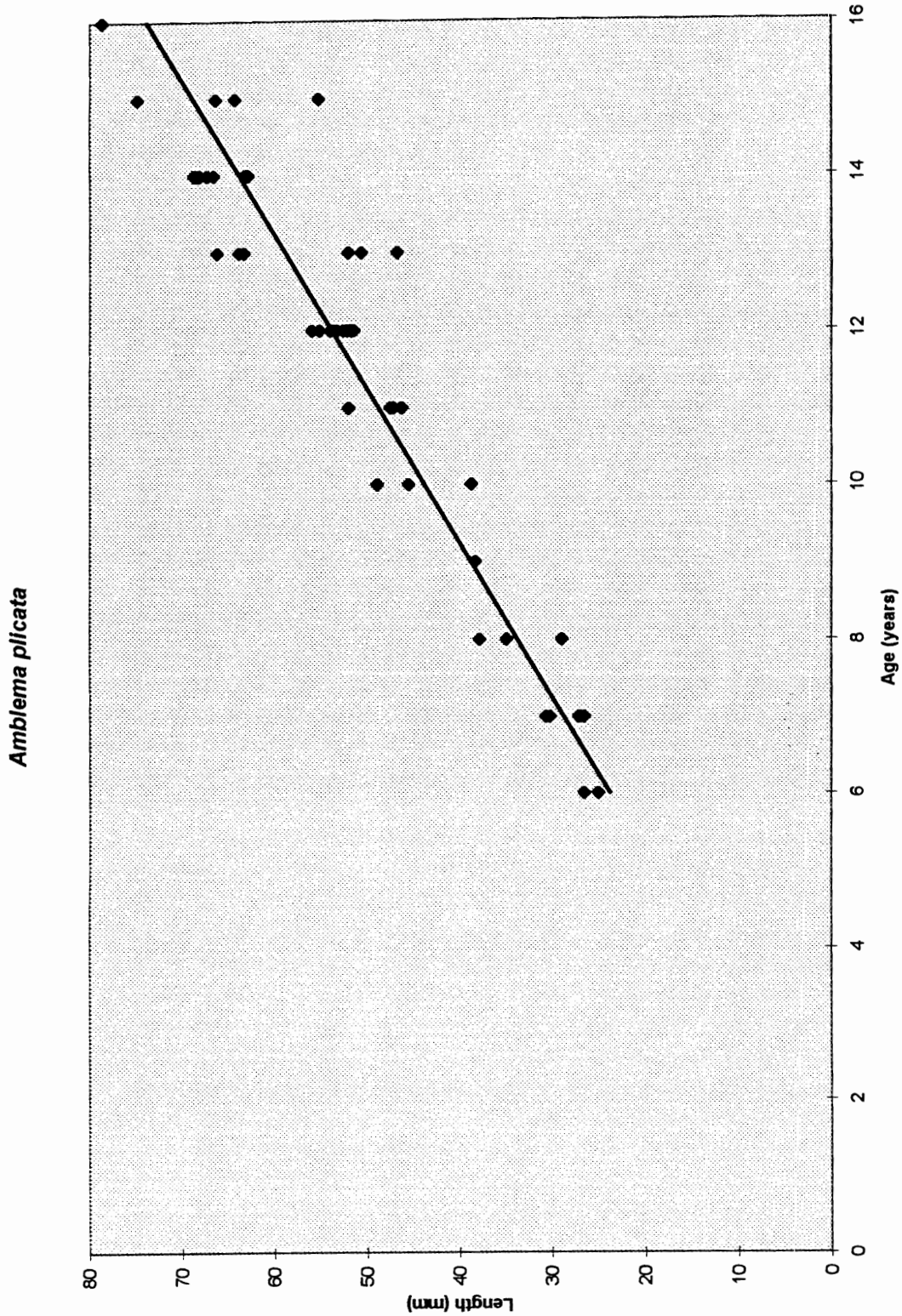
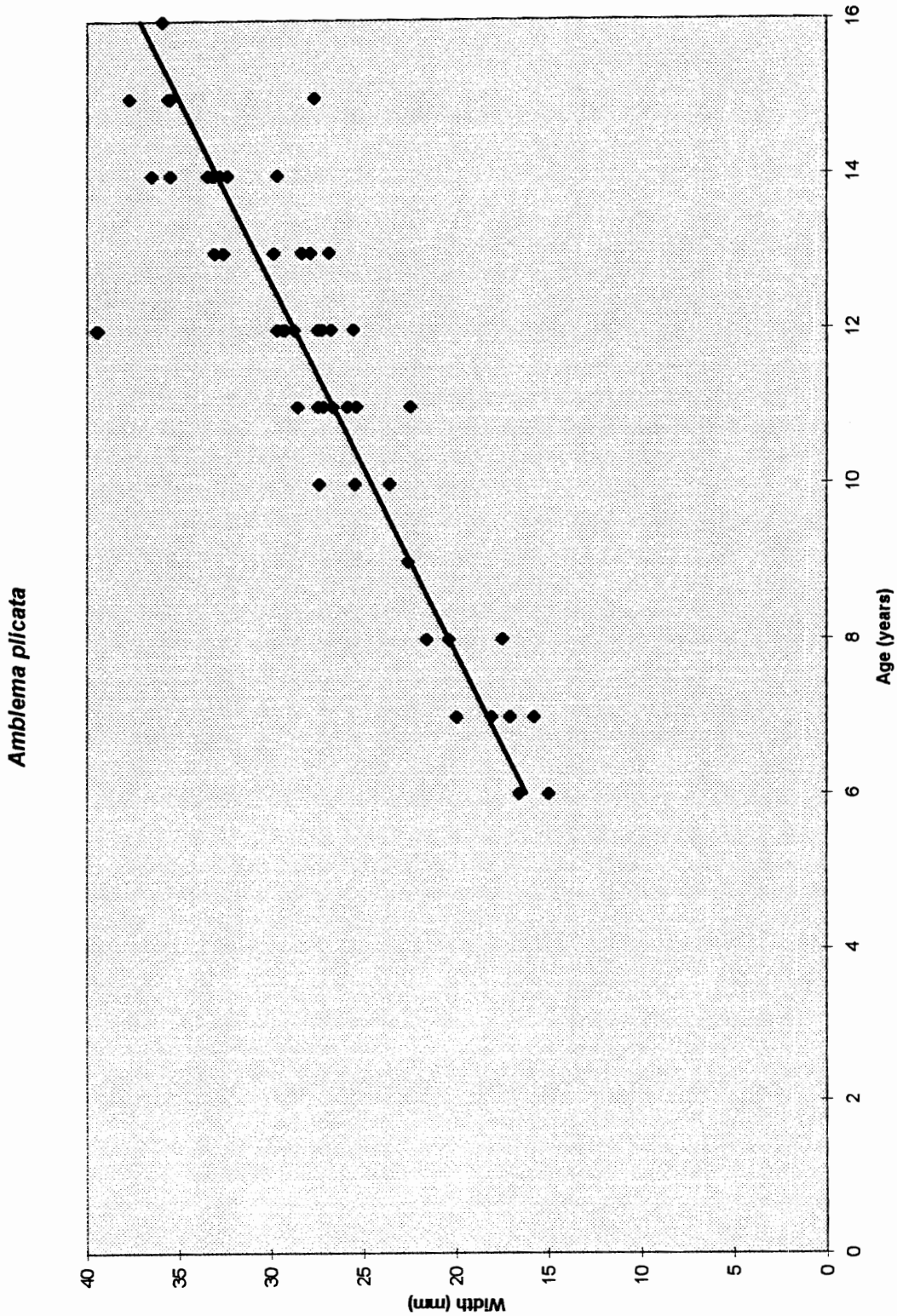

Amblema plicata
Length (mm)
Age (years)
0
10
20
30
40
50
60
70
80
0
2
4
6
8
10
12
14
16

Figure 12. Scatterplot of width and age correlations for *Amblema plicata*, Saline River, Arkansas.


Amblema plicata
Width (mm)
Age (years)
0
5
10
15
20
25
30
35
40
0
2
4
6
8
10
12
14
16

Deeper depths (e.g. below the thermocline) may produce slower growth rates in response to stress (e.g. low dissolved oxygen and other physicochemical parameters) at these depths.

Knowing the age at MLHS is an integral part of successfully managing Unionacea. These age-growth data provide baseline data for this species in the Saline River. These also allow for comparisons with Posey's (1997) data on the Ouachita River. Together these data will provide a better understanding of growth rates in the lower Ouachita River drainage basin.

There is currently a lack of literature addressing the onset of reproduction for this species. When data become available on reproductive maturity, researchers will be able to use the age-growth data presented in this study to determine if the MLHS set by the Arkansas Game and Fish Commission is providing adequate protection for reproductive individuals.

Ozark Lake

Results

Fifty specimens of *Quadrula quadrula* were collected for age-growth analysis. These specimens ranged in depth from 14.0-84.1 mm, and in age from 3 to 18 years (Table 28, Figures 13-15). Von Bertalanffy growth parameters were successively determined for this species. The von Bertalanffy equations were as follows:

$$L_t = 131.51\left(1 - e^{-0.06(t+2.26)}\right) Depth$$
$$L_t = 166.19\left(1 - e^{-0.06(t+2.09)}\right) Length$$
$$L_t = 76.36\left(1 - e^{-0.07(t+2.31)}\right) Width$$

The r value was calculated to be 0.97 with a r^2 of 0.95. Age at 69.85 mm (2.75 in.), MLHS, was determined to be 14.74 years with a standard error of prediction of 0.96 year.

Table 28. Length, depth, width, dry weight and age of selected *Quadrula quadrula* used for thin-sectioned age determinations from Ozark Lake, Arkansas.

Number	Length	Depth	Width	Shell Weight	Age
1*	88.0	69.6	46.1	141	15
2	86.1	68.4	44.9	110	11
3	87.1	70.7	42.3	111	16
4	75.6	60.6	40.4	81	12
5	78.4	61.4	41.3	88	13
6	94.3	70.8	45.9	130	16
7	75.6	57.3	39.2	88	12
8	90.0	73.0	46.2	146	17
9	82.9	70.3	48.7	129	16
10	78.1	61.2	42.8	114	12
11	72.0	56.2	40.2	87	11
12	67.6	51.8	38.3	67	11
13	56.9	46.9	29.1	39	9
14	43.4	35.6	25.8	23	8
15	52.0	43.0	30.3	39	9
16	45.7	36.2	24.7	24	8
17	47.4	35.6	24.7	21	7
18	42.3	35.0	23.4	19	8
19	39.4	29.6	20.9	14	7
20	42.0	32.2	22.2	17	7
21	30.4	22.6	16.9	6	6
22	27.8	20.2	14.3	4	6
23	23.5	18.5	11.6	3	5
24	21.7	15.9	10.7	2	4
25	18.9	14.0	10.2	2	3
26	95.5	73.4	47.9	165	17
27	104.1	81.1	52.8	193	18
28	96.2	74.3	51.0	172	16
29	100.1	78.0	49.8	178	18
30	95.1	80.7	53.2	194	18
31	100.7	84.1	51.4	194	17
32	85.3	71.5	51.2	159	15
33	94.5	74.8	54.8	192	16
34	63.9	52.5	36.9	68	10
35	82.2	63.4	42.0	108	12
36	69.7	56.0	39.3	73	11
37	87.9	66.3	45.3	127	14
38	53.5	42.7	31.2	42	9

Table 28. Cont'd

39	45.9	36.6	24.0	21	8
40	54.1	41.7	25.4	28	9
41	72.7	56.6	36.0	67	10
42	65.8	54.5	36.6	66	11
43	49.0	42.9	30.2	36	9
44	45.4	37.6	23.5	21	9
45	43.2	34.3	23.5	21	8
46	38.7	28.7	17.9	12	7
47	34.5	25.3	18.9	10	6
48	33.4	24.9	19.1	9	6
49	32.4	26.8	19.1	10	6
50	24.5	19.8	13.3	4	5

*1-25 = Main channel
26-50 = Bay areas

Figure 13. Scatterplot and growth curve, predicted limits of depth and age correlations for *Quadrula quadrula*, Ozark Lake, Arkansas.

Quadrula quadrula

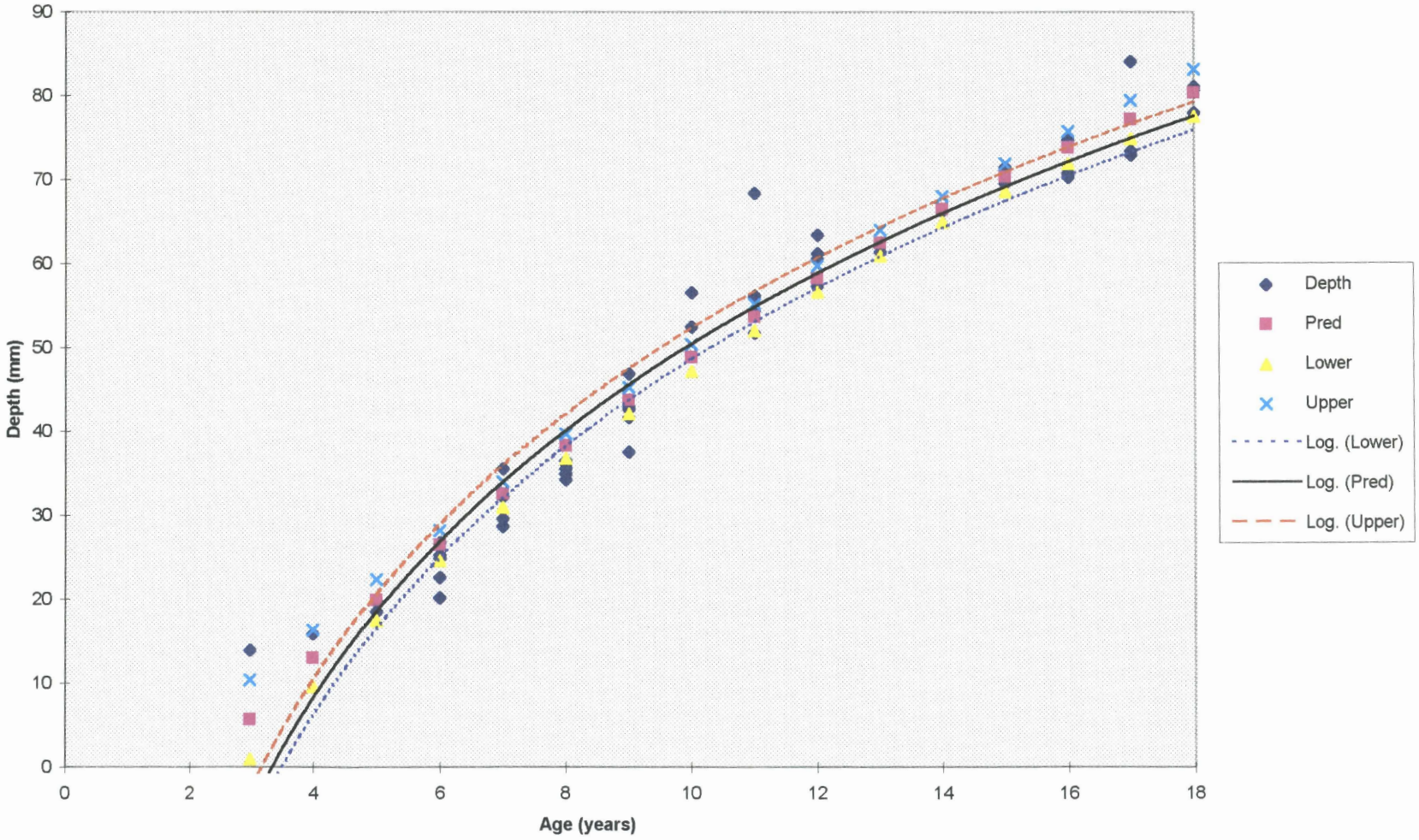


Figure 14. Scatterplot and growth curves, predicted limits of length and age correlations for *Quadrula quadrula*, Ozark Lake, Arkansas.

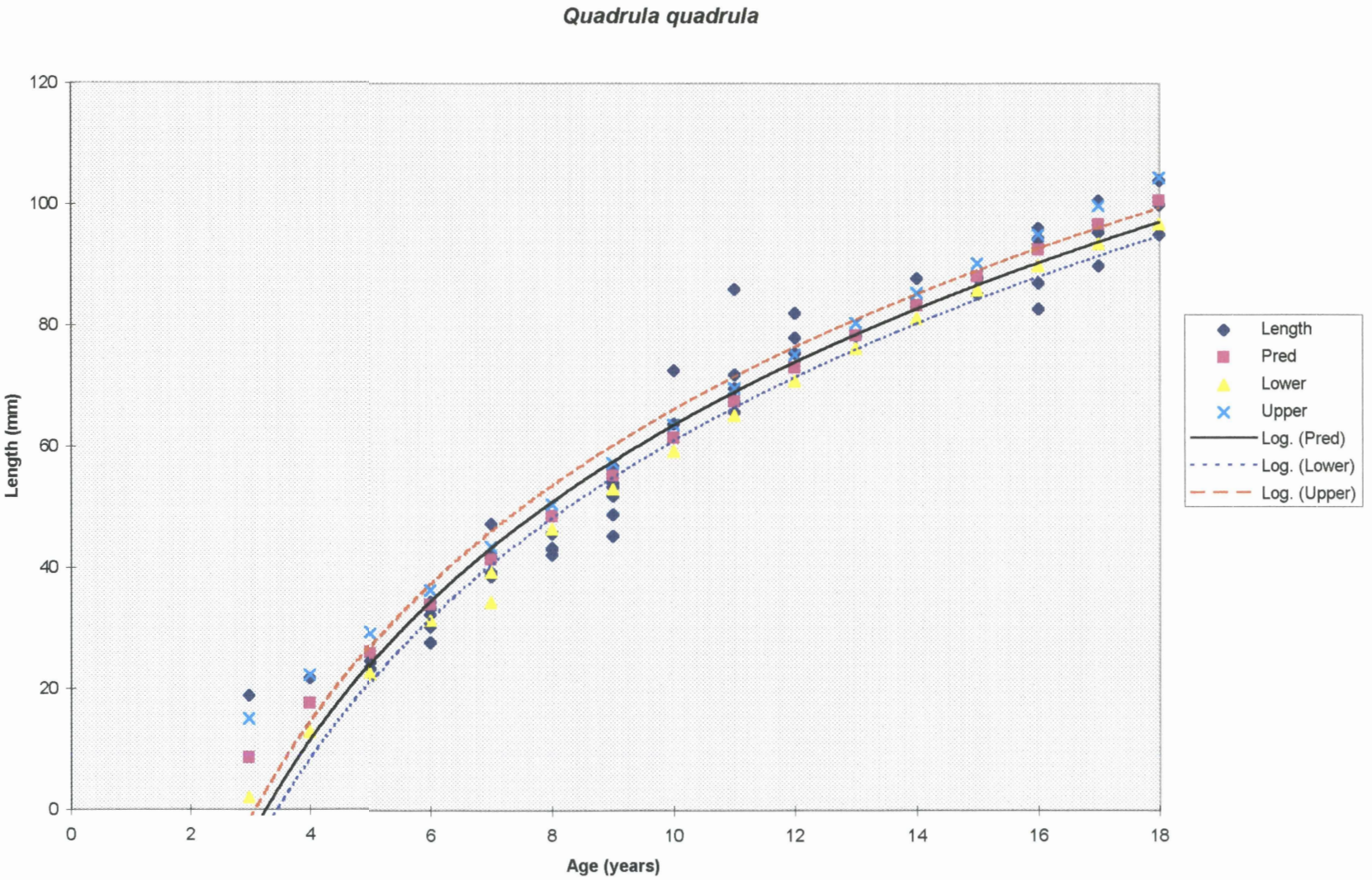
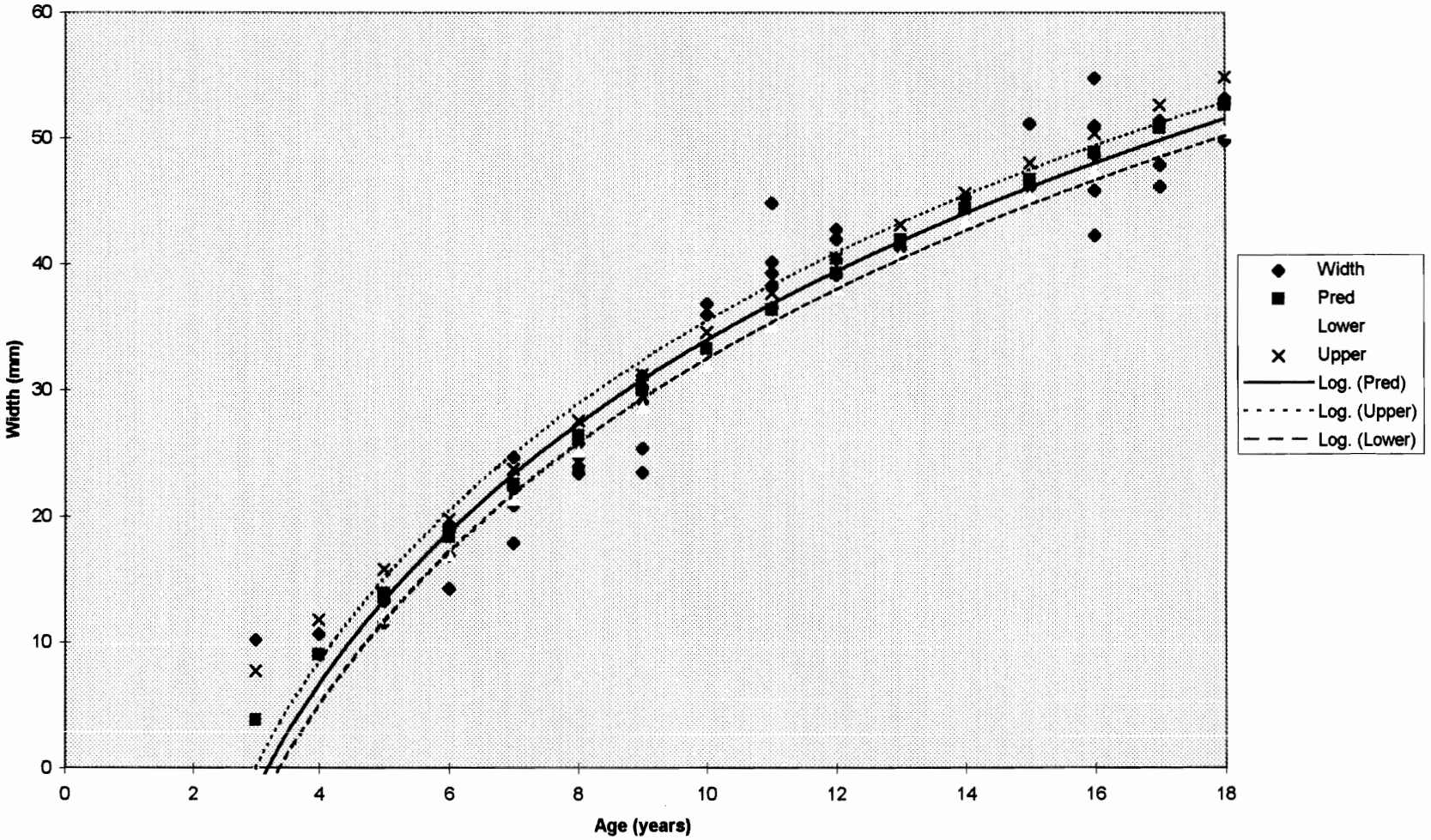

Quadrula quadrula
Length (mm)
Age (years)
0
20
40
60
80
100
120
0
2
4
6
8
10
12
14
16
18
Length
Pred
Lower
Upper
Log. (Pred)
Log. (Lower)
Log. (Upper)

Figure 15. Scatterplot and growth curve, predicted limits of width and age correlations for *Quadrula quadrula*, Ozark Lake, Arkansas.


Quadrula quadrula
Width (mm)
Age (years)
0
10
20
30
40
50
60
0
2
4
6
8
10
12
14
16
18
Width
Pred
Lower
Upper
Log. (Pred)
Log. (Upper)
Log. (Lower)

Discussion

The high r^2 value indicates a strong correlation between depth and age, with only 5% of the variability unexplained. As with *A. plicata*, this variation may result from individuals being collected from different sites or could be a measurement error. The recent fouling caused by zebra mussels is likely to cause stress (e.g. reduced resources), which also may result in slower growth rates, if not mortality.

As with *A. plicata*, reproductive maturity data are needed for *Quadrula quadrula*. We can not assume that we are protecting the reproductive individuals until we compare these age-growth data with reproductive maturity data. Considering the high harvest rates on this lake, it is critical that MLHS restrictions be supported by scientific data.

Lake Dardanelle

Results

Forty-eight specimens of *Quadrula quadrula* were collected for age-growth analysis. These specimens ranged in depth from 16.3-76.5 mm, and in age from 3 to 17 years (Table 29, Figures 16-18). Von Bertalanffy growth parameters were successively determined for this species. The von Bertalanffy equations were as follows:

$$L_t = 163.77\left(1 - e^{-0.04(t+7.82)}\right) Depth$$
$$L_t = 628.49\left(1 - e^{-0.01(t+1.10)}\right) Length$$
$$L_t = 106.87\left(1 - e^{-0.04(t+1.82)}\right) Width$$

The r value was calculated to be 0.96 with a r^2 value of 0.93. This correlation was similar to that for Ozark Lake. Age at 69.85 mm (2.75 in.), MLHS, was determined to be 15.42 years with a standard error of prediction of 0.88 year.

Table 29. Length, depth, width, dry weight and age of selected *Quadrula quadrula* used for thin-sectioned age determinations from Lake Dardanelle, Arkansas.

Number	Length	Depth	Width	Shell Weight	Age
1	81.9	65.5	41.2	104	16
2	91.9	75.3	48.0	177	17
3	77.4	61.6	46.0	119	15
4	39.8	32.6	22.3	16	9
5	65.7	55.9	37.3	72	13
6	76.8	66.1	41.6	105	15
7	73.5	66.2	41.1	102	15
8	82.9	65.5	43.5	117	16
9	81.1	68.3	42.0	115	15
10	78.8	65.5	44.7	117	14
11	83.7	69.0	43.4	122	15
12	66.6	56.1	39.0	75	12
13	80.4	64.7	43.7	127	16
14	82.4	67.3	43.0	115	16
15	86.2	72.9	44.2	137	16
16	70.0	58.2	38.1	91	13
17	74.3	59.0	42.5	86	12
18	81.1	68.3	40.8	113	15
19	83.0	65.8	44.2	98	16
20	70.9	61.5	40.3	89	13
21	74.5	60.0	41.8	95	14
22	72.8	62.5	41.6	97	13
23	78.8	64.3	44.3	126	15
24	74.5	63.4	40.2	94	14
25	79.0	62.8	43.1	103	15
26	70.0	60.6	37.6	81	12
27	76.2	63.3	37.6	83	-
28	62.1	53.0	35.6	64	12
29	70.7	56.8	38.8	77	13
30	87.0	69.6	49.1	168	16
31	72.3	62.9	40.8	97	13
32	65.5	52.1	37.1	67	11
33	58.5	50.8	32.8	54	10
34	97.5	76.5	49.5	194	17

Table 29. Cont'd

35	70.6	60.3	42.5	92	13
36	73.2	62.0	40.1	89	13
37	49.0	40.5	27.3	30	9
38	97.0	73.0	56.3	226	17
39	74.6	63.3	40.1	101	13
40	64.0	58.7	38.3	72	12
41	72.5	62.3	41.2	101	13
42	38.0	30.4	19.7	13	8
43	38.7	31.7	22.4	14	8
44	34.7	28.3	19.3	11	8
45	32.0	26.8	16.9	8	7
46	37.6	32.3	21.3	13	8
47	35.3	30.4	18.3	11	7
48	24.8	20.1	12.6	3	6
49	19.9	16.3	11.4	2	3

Figure 16. Scatterplot and growth curve, predicted limits of depth and age correlations for *Quadrula quadrula*, Lake Dardanelle, Arkansas.

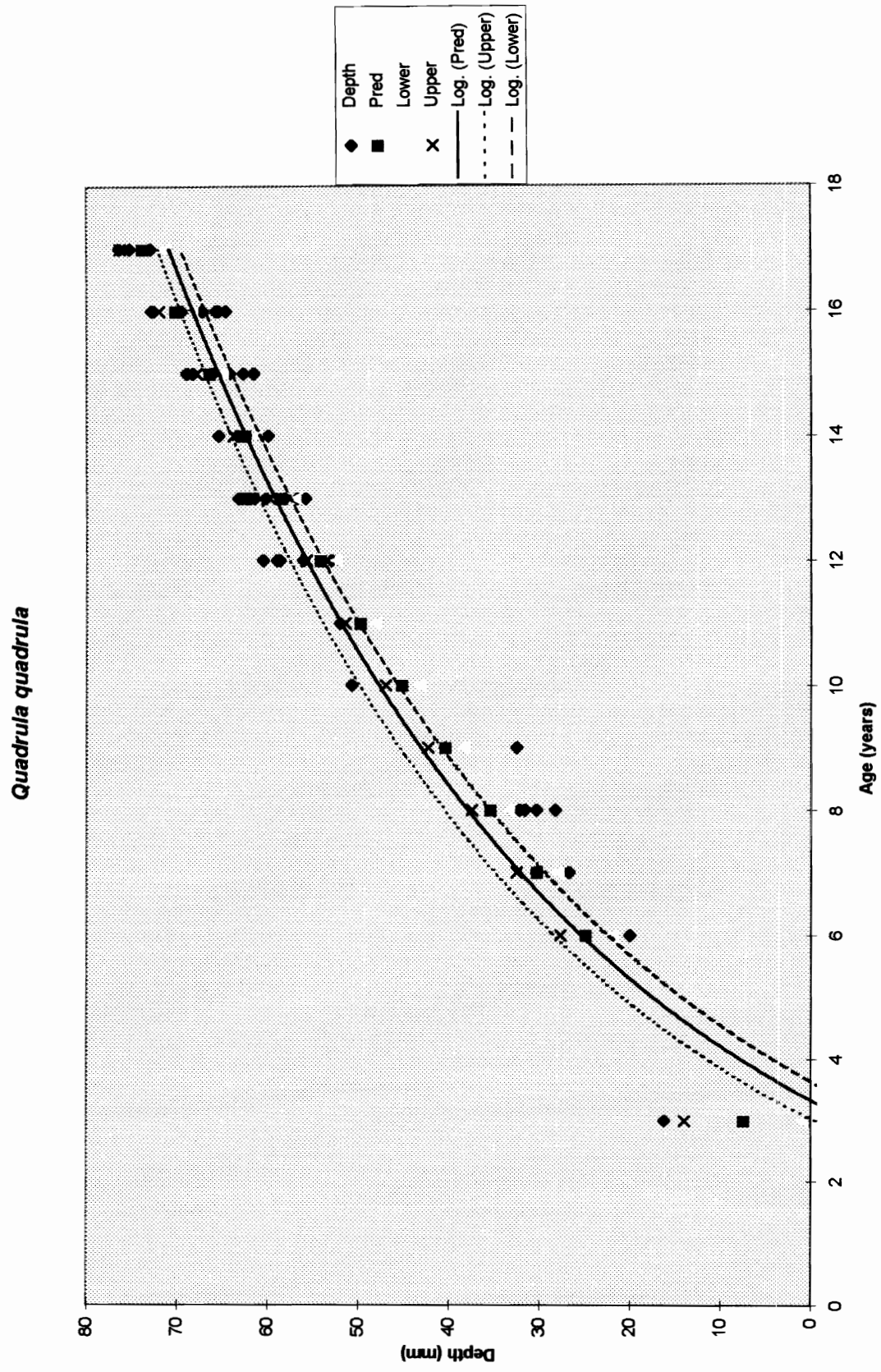

Quadrula quadrula
Depth
Pred
Lower
Upper
Log. (Pred)
Log. (Upper)
Log. (Lower)
Depth (mm)
Age (years)
0
10
20
30
40
50
60
70
80
0
2
4
6
8
10
12
14
16
18

Figure 17. Scatterplot and growth curve, predicted limits of length and age correlations for *Quadrula quadrula*, Lake Dardanelle, Arkansas.

Quadrula quadrula

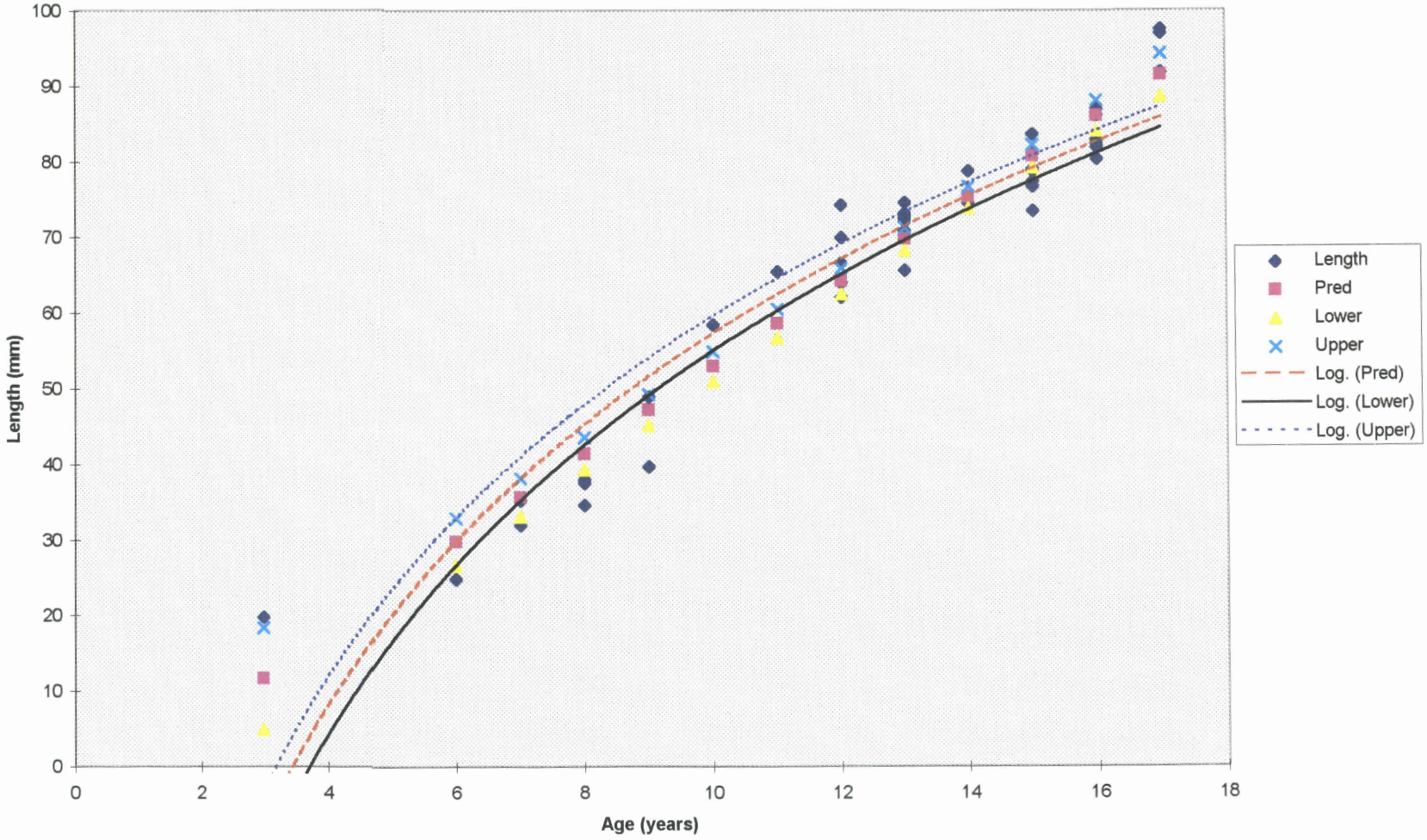
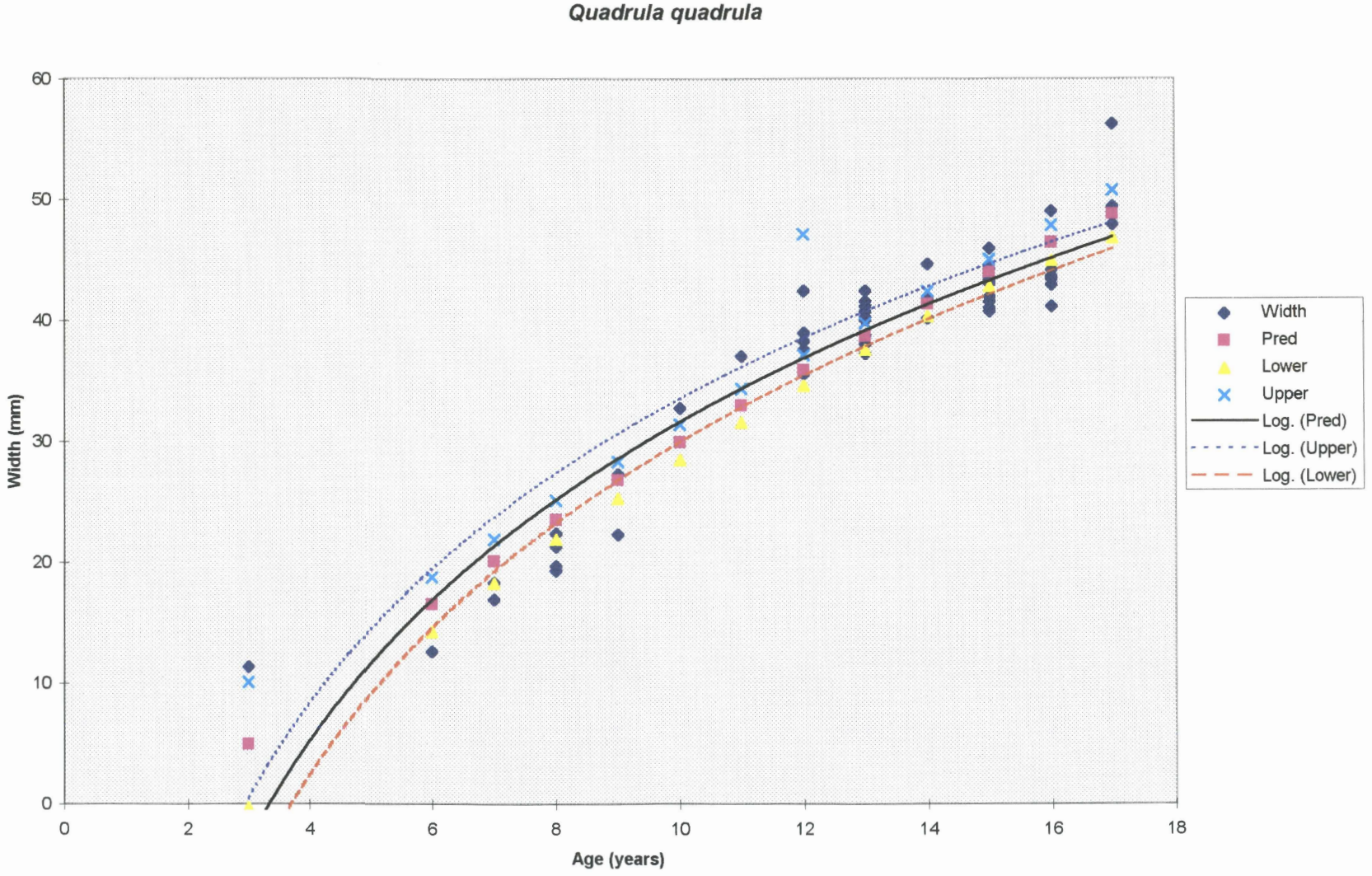


Figure 18. Scatterplot and growth curve, predicted limits of width and age correlations for *Quadrula quadrula*, Lake Dardanelle, Arkansas.


Quadrula quadrula
Width (mm)
Age (years)
0
10
20
30
40
50
60
0
2
4
6
8
10
12
14
16
18
Width
Pred
Lower
Upper
Log. (Pred)
Log. (Upper)
Log. (Lower)

Discussion

The high r^2 value indicates a strong correlation between age and depth with only 7% of the variability unexplained. Therefore, the same variability may be acting on both sets of data. I would expect the physicochemical profile for these two lakes to be different. Lake Dardanelle is comprised primarily of shallow flats and bays, whereas Ozark Lake is more riverine in its characteristics. Due to these differences in expected physicochemical profiles within the lakes and from site to site within the lakes, variability would seem to be a result of one or both of these two factors; first, the differences in growth rates from site to site as a result of varying physicochemical parameters (e.g. dissolved oxygen); second, resource limitation, which could be attributed to differences in physiochemical parameters (e.g. light penetration) or both inter- and intraspecific competition.

Chapter V

MANAGEMENT RECOMMENDATIONS

During the past six years, an enormous database has been established concerning Arkansas' freshwater mussel fauna. Establishing location and community structure of Unionacea assemblages in five of six of the major drainage basins in Arkansas was the first step in a long process of establishing a statewide management plan for the sustained viability of these organisms. However, a need still exists to establish baseline data on small to medium size streams within Arkansas, age at reproductive maturity and fish host relationships. Due to the recent introduction of the zebra mussel into our state, it is particularly imperative that both natural and man-made refugia be created.

Baseline age-growth data exist for four major commercial species. However, these data are of limited value until they are compared with age at reproductive maturity. The mussel communities of Dardanelle and Ozark lakes have been particularly threatened by the recent introduction of the zebra mussel. The decline of these mussels may be irreversible. Based on the findings of this study, I propose that some bays within both of these reservoirs may serve as natural refugia from the high zebra mussel densities prevalent in the main lake. It is already obvious that these areas will not be zebra mussel free, but the areas may not support the high densities which lead to mortality of native mussels. Further research is required to determine why and if these areas will potentially serve as refugia.

The AGFC has implemented a plan to retard the spread of the zebra mussel within Arkansas. As seen with the 1996 incident of the transport of zebra mussels from the Arkansas River to Lake Ouachita, the public has not fully grasped either the ecological or socioeconomic impacts of this pest. Wash stations must be established at all access ramps to bodies of water infested with zebra mussels, and any watercraft using these access points must use wash stations properly. It is important that resource managers are able to communicate to the public that the only way to stop further spread of this organism is through their willingness to support and participate in measures deemed necessary to prevent this invasion.

It is obvious from harvest reports that Blue Mountain Lake once supported a higher numerical standing crop of mussels. The greatest threat to the reintroduction and survival of mussels in this reservoir is the periodic drawdowns of the lake by the USACE. If drawdowns cease it is possible that populations of mussels from the Petit Jean River may reestablish themselves in the lake.

Commercial shelling does not occur on either the Saline or the Little Missouri rivers. Since neither of these rivers supports a viable mussel industry, shelling should not be permitted in these rivers. As a result of finding the federally endangered *Lampsilis abrupta* and *Quadrula fragosa* in the Little Missouri River, this river should receive protection from anthropogenic influences. The portion of the Saline River surveyed in this study is already protected because it lies within the boundaries of Felsenthal National Wildlife Refuge.

LITERATURE CITED

ADPCE. 1987. Physical, chemical and biological characteristics of least disturbed stream in Arkansas Ecoregions. Volume I: Data compilation. 683 pp.

ADPCE. 1994. Water quality inventory report. 152 pp.

Anderson, H. M. 1993. Conserving America's freshwater ecosystems: The Wilderness Society's approach. J.N. Am. Benthol. Soc., 12(2)194-196.

Burnley, T. 1996. 1995 mussel harvest report. AGFC, Little Rock. 17 pp.

Call, R. E. 1895. A study of the Unionidae of Arkansas, with incidental reference to their distribution in the Mississippi Valley. Trans. Acad. Sci. St. Louis. 7:1-65.

Christian, A. D. 1995. Analysis of the commercial mussel beds in the Cache and White rivers in Arkansas. Unpubl. M. S. thesis, Arkansas State University, State University, AR. 197 pp.

Cox, G. W. 1980. A laboratory manual of general ecology. 5th Ed. William C. Brown Co. Publishers. Dubuque, IA. 237 pp.

Croneis, C. 1930. Geology of the Arkansas Paleozoic Area. Ark. Geol. Survey Bull. 457 pp.

Cummings, K. S. and C. A. Mayer. 1992. Field guide to freshwater mussels of the Midwest. Illinois Natural History Survey publ. 194 pp.

Cvancara, A. M. and P. G. Freeman. 1978. Diversity and distribution of mussels (Bivalvia: Unionacea) in a eutrophic reservoir, Lake Ashtabula, North Dakota. Nautilus. 92(1)1-9.

Dopplet, R. 1993. The vital role of the scientific community in new river conservation strategies. J. N. Am. Benthol. Soc. 12(2):189-193.

Gagen, C. J. and J. N. Stoeckel. 1994/1995. Effects of zebra mussel, *Dreissena polymorpha*, infestation of Lake Dardanelle water quality, 1994-1995. Unpubl. 84 pp.

Gilbert, C. R. 1980. Zoogeographic factors in relation to biological monitoring of fish, pp. 309-355. *In:* C. H. Hocutt and J. R. Stauffer, Jr., eds. Biological monitoring of fish. Lexington Books, D. C. Heath and Co., Lexington, MA. 416 pp.

Gordon, M. E. 1980. Recent Mollusca of Arkansas with annotations to systematics and zoogeography. Proc. Ark. Acad. Sci. 34:58-62.

Gordon, M. E. 1985. Mollusca of Frog Bayou, Arkansas. Nautilus. 99(1):6-9.

Gordon, M. E., L. R. Kraemer and A. V. Brown. 1979. Unionaceae of Arkansas: Historical review, checklist, and observations on distribution patterns. American Malacological Bulletin 1979:31-37.

Gordon, M. E., S. W. Chordas, G. L. Harp and A. V. Brown. 1994. Aquatic Mollusca of the White River National Wildlife Refuge, Arkansas, U.S.A. Walkerana 7(17/18):1-9.

Haag, W. R., D. J. Berg, D. W. Garton and J. L. Farris. 1993. Reduced survival and fitness in native bivalves in response to fouling by the introduced zebra mussel (*Dreissena polymorpha*) in western Lake Erie. Can. J. Fish. Aq. Sci. 50(1):13-19.

Harris, J. L. 1992. A mussel survey of Lake Dardanelle in the vicinity of the proposed River Mountain Pumped Storage Project. Unpubl. Report. Welch/Harris Inc. 20 pp.

Harris, J. L. and M. E. Gordon. 1987. Distribution and status of rare and endangered mussels (Mollusca: Margaritiferidae, Unionidae) in Arkansas. Proc. Ark. Acad. Sci. 41:49-55.

Harris, J. L. and M. E. Gordon. Undated. Arkansas mussels. AGFC, Little Rock, Arkansas. 32 pp.

Harris, J. L., P. Rust, S. W. Chordas and G. L. Harp. 1993. Distribution and population structure of freshwater mussels (Unionidae) in Lake Chicot, Arkansas. Proc. Ark. Acad. Sci. 47:38-43.

Hellawell, J. M. 1986. Biological indicators of freshwater pollution and environmental management. Elsevier Applied Science Publishers, London, NY. 546 pp.

Huebner, J. D., D. F. Malley and K. Donkersloot. 1990. Population ecology of the freshwater mussel *Anodonta grandis grandis* in a Precambrian Shield lake, Can. J. Zool. 68:1931-1941.

Hynes, H. D. N. 1979. The ecology of running waters. University of Toronto Press. 555 pp.

Isom, B. G. and C. Gooch. 1983. Rationale and sampling designs for freshwater mussels (Unionidae) in streams, large rivers, impoundments and lakes. ASTI Special Technical Publication 894:46-59.

Justus, B. G. and G. L. Harp. 1992. Boat clarifier effectiveness. Water Environment & Technology Magazine. 4(3):5 pp.

Kohler, C. C. and W. A. Hubert. 1993. Inland fisheries management in North America. Am. Fish. Soc. 594 pp.

Layzer, J. B. and L. M. Madison. 1995. Microhabitat use by freshwater mussels and recommendations for determining their instream flow needs. Regul. Riv. 10:329-345.

Layzer, J. B., M. E. Gordon and R. M. Anderson. 1993. 'Mussels: the forgotten fauna of regulated rivers. A case study of the Caney Fork River'. Regul. Riv. 8:63-71.

Meglitsch, P. H. and F. R. Schram. 1991. Invertebrate zoology. 3rd Ed. Oxford University Press. 623 pp.

Neves, R. J. and S. N. Moyer. 1988. Evaluation of techniques for age determination of freshwater mussels (Unionidae). Am. Malacol. Bull. 6(2):179-188.

Oesch, R. D. 1984. Missouri naiades. Missouri Dept. of Consevation, Jefferson City. 270 pp.

Ohio Sea Grant. 1996. Sea Grant zebra mussel update. A 1995 report of research. Ohio State University. 92 pp.

Petts, G. E. 1984. Impounded rivers, perspectives for ecological management. John Wiley & Sons Ltd. 326 pp.

Posey, W. R. II, J. L. Harris and G. L. Harp. 1996. New distributional records for freshwater mussels in the Ouachita River, Arkansas. Proc. Ark. Acad. Sci. 50:96-98.

Posey, W. R. II. 1997. Location, species composition and community estimates for mussel beds in the St. Francis and Ouachita rivers in Arkansas. Unpubl. M.S. thesis, Arkansas State University, State University, AR. 178 pp.

Riccardi, A., F. G. Whoriskey and J. B. Rasmussen. 1995. Predicting the intensity and impact of *Dreissena polymorpha* infestation on native unionid bivalves from *Dreissena* field density. Can J. Fish. Aq. Sci. 52:1449-1461.

Richter, B. D. 1993. Ecosystem-level conservation at The Nature Conservancy: growing needs for applied research in conservation biology. J. N. Am. Benthol. Soc. 12(2):197-200.

Ritchie, J. C. 1972. Sediment, fish, and fish habitat. J. Soil Water Cons. 27:124-125.

Robison, H. W. and T. M. Buchanan. 1988. Fishes of Arkansas. The University of Arkansas Press, Fayetteville. 536 pp.

Rust, P. J. 1993. Analysis of the commercial mussel beds in the Black, Spring, Strawberry and Current rivers in Arkansas. Unpubl. M.S. thesis, Arkansas State University, State University, AR. 125 pp.

Sampford, M. R. 1962. An introduction to sampling theory with applications to agriculture. Oliver & Boyd Ltd., Edinburgh.

Stoeckel, J. N., C. J. Gagen, C. S. Howell and J. Phillips. 1995. 1995 annual report: monitoring of the Lake Dardanelle zebra mussel population and collection of radiological samples. Unpubl. Arkansas Tech University, Russellville. 20 pp.

Stoeckel, J.N., L. Lewis and J. Harlan. 1996. Mulberry River freshwater mussel survey. Unpubl. Arkansas Tech University, Russellville. 52 pp.

Tucker, J. K. and E. R. Atwood. 1995. Contiguous backwater lakes as possible refugia for unionid mussels in areas of heavy zebra mussel (*Dreissena polymorpha*) colonization. J. Freshwater Ecol. 10(1):43-47.

USACE. 1974. Blue Mountain Lake. Dept. of the Army. Little Rock District, Corps of Engineers. Little Rock, AR. Pamphlet. 2 pp.

USACE. 1987. Ozark Lake. Dept. of the Army, Little Rock District, Corps of Engineers. Little Rock, AR. Pamphlet. 2 pp.

USGS. 1970. Drainage areas of streams in the Arkansas River Basin. Open file report. Little Rock, AR. 75 pp.

USGS. 1979. Drainage areas of streams in Arkansas in the Ouachita River Basin. Open file report. Little Rock, AR. 87 pp.

Wheeler, H. E. 1918. The Mollusca of Clark County, Arkansas. Nautilus. 80(4):111-125.

Williams, J. D., M. L. Warren, Jr., K. S. Cummings, J. L. Harris and R. J. Neves. 1993. Conservation status of freshwater mussels of the United States and Canada. Am. Fish. Soc. 18(9):6-22.

Witte, R. S. 1993. Statistics. 4th Ed. Harcourt Brace Jonanovich Publishers. Orlando, FL. 551 pp.

Appendix 1.1 Physical parameters, species composition, percent legal harvestable mussels and mean densities for minor beds of the Little Missouri River, Arkansas, 1996.

Location: 1*m*, LHBFDS*, SW ¼ SW ¼ S3, T11S, R18W, Clark Co.
Stratum Size: 1.) 5 m x 50 m = 250 m^2
Substrate: 1.) Silt/sand to gravel/silt/sand
Total Samples: 5
Minimum - Maximum density (#/m^2): 15.0 - 72.0
Mean density #/m^2 (Standard deviation): 1.) 38.8 (24.2)

Species	Number Collected	Percent of Total	Percent legal	Mean Density
Actinonaias ligamentina	2	1.0	NA**	0.4
Amblema plicata	13	6.7	0.0	2.6
✓*Cyprogenia aberti*	1	0.5	0.0	0.2
Elliptio dilatata	9	4.6	0.0	0.6
Ellipsaria lineolata	3	1.5	0.0	1.8
Fusconaia flava	21	10.8	0.0	4.2
Lampsilis satura	1	0.5	NV***	0.2
Megalonaias nervosa	4	2.1	0.0	0.8
Obovaria jacksoniana	1	0.5	NV	0.2
Pleurobema pyramidatum	111	57.2	0.0	22.2
Potamilus purpuratus	1	0.5	0.0	0.2
Quadrula pustulosa	21	10.8	0.0	4.2
Quadrula metanevra	6	3.1	0.0	1.2
Totals	194	99.8	0.0	38.8

* LHBFDS = Left hand bank facing downstream
** NA = Wrong measurement taken
*** NV = No Commercial Value

Appendix 1.1 cont'd

Location: 2*m*, LHBFDS*, NE ¼ NE ¼ S9, T11S, R18W, Clark Co.
Stratum Size: 1.) 3 m x 50 m = 150 m^2
Substrate: 1.) Sand/gravel to sand
Total Samples: 5
Minimum - Maximum density (#/m^2): 15.0 - 37.0
Mean density #/m^2 (Standard deviation): 1.) 26.6 (8.62)

Species	Number Collected	Percent of Total	Percent legal	Mean Density
Actinonaias ligamentina	3	2.3	NA**	0.6
Amblema plicata	7	5.3	0.0	1.4
Cyprogenia aberti	1	0.8	0.0	0.2
Elliptio dilatata	34	25.6	0.0	6.8
Ellipsaria lineolata	1	0.8	0.0	0.2
Fusconaia ebena	1	0.8	0.0	0.2
Fusconaia flava	14	10.5	0.0	2.8
Plectomerus dombeyanus	1	0.8	0.0	0.2
Pleurobema pyramidatum	38	28.6	0.0	7.6
Potamilus purpuratus	2	1.5	0.0	0.4
Quadrula pustulosa	10	7.5	0.0	2.0
Quadrula metanevra	20	15.0	0.0	4.0
Tritogonia verrucosa	1	0.8	0.0	0.2
Totals	133	100.3	0.0	26.6

* LHBFDS = Left hand bank facing downstream
** NA = Wrong measurement taken

Appendix 1.1 cont'd

Location: 3*m*, LHBFDS*, NE ¼ SW ¼ S8, T11S, R18W, Clark Co.
Stratum Size: 1.) 4 m x 50 m = 200 m^2
Substrate: 1.) Sand/gravel
Total Samples: 5
Minimum - Maximum density (#/m^2): 16.0 - 42.0
Mean density #/m^2 (Standard deviation): 1.) 24.2 (10.69)

Species	Number Collected	Percent of Total	Percent legal	Mean Density
Amblema plicata	7	5.8	0.0	1.4
Elliptio dilatata	9	7.4	0.0	1.8
Ellipsaria lineolata	3	2.8	0.0	0.6
Fusconaia flava	9	7.4	0.0	1.8
Ligumia recta	1	0.8	100.0	0.2
Megalonaias nervosa	1	0.8	0.0	0.8
Plectomerus dombeyanus	4	3.3	0.0	10.2
Pleurobema pyramidatum	51	42.1	0.0	0.4
Potamilus purpuratus	2	1.7	0.0	0.2
Quadrula cylindrica	1	0.8	0.0	1.4
Quadrula pustulosa	7	5.8	0.0	4.2
Quadrula metanevra	21	17.4	NV**	0.4
Truncilla truncata	2	1.7	0.0	0.4
Tritogonia verrucosa	1	1.7	0.0	0.2
Totals	121	100.3	0.8	24.2

* LHBFDS = Left hand bank facing downstream
** NV = No commercial value

Appendix 1.1 cont'd

Location: 4*m*, LHBFDS*, SW ¼ SW ¼ S8, T11S, R18W, Clark Co.
Stratum Size: 1.) 8 m x 50 m = 400 m^2
Substrate: 1.) Silt over gravel/sand
Total Samples: 5
Minimum - Maximum density (#/m^2): 9.0 - 31.0
Mean density #/m^2 (Standard deviation): 1.) 20.0 (8.28)

Species	Number Collected	Percent of Total	Percent legal	Mean Density
Amblema plicata	27	27.0	0.0	5.4
Elliptio dilatata	1	1.0	0.0	0.2
Ellipsaria lineolata	1	1.0	0.0	0.2
Fusconaia flava	9	9.0	0.0	1.8
Lampsilis satura	2	2.0	NV**	0.4
Lampsilis teres	3	3.0	0.0	0.6
Megalonaias nervosa	1	1.0	0.0	0.2
Obovaria jacksoniana	1	1.0	0.0	0.2
Plectomerus dombeyanus	4	4.0	0.0	0.8
Pleurobema pyramidatum	30	30.0	0.0	0.2
Ptychobranchus occidentalis	1	1.0	0.0	6.0
Potamilus purpuratus	4	4.0	0.0	0.8
Quadrula pustulosa	15	15.0	0.0	3.0
Toxolasma lividus	1	1.0	0.0	0.2
Totals	100	100.0	0.0	20.0

* LHBFDS = Left hand bank facing downstream
** NV = No commercial value

Appendix 1.2 Physical parameters, species composition, percent legal harvestable mussels and population and community estimates for major beds of the Saline River, Arkansas, 1996.

Location: 3*M*, LHBFDS*, SW ¼ SW ¼ S27, T17S, R10W, Bradley Co.
Stratum Size: 1.) 18 m x 50 m = 900 m^2
Substrate: 1.) Gravel with sand
Total Samples: 9
Minimum - Maximum density (#/m^2): 1.0 - 61.0
Mean density #/m^2 (Standard deviation): 1.) 27.2 (19.76)

Species	Number Collected	Percent of Total	Percent legal	Population Estimate
Amblema plicata	110	44.9	0.0	11000±6993
Ellipsaria lineolata	3	1.2	0.0	300±344
Fusconaia ebena	5	2.0	0.0	500±363
Fusconaia flava	14	5.7	0.0	1400±1338
Leptodea fragilis	2	0.8	NV**	200±303
Lampsilis teres	1	0.4	0.0	100±229
Megalonaias nervosa	3	1.2	0.0	300±487
Obovaria jacksoniana	5	2.0	NV	500±500
Obliquaria reflexa	3	1.2	0.0	300±487
Plectomerus dombeyanus	2	0.8	0.0	200±303
Pyganodon grandis	2	0.8	NV	200±459
Potamilus purpuratus	3	1.2	0.0	300±344
Quadrula nodulata	1	0.4	0.0	100±229
Quadrula pustulosa	74	30.2	0.0	7400±3606
Quadrula quadrula	11	4.5	0.0	1100±827
Truncilla truncata	2	0.8	0.0	200±303
Tritogonia verrucosa	4	1.6	0.0	400±698
Totals	245	99.7	0.0	24500±13600

* LHBFDS = Left hand bank facing downstream
** NV = No Commercial Value

Appendix 1.2 cont'd

Location: 5*M*, RHBFDS*, NE ¼ SE ¼ S22, T17S, R10W, Bradley Co.
Stratum Size: 1.) 20 m x 90 m = 1800 m^2
2.)10 m x 50 m = 500 m^2
Total: 2300 m^2
Substrate: 1.) Small cobble and sand
2.) Small cobble and sand
Total Samples: 23
Minimum - Maximum density (#/m^2): 6.0 - 60.0
Mean density #/m^2 (Standard deviation): 1.) 29.1 (12.16)
2.) 32.0 (19.34)
Total: 29.7 (13.56)

Species	Number Collected	Percent of Total	Percent legal	Population Estimate
Actinonaias ligamentina	1	0.1	0.0	100±207
Amblema plicata	282	41.2	0.7	28200±6678
Ellipsaria lineolata	11	1.6	0.0	1100±770
Fusconaia ebena	71	10.4	0.0	7100±4640
Fusconaia flava	52	7.6	0.0	5200±2103
Lampsilis satura	4	0.6	NV**	400±639
Leptodea fragilis	3	0.4	NV	300±335
Megalonaias nervosa	15	2.2	0.0	1500±719
Obovaria jacksoniana	10	1.5	NV	1000±570
Obliquaria reflexa	5	0.7	0.0	500±402
Plectomerus dombeyanus	7	1.0	0.0	700±541
Pyganodon grandis	2	0.3	NV	200±283
Ptychobranchus occidentalis	2	0.3	0.0	200±283
Potamilus purpuratus	6	0.9	NV	600±443
Pleurobema pyramidatum	1	0.1	0.0	100±207
Quadrula pustulosa	168	24.6	0.0	16800±3782
Quadrula quadrula	18	2.6	0.0	1800±838
Truncilla truncata	5	0.7	0.0	500±502
Tritogonia verrucosa	21	3.1	0.0	2100±945
Totals	684	99.9	0.3	68400±13500

* RHBFDS = Right hand bank facing downstream
** NV = No Commercial Value

Appendix 1.2 cont'd

Location: 7*M*, RHBFDS*, SE ¼ NW ¼ S23, T17S, R10W, Bradley Co.
Stratum Size: 1.) 10 m x 100 m = 1000 m^2
Substrate: 1.) Gravel with sand
Total Samples: 10
Minimum - Maximum density (#/m^2): 5.0 - 74.0
Mean density #/m^2 (Standard deviation): 1.) 32.8 (18.94)

Species	Number Collected	Percent of Total	Percent legal	Population Estimate
Amblema plicata	114	34.8	0.0	11400±7174
Ellipsaria lineolata	6	1.8	0.0	600±600
Fusconaia ebena	24	7.3	0.0	2400±1813
Fusconaia flava	16	4.9	0.0	1600±835
Lampsilis satura	5	1.5	NV**	500±503
Lampsilis teres	2	0.6	NV	200±300
Leptodea fragilis	2	0.6	NV	200±300
Megalonaias nervosa	11	3.4	9.1	1100±783
Obovaria jacksoniana	4	1.2	NV	400±368
Obliquaria reflexa	3	0.9	0.0	300±481
Plectomerus dombeyanus	4	1.2	0.0	400±498
Quadrula pustulosa	117	35.6	0.0	11700±5231
Quadrula quadrula	4	1.2	0.0	400±498
Truncilla truncata	7	2.1	NV	700±825
Tritogonia verrucosa	9	2.7	0.0	900±708
Totals	328	99.8	0.3	32800±13478

* RHBFDS = Right hand bank facing downstream
** NV = No Commercial Value

Appendix 1.2 cont'd

Location: 8*M*, RHBFDS*, NE ¼ SE ¼ S13, T17S, R10W, Bradley Co.
Stratum Size: 1.) 10 m x 230 m = 2300 m^2
Substrate: 1.) Sandy gravel
Total Samples: 23
Minimum - Maximum density (#/m^2): 5.0 - 39.0
Mean density #/m^2 (Standard deviation): 1.) 15.3 (9.86)

Species	Number Collected	Percent of Total	Percent legal	Population Estimate
Amblema plicata	134	38.2	0.7	13400±4648
Fusconaia ebena	10	2.8	0.0	1000±780
Fusconaia flava	42	12.0	0.0	4200±2526
Lampsilis hydiana	3	0.9	NV**	300±341
Lampsilis satura	4	1.1	0.0	400±486
Megalonaias nervosa	10	2.8	0.0	1000±720
Obovaria jacksoniana	10	2.8	NV	1000±720
Obliquaria reflexa	6	1.7	0.0	600±612
Plectomerus dombeyanus	11	3.1	0.0	1100±782
Ptychobranchus occidentalis	2	0.6	0.0	200±285
Pleurobema pyramidatum	2	0.6	0.0	200±285
Quadrula pustulosa	102	29.1	0.0	10200±3934
Quadrula quadrula	5	1.4	0.0	500±418
Strophitus undulatus	1	0.3	0.0	100±205
Truncilla truncata	5	1.4	0.0	500±418
Tritogonia verrucosa	4	1.1	0.0	400±486
Totals	351	99.9	0.3	35100±9762

* RHBFDS = Right hand bank facing downstream
** NV = No Commercial Value

Appendix 1.2 cont'd

Location: 10*M*, RHBFDS*, SW ¼ NE ¼ S13, T17S, R10W, Bradley Co.
Stratum Size: 1.) 20 m x 130 m = 2600 m^2
2.) 20 m x 170 m = 3400 m^2
Total: 6000 m^2
Substrate: 1.) Sand with gravel
2.) Sand with gravel
Total Samples: 25
Minimum - Maximum density (#/m^2): 16.0 - 94.0
Mean density #/m^2 (Standard deviation): 1.) 31.0 (10.23)
2.) 46.1 (21.27)
Total: 39.5 (18.64)

Species	Number Collected	Percent of Total	Percent legal	Population Estimate
Actinonaias ligamentina	7	0.7	0.0	1694±1283
Amblema plicata	203	20.5	0.5	48781±10435
Ellipsaria lineolata	17	1.7	0.0	4077±2627
Elliptio dilatata	3	0.3	0.0	729±795
Fusconaia ebena	48	4.9	0.0	11495±4853
Fusconaia flava	167	16.9	0.0	40096±9070
Lampsilis hydiana	8	0.8	NV**	1936±1296
Lampsilis teres	9	0.9	0.0	2186±1573
Lampsilis satura	15	1.5	NV	3630±2435
Leptodea fragilis	4	0.4	NV	971±876
Megalonaias nervosa	28	2.8	0.0	6722±3422
Obovaria jacksoniana	14	1.4	NV	3394±2048
Obliquaria reflexa	10	1.0	0.0	2422±1789
Plectomerus dombeyanus	44	4.4	0.0	10627±4555
Ptychobranchus occidentalis	6	0.6	0.0	1457±1591
Potamilus purpuratus	5	0.5	0.0	1214±1183
Pleurobema pyramidatum	69	7.0	0.0	16621±7582
Quadrula metanevra	12	1.2	0.0	2901±1950
Quadrula nodulata	1	0.1	0.0	243±498
Quadrula pustulosa	257	26.0	0.0	61765±16316
Quadrula quadrula	4	0.4	0.0	971±876
Strophitus undulatus	2	0.2	NV	479±679
Truncilla donaciformis	4	0.4	NV	971±876
Truncilla truncata	27	2.7	NV	6544±2911
Tritogonia verrucosa	24	2.4	0.0	5809±3779

Appendix 1.2 cont'd

Utterbackii imbecillis	1	0.1	NV	236±486
Totals	989	99.8	0.1	237486±42385

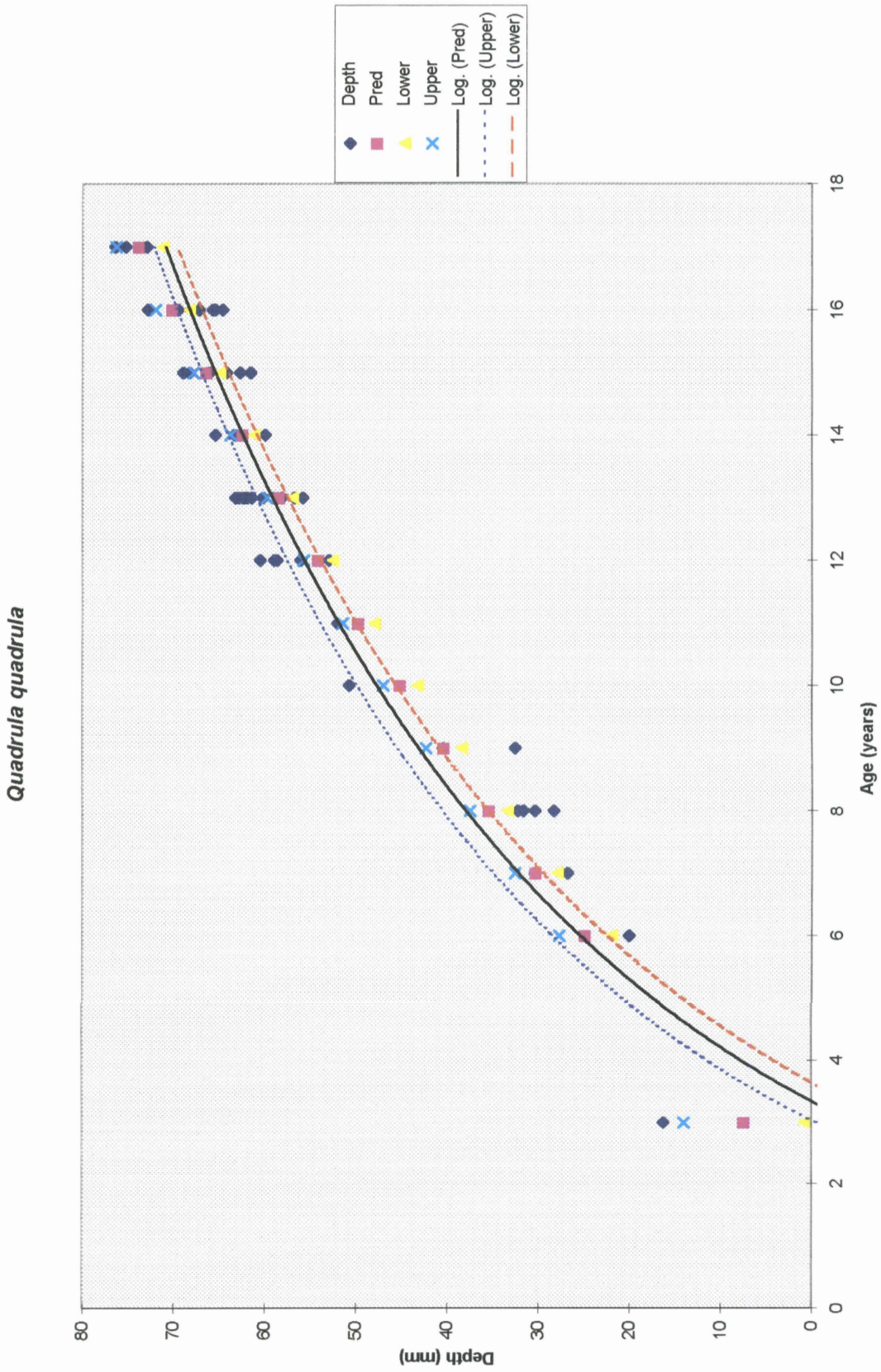
* RHBFDS = Right hand bank facing downstream
** NV = No Commercial Value

Appendix 1.3 Physical parameters, species composition, percent legal harvestable mussels and mean densities for minor beds of the Saline River, Arkansas, 1996.

Location: 1*m*, RHBFDS*, SW ¼ SW ¼ S27, T17S, R10W, Bradley Co.
Stratum Size: 1.) 5 m x 90 m = 450 m^2
Substrate: 1.) Gravel with sand
Total Samples: 5
Minimum - Maximum density (#/m^2): 17.0 - 38.0
Mean density #/m^2 (Standard deviation): 1.) 24.6 (8.79)

Species	Number Collected	Percent of Total	Percent legal	Mean Density
Amblema plicata	58	47.2	0.0	11.6
Fusconaia ebena	3	2.4	0.0	0.6
Fusconaia flava	2	1.6	0.0	0.4
Megalonaias nervosa	1	0.8	0.0	0.2
Obovaria jacksoniana	4	3.3	NV**	0.8
Obliquaria reflexa	1	0.8	0.0	0.2
Potamilus purpuratus	1	0.8	0.0	0.2
Quadrula pustulosa	53	43.1	0.0	10.6
Totals	123	100.0	0.0	24.6

* RHBFDS = Left hand bank facing downstream
** NV = No Commercial Value


Quadrula quadrula
Depth
Pred
Lower
Upper
Log. (Pred)
Log. (Upper)
Log. (Lower)
Depth (mm)
Age (years)
0
10
20
30
40
50
60
70
80
0
2
4
6
8
10
12
14
16
18

Appendix 1.3 cont'd

Location: 2*m*, LHBFDS*, NW ¼ SW ¼ S27, T17S, R10W, Bradley Co.
Stratum Size: 1.) 7 m x 65 m = 455 m^2
Substrate: 1.) Gravel
Total Samples: 5
Minimum - Maximum density (#/m^2): 10.0 - 51.0
Mean density #/m^2 (Standard deviation): 1.) 26.6 (16.92)

Species	Number Collected	Percent of Total	Percent legal	Mean Density
Amblema plicata	58	43.6	0.0	11.6
Fusconaia ebena	1	0.8	0.0	0.2
Fusconaia flava	3	2.3	0.0	0.6
Lampsilis hydiana	1	0.8	NV**	0.2
Lampsilis satura	2	1.5	NV	0.4
Lampsilis teres	2	1.5	0.0	0.4
Megalonaias nervosa	1	0.8	0.0	0.2
Obovaria jacksoniana	6	4.5	NV	1.2
Obliquaria reflexa	2	1.5	0.0	0.4
Plectomerus dombeyanus	8	6.0	0.0	1.6
Potamilus purpuratus	1	0.8	0.0	0.2
Quadrula pustulosa	48	36.1	0.0	9.6
Totals	133	100.2	0.0	26.6

* LHBFDS = Left hand bank facing downstream
** NV = No commercial value

Appendix 1.3 cont'd

Location: 4*m*, LHBFDS*, SE ¼ NW ¼ S27, T17S, R10W, Bradley Co.
Stratum Size: 1.) 5 m x 60 m = 300 m^2
Substrate: 1.) Gravel
Total Samples: 5
Minimum - Maximum density (#/m^2): 9.0 - 37.0
Mean density #/m^2 (Standard deviation): 1.) 24.2 (13.61)

Species	Number Collected	Percent of Total	Percent legal	Mean Density
Actinonaias ligamentina	1	0.8	0.0	0.2
Amblema plicata	48	36.7	0.0	9.6
Ellipsaria lineolata	1	0.8	0.0	0.2
Fusconaia ebena	2	1.6	0.0	0.4
Fusconaia flava	4	3.3	0.0	0.8
Lampsilis hydiana	1	0.8	NV**	0.2
Megalonaias nervosa	4	3.3	0.0	0.8
Plectomerus dombeyanus	1	0.8	0.0	0.2
Potamilus purpuratus	1	0.8	0.0	0.2
Quadrula nodulata	1	0.8	0.0	0.2
Quadrula pustulosa	50	41.3	0.0	10.0
Quadrula quadrula	3	2.5	0.0	0.6
Truncilla truncata	1	0.8	NV	0.2
Tritogonia verrucosa	3	2.5	0.0	0.6
Totals	121	96.8	0.0	24.2

* LHBFDS = Left hand bank facing downstream
** NV = No commercial value

Appendix 1.3 cont'd

Location: 6*m*, RHBFDS*, SW ¼ NW ¼ S23, T17S, R10W, Bradley Co.
Stratum Size: 1.) 5 m x 85 m = 425 m^2
Substrate: 1.) Gravel with sand
Total Samples: 5
Minimum - Maximum density (#/m^2): 9.0 - 28.0
Mean density #/m^2 (Standard deviation): 1.) 21.6 (7.44)

Species	Number Collected	Percent of Total	Percent legal	Mean Density
Amblema plicata	43	39.8	0.0	8.6
Ellipsaria lineolata	1	0.9	0.0	0.2
Fusconaia ebena	2	1.9	0.0	0.4
Fusconaia flava	6	5.6	0.0	1.2
Lampsilis satura	2	1.9	NV**	0.4
Leptodea fragilis	1	0.9	NV	0.2
Megalonaias nervosa	3	2.8	0.0	0.6
Obovaria jacksoniana	7	6.5	NV	1.4
Obliquaria reflexa	1	0.9	0.0	0.2
Plectomerus dombeyanus	3	2.8	0.0	0.6
Quadrula pustulosa	33	30.6	0.0	6.6
Quadrula quadrula	4	3.7	0.0	0.8
Tritogonia verrucosa	2	1.9	0.0	0.4
Totals	108	100.2	0.0	21.6

* RHBFDS = Right hand bank facing downstream
** NV = No commercial value

Appendix 1.3 cont'd

Location: 9*m*, LHBFDS*, NW ¼ SE ¼ S13, T17S, R10W, Bradley Co.
Stratum Size: 1.) 5 m x 90 m = 450 m^2
Substrate: 1.) Gravel to gravel with sand
Total Samples: 5
Minimum - maximum density (#/m^2): 26.0 - 50.0
Mean density #/m^2 (Standard deviation): 1.) 39.2 (10.55)

Species	Number Collected	Percent of Total	Percent legal	Mean Density
Amblema plicata	93	47.4	0.0	18.6
Fusconaia ebena	1	0.5	0.0	0.2
Fusconaia flava	20	10.2	0.0	4.0
Lampsilis satura	3	1.5	NV**	0.6
Leptodea fragilis	2	1.0	NV	0.4
Megalonaias nervosa	6	3.1	0.0	1.2
Obovaria jacksoniana	2	1.0	NV	0.4
Obliquaria reflexa	4	2.0	0.0	0.8
Plectomerus dombeyanus	4	2.0	0.0	0.8
Quadrula pustulosa	53	27.0	0.0	10.6
Quadrula quadrula	2	1.0	0.0	0.4
Truncilla truncata	4	2.0	NV	0.8
Tritogonia verrucosa	2	1.0	0.0	0.4
Totals	196	99.7	0.0	39.2

* LHBFDS = Left hand bank facing downstream
** NV = No commercial value

Appendix 1.4 Physical parameters, species composition, percent legal harvestable mussels and population and community estimates for major beds of Ozark Lake, Arkansas, 1996.

Location: 1*M*, LHBFDS*, SW ¼ SW ¼ S36, T10N, R27W, Franklin Co.
Stratum Size: 1.) 10 m x 500 m = 5000 m^2
2.) 10 m x 100 m = 1000 m^2
Total: 6000 m^2
Substrate: 1.) Silty clay
2.) Silty clay
Total Samples: 25
Minimum - Maximum density (#/m^2): 0.0 - 7.0
Mean density #/m^2 (Standard deviation): 1.) 0.8 (1.59)
2.) 0.6 (0.89)
Total: 0.7 (1.46)

Species	Number Collected	Percent of Total	Percent legal	Population Estimate
Arcidens confragosus	1	5.6	NV**	250±514
Obliquaria reflexa	7	38.8	0.0	1600±1224
Potamilus ohiensis	1	5.6	NV	250±514
Quadrula quadrula	9	50.0	11.1	2250±3210
Totals	18	100.0	6.3	4350±3697

* LHBFDS = Left hand bank facing downstream
** NV = No Commercial Value

Appendix 1.4 cont'd.

Location: 2*M*, LHBFDS*, NW ¼ NW ¼ S18, T9N, R29W, Crawford Co.
Stratum Size: 1.) 15 m x 200 m = 3000 m^2
2.) 8 m x 600 m = 4800 m^2
Total: 7800 m^2
Substrate: 1.) Silty clay with sand
2.) Silty clay with sand
Total Samples: 30
Minimum - Maximum density (#/m^2): 0.0 - 2.0
Mean density #/m^2 (Standard deviation): 1.) 0.8 (0.86)
2.) 0.6 (0.84)
Total: 0.8 (0.86)

Species	Number Collected	Percent of Total	Percent legal	Population Estimate
Arcidens confragosus	2	9.5	NV**	375±522
Obliquaria reflexa	1	4.8	0.0	188±383
Pyganodon grandis	4	19.0	NV	1216±1563
Quadrula quadrula	14	66.7	35.7	3557±1994
Totals	21	100.0	33.3	5336±2521

* LHBFDS = Left hand bank facing downstream
** NV = No Commercial Value

Appendix 1.4 cont'd.

Location: 3*M*, LHBFDS*, NW ¼ NW ¼ S25, T9N, R30W, Crawford Co.
Stratum Size: 1.) 10 m x 50 m = 500 m^2
Substrate: 1.) Sand/sand with gravel
Total Samples: 10
Minimum - Maximum density (#/m^2): 0.0 - 2.0
Mean density #/m^2 (Standard deviation): 1.) 0.6 (0.84)

Species	Number Collected	Percent of Total	Percent legal	Population Estimate
Quadrula quadrula	6	100.0	0.0	300±298
Totals	6	100.0	0.0	300±298

* LHBFDS = Left hand bank facing downstream

Appendix 1.5 Physical parameters, species composition, percent legal harvestable mussels and population and community estimates for major beds of Lake Dardanelle, Arkansas, 1996.

Location: 1*M*, LHBFDS*, NE ¼ NE ¼ S26, T8N, R21W, Pope Co.
Stratum Size: 1.) 3 m x 250 m = 750 m^2
2.) 3 m x 125 m = 375 m^2
3.) 5 m x 200 m = 1000 m^2
Total: 2125 m^2
Substrate: 1.) Silt over gravel
2.) Silt over gravel
3.)Silt over sand/gravel
Total Samples: 22
Minimum - Maximum density (#/m^2): 0.0 - 13.0
Mean density #/m^2 (Standard deviation): 1.) 7.4 (4.10)
2.) 4.0 (1.73)
3.) 1.9 (1.97)
Total: 4.5 (3.91)

Species	Number Collected	Percent of Total	Percent legal	Population Estimate
Fusconaia flava	2	2.0	0.0	167±224
Leptodea fragilis	1	1.0	NV**	83±171
Obliquaria reflexa	8	8.2	0.0	742±692
Plectomerus dombeyanus	39	39.8	25.6	3550±1641
Pyganodon grandis	1	1.0	NV	83±171
Quadrula pustulosa	1	1.0	0.0	83±171
Quadrula quadrula	46	47.0	8.7	4275±1876
Totals	98	100.0	14.6	8983±2524

* LHBFDS = Left hand bank facing downstream
** NV = No Commercial Value

Appendix 1.5 cont'd

Location: 2*M*, LHBFDS*, NW ¼ NW ¼ S26, T8N, R21W, Pope Co.
Stratum Size: 1.) 3 m x 250 m = 750 m^2
2.) 5 m x 175 m = 875 m^2
Total: 1625 m^2
Substrate: 1.) Silt/gravel and silt over sand
2.) Silt/gravel and silt over sand
Total Samples: 17
Minimum - Maximum density (#/m^2): 1.0 - 16.0
Mean density #/m^2 (Standard deviation): 1.) 5.6 (5.48)
2.) 9.0 (3.78)
Total: 7.4 (4.82)

Species	Number Collected	Percent of Total	Percent legal	Population Estimate
Arcidens confragosus	1	0.8	NV**	97±203
Fusconaia flava	3	2.4	0.0	268±321
Obliquaria reflexa	11	8.7	0.0	1056±715
Plectomerus dombeyanus	61	48.4	16.4	5889±2381
Pyganodon grandis	1	0.8	NV	97±203
Quadrula pustulosa	7	5.6	0.0	670±599
Quadrula quadrula	42	33.3	9.5	3997±2069
Totals	126	100.0	11.2	12074±3679

* LHBFDS = Left hand bank facing downstream
** NV = No Commercial Value

Appendix 1.5 cont'd

Location: 3*M*, LHBFDS*, NW ¼ NW ¼ S2, T7N, R21W, Pope Co.
Stratum Size: 1.) 10 m x 150 m = 1500 m^2
Substrate: 1.) Crumbly clay to silt over clay
Total Samples: 15
Minimum - Maximum density (#/m^2): 1.0 - 21.0
Mean density #/m^2 (Standard deviation): 1.) 4.7 (4.84)

Species	Number Collected	Percent of Total	Percent legal	Population Estimate
Leptodea fragilis	1	1.4	NV**	100±219
Obliquaria reflexa	2	2.9	0.0	200±287
Plectomerus dombeyanus	54	77.1	25.9	5400±3296
Pyganodon grandis	2	2.9	NV	200±287
Quadrula quadrula	11	15.7	9.1	1100±795
Totals	70	100.0	22.4	7000±4006

* LHBFDS = Left hand bank facing downstream
** NV = No Commercial Value

Appendix 1.5 cont'd

Location: 5*M*, SBNB*, NW ¼ NW ¼ S12, T7N, R22W, Logan Co.
Stratum Size: 1.) 1 m x 90 m = 90 m^2
2.)1 m x 350 m = 350 m^2
3.)4 m x 125 m = 500 m^2
Total: 1040 m^2
Substrate: 1.) Silt over loose slate/gravel
2.) Silt over loose slate/gravel
3.) Silt over loose slate/gravel
Total Samples: 15
Minimum - Maximum density (#/m^2): 0.0 - 4.0
Mean density #/m^2 (Standard deviation): 1.) 1.6 (1.14)
2.) 1.4 (0.89)
3.) 1.8 (1.79)
Total: 1.6 (1.24)

Species	Number Collected	Percent of Total	Percent legal	Population Estimate
Plectomerus dombeyanus	1	4.2	0.0	100±212
Quadrula quadrula	23	95.8	0.0	1434±898
Totals	24	100.0	0.0	1534±886

* SBNB = South bank of north bay

Appendix 1.5 cont'd

Location: 6*M*, NE ¼ NE ¼ S3, T8N, R24W, Logan Co.
Stratum Size: 1.) 10 m x 150 m = 1500 m^2
Substrate: 1.) Silty clay and crumbly clay
Total Samples: 15
Minimum - Maximum density (#/m^2): 0.0 - 2.0
Mean density #/m^2 (Standard deviation): 1.) 0.3 (0.59)

Species	Number Collected	Percent of Total	Percent legal	Population Estimate
Pyganodon grandis	2	50.0	NV*	200±299
Quadrula quadrula	2	50.0	50.0	200±446
Totals	4	100.0	50.0	400±490

* NV = No Commercial Value

Appendix 1.6 Physical parameters, species composition, percent legal harvestable mussels and mean densities for minor beds of Lake Dardanelle, Arkansas, 1996.

Location: 4*m*, NBSB*, SE ¼ SE ¼ S16, T7N, R21W, Yell Co.
Stratum Size: 1.) 1 m x 100 m = 100 m^2
Substrate: 1.) Silt over loose clay
Total Samples: 5
Minimum - Maximum density (#/m^2): 0.0 - 5.0
Mean density #/m^2 (Standard deviation): 1.) 1.0 (2.24)

Species	Number Collected	Percent of Total	Percent legal	Mean Density
Quadrula quadrula	5	100.0	0.0	1.0
Totals	5	100.0	0.0	1.0

* NBSB = North Bank of South Bay

Appendix 1.7 Physical parameters, species composition, percent legal harvestable mussels and population and community estimates for a follow-up on selected major beds of Lake Dardanelle, Arkansas, 1997.

Location: 1*M*, LHBFDS*, NE ¼ NE ¼ S26, T8N, R21W, Pope Co.
Stratum Size: 1.) 3 m x 250 m = 750 m^2
2.) 3 m x 125 m = 375 m^2
3.) 5 m x 200 m = 1000 m^2
Total: 2125 m^2
Substrate: 1.) Silt over gravel
2.) Silt over gravel
3.) Silt over sand/gravel
Total Samples: 22
Minimum - Maximum density (#/m^2): 0.0 - 13.0
Mean density #/m^2 (Standard deviation): 1.) 5.9 (2.42)
2.) 5.7 (3.51)
3.) 5.3 (4.16)
Total: 4.8 (3.38)

Species	Number Collected	Percent of Total	Percent legal	Population Estimate
Amblema plicata	1	0.9	100.0	100±205
Lasmigona complanata	1	0.9	NV**	125±256
Lampsilis teres	1	0.9	0.0	100±205
Obliquaria reflexa	1	0.9	0.0	83±171
Plectomerus dombeyanus	70	66.7	12.8	7067±2649
Quadrula pustulosa	2	1.9	0.0	183±260
Quadrula quadrula	29	27.6	10.3	2683±1183
Totals	105	99.8	12.4	10341±3289

* LHBFDS = Left hand bank facing downstream
** NV = No Commercial Value

Appendix 1.7 cont'd

Location: 2*M*, LHBFDS*, NW ¼ NW ¼ S26, T8N, R21W, Pope Co.
Stratum Size: 1.) 3 m x 250 m = 750 m^2
2.) 5 m x 175 m = 875 m^2
Total: 1625 m^2
Substrate: 1.) Silt/gravel and silt over sand
2.) Silt/gravel and silt over sand
Total Samples: 17
Minimum - Maximum density (#/m^2): 2.0 - 12.0
Mean density #/m^2 (Standard deviation): 1.) 6.1 (3.04)
2.) 4.9 (7.83)
Total: 5.5 (2.48)

Species	Number Collected	Percent of Total	Percent legal	Population Estimate
Anondonta suborbiculata	1	1.1	NV**	97±203
Fusconaia flava	2	2.2	0.0	188±255
Leptodea fragilis	1	1.1	0.0	94±201
Obliquaria reflexa	3	3.2	0.0	292±352
Plectomerus dombeyanus	51	54.8	23.5	4896±1299
Pyganodon grandis	2	2.2	NV	188±255
Quadrula pustulosa	10	10.7	10.0	938±713
Quadrula quadrula	23	24.7	17.4	2181±864
Totals	93	100.0	18.3	8874±1965

* LHBFDS = Left hand bank facing downstream
** NV = No Commercial Value

Appendix 1.7 cont'd

Location: 3*M*, LHBFDS*, NW ¼ NW ¼ S2, T7N, R21W, Pope Co.
Stratum Size: 1.) 10 m x 150 m = 1500 m^2
Substrate: 1.) Crumbly clay to silt over clay
Total Samples: 15
Minimum - Maximum density (#/m^2): 0.0 - 5.0
Mean density #/m^2 (Standard deviation): 1.) 2.9 (2.21)

Species	Number Collected	Percent of Total	Percent legal	Population Estimate
Plectomerus dombeyanus	37	84.1	13.5	3700±1119
Quadrula quadrula	7	15.9	0.0	700±751
Totals	44	100.0	11.4	4400±1232

* LHBFDS = Left hand bank facing downstream

Appendix 2.0 Scientific nomen and common names of Unionacea known to occur in Arkansas, 1997.

Scientific Nomen	Common Name
Actinonaias ligamentina	Mucket
Alasmidonta marginata	Elktoe
Alasmidonta viridis	Slippershell mussel
Amblema plicata	Threeridge
Anodonta suborbiculata	Flat floater
Anodontoides ferussacianus	Cylindrical papershell
Arcidens confragosus	Rock-pocketbook
Arkansia wheeleri	Arkansas rock-pocketbook
Cumberlandia monodonta	Spectaclecase
Cyclonaias tuberculata	Purple wartyback
Cyprogenia aberti	Western fanshell
Ellipsaria lineolata	Butterfly
Elliptio dilatata	Spike
Epioblasma florentina	Curtis pearlymussel
Epioblasma triquetra	Snuffbox
Epioblasma turgidula	Turgid blossom
Fusconaia ebena	Ebonyshell
Fusconaia flava	Wabash pigtoe
Fusconaia ozarkensis	Ozark pigtoe
Fusconaia subrotunda	Long-solid
Glebula rotundata	Round pearlymussel
Lampsilis abrupta	Pink mucket
Lampsilis cardium	Plain pocketbook
Lampsilis hydiana	Louisiana fatmucket
Lampsilis powelli	Arkansas fatmucket
Lampsilis rafinesqueana	Neosho mucket
Lampsilis reeviana	Arkansas broken-ray
Lampsilis satura	Sandbank pocketbook
Lampsilis siliquoidea	Fatmucket
Lampsilis strekeri	Speckled pocketbook
Lampsilis teres	Yellow sandshell
Lasmigona complanata	White heelsplitter
Lasmigona costata	Fluted-shell
Leptodea fragilis	Fragile papershell
Leptodea leptodon	Scaleshell
Ligumia recta	Black sandshell
Ligumia subrostrata	Pondmussel
Megalonaias nervosa	Washboard
Obliquaria reflexa	Threehorn wartyback
Obovaria jacksoniana	Southern hickorynut

Appendix 2.0 Cont'd

Scientific Nomen	Common Name
Obovaria olivaria	Hickorynut
Plectomerus dombeyanus	Bankclimber
Pleurobema coccineum	Round pigtoe
Pleurobema cordatum	Ohio pigtoe
Pleurobema pyramidatum	Pyramid pigtoe
Potamilus alatus	Pink heelsplitter
Potamilus capax	Fat pocketbook
Potamilus ohiensis	Pink papershell
Potamilus purpuratus	Bleufer
Ptychobranchus occidentalis	Ouachita kidneyshell
Pyganodon grandis	Giant floater
Quadrula apiculata	Warty pimpleback
Quadrula cylindrica	Rabbitsfoot
Quadrula fragosa	Winged mapleleaf
Quadrula metanevra	Monkeyface
Quadrula nodulata	Wartyback
Quadrula pustulosa	Pimpleback
Quadrula quadrula	Mapleleaf
Simpsonaias ambigua	Salamander mussel
Strophitus undulatus	Squawfoot
Toxolasma lividus	Purple lilliput
Toxolasma parvus	Lilliput
Toxolasma texasensis	Texas lilliput
Tritogonia verrucosa	Pistolgrip
Truncilla donaciformis	Fawnsfoot
Truncilla truncata	Deertoe
Uniomerus declivis	Tapered pondhorn
Uniomerus tetralasmus	Pondhorn
Utterbackii imbecillis	Paper pondshell
Venustaconcha ellipsiformis	Ellipse
Venustaconcha pleasii	Bleedingtooth mussel
Villosa arkansasensis	Ouachita creekshell
Villosa iris	Rainbow
Villosa lienosa	Little spectaclecase